Supplementary material for: Settling 648 unsolved BOBILib instances: published optima, new optimality proofs and infeasibility certificates

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This document supplements the paper. It contains the proofs behind the paper's methods, the verification protocol, a map from each result to its machine-checked (Lean) proof, a description of the data package, and the per-instance tables. Section A.4 sorts every result into three certificate classes (exact certificates, solver-certified numerical results, benchmark-accepted points). Notation follows the paper. Theorems, lemmas and equations are numbered by section (Theorem B.3, equation (A.1)); tables are numbered S1, S2, ... The counts quoted in the text are macros generated from the data package (`bobilib/results/`), and so are the tables of Section I. Sections B to F say for each result whether it is formalised in Lean ("Formalised:") or proved in prose only ("Not formalised").

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A. NOTATION AND VERIFICATION PROTOCOL

A.1 Notation and conventions

Bilevel problems in BOBILib form. The leader controls $x \in \mathbf{R}^{n_x}$ and the follower controls $y \in \mathbf{R}^{n_y}$. As in BOBILib,[1] the leader minimises:

$$\begin{aligned} \min_{x,y} \quad & c^\top x + d^\top y \\ \text{s.t.} \quad & Ax + By \geq a, \qquad x_i \in \mathbf{Z}\ (i \in I_u), \qquad y \in S(x), \\ S(x) = \ & \arg\min_{y'} \{ f^\top y' : Cx + Dy' \geq b,\ y'_i \in \mathbf{Z}\ (i \in I_l) \}. \end{aligned} \tag{A.1}$$

All data are rational. Variable bounds, equality rows and ranged rows of the instance files are written as pairs of rows of the form "$\geq$"; when an argument depends on the side of a row we say so. For a leader choice x the follower's problem is $P(x) = \min\{f^\top y : Cx + Dy \geq b,\ y_i \in \mathbf{Z}\ (i \in I_l)\}$ and its value is $\varphi(x) \in \mathbf{R} \cup \{-\infty, +\infty\}$ ($+\infty$ when $P(x)$ is infeasible). So $S(x) \neq \emptyset$ exactly when $\varphi(x)$ is finite and attained.

An upper-level row is a *coupling row* if it has a nonzero coefficient on a follower variable. A leader variable is a *linking variable* if its column in C is nonzero; x_L denotes the linking variables and C_L their columns, so $Cx = C_L x_L$, and $P(x)$, $\varphi(x)$ and $S(x)$ depend on x only through x_L.

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Definition A.1 (Bilevel feasible point, optimistic convention)**.** A point (x,y) is *bilevel feasible* if $Ax+By \geq a$, $x_i \in \mathbf{Z}$ for $i \in I_u$, and $y \in S(x)$. The bilevel optimum OPT is the infimum of $c^\top x + d^\top y$ over bilevel feasible points ($+\infty$ if there is none).

Problem (A.1) minimises over the pair (x,y), so when $S(x)$ has several elements the follower is assumed to pick the one that is best for the leader. This is the optimistic convention, which BOBILib uses for all of its reported values. For a fixed x the optimistic choice solves the *tie-break problem*

$$\min\bigl\{c^\top x + d^\top y : Ax+By \geq a,\ y \text{ feasible for } P(x),\ f^\top y \leq \varphi(x)\bigr\}.$$

High point relaxation. The *high point relaxation* (HPR) of (A.1) drops the condition $y \in S(x)$ and keeps every row, bound and integrality requirement of both levels and the leader's objective:

$$\text{HPR} = \min\bigl\{c^\top x + d^\top y : Ax+By \geq a,\ Cx+Dy \geq b,\ x_i \in \mathbf{Z}\ (i \in I_u),\ y_i \in \mathbf{Z}\ (i \in I_l)\bigr\}.$$

Packing interdiction. Items are $N = \{1,\dots,n\}$. The leader chooses $x \in \{0,1\}^n$ subject to budget rows $Gx \leq g$ with $G \in \mathbf{Z}^{m_u \times n}_{\geq 0}$ and $g \in \mathbf{Z}^{m_u}_{\geq 0}$. The follower then solves

$$\Phi(x) = \max\bigl\{p^\top y : Wy \leq c,\ y \leq \mathbf{1} - x,\ y \in \{0,1\}^n\bigr\}$$

with profits $p \in \mathbf{Z}^n_{\geq 0}$, weights $W \in \mathbf{Z}^{m \times n}_{\geq 0}$ and capacities $c \in \mathbf{Z}^m_{\geq 0}$, and the leader solves OPT $= \min\{\Phi(x) : Gx \leq g,\ x \in \{0,1\}^n\}$. In BOBILib form the leader's objective is $p^\top y$ and the follower's is $-p^\top y$, so the game is zero-sum: every $y \in S(x)$ gives the leader the same value $\Phi(x)$, and the optimistic and pessimistic readings coincide. We identify x with the set $X = \{j : x_j = 1\}$ of interdicted items and write $q(Y) = \sum_{j \in Y} q_j$ for a vector q and a set $Y \subseteq N$. With $K(T) = \max\{p(Y) : Y \subseteq T,\ \sum_{j \in Y} W_{\cdot j} \leq c\}$ the best packing of the items in T, we have $\Phi(X) = K(N \setminus X)$.

KIP (the knapsack interdiction problem) is the case $m_u = m = 1$. There we write $\gamma_j = G_{1j}$ for the cost of interdicting item j, $\beta = g_1$ for the leader's budget, $w_j = W_{1j}$ for the follower's weights and $c = c_1$ for its capacity, so $\Phi(X) = \max\{p(Y) : Y \subseteq N \setminus X,\ w(Y) \leq c\}$ and the leader's sets satisfy $\gamma(X) \leq \beta$. MKIP (multidimensional KIP) allows $m_u \geq 1$ and $m \geq 1$. In BOBILib's OR set, variant -100 has one follower row ($m = 1$) and several leader rows; variants -0 and -50 have several follower rows. The *subset-sum class* of KIP is $p = w$; then $\Phi(X)$ is the largest subset sum of $\{w_j : j \notin X\}$ that does not exceed c.

Clique and network interdiction. For edge clique interdiction, $G = (V,E)$ is a simple graph with $n = |V|$ and $m = |E|$; the leader deletes $D \subseteq E$ with $|D| \leq k$, and the follower takes a maximum clique of $G - D = (V, E \setminus D)$, of size $\omega(G-D)$. In the attack-protect network game the leader attacks at most Φ nodes and the follower then protects at most Λ nodes (Section F; there Φ is the attack budget, as in BOBILib's row name, not a value function).

Letters with two meanings. Following the paper, a few letters are reused; the second meaning is local to one section, and the context always says which is meant: c is the leader's cost vector in (A.1) and the follower's capacity in packing interdiction; D is the follower's matrix in (A.1) and the deleted edges in Section D; G is the leader's budget matrix and the graph of Section D; m is the number of follower rows in Section C and the number of edges in Section D; k is an index or a count elsewhere (the depth of a search node in Section B, a row in Section C) and the deletion budget in Section D; w denotes follower weights, the width of a band table (Section B) and node weights (Section F); Φ is the follower's value in packing interdiction and the attack budget in Section F. The value of the online game is Ω (Sections B and C), and ω is the clique number only; a recession direction of the follower is r (Section E), not to be confused with the cost vector d.

A.2 Tolerances of the two checkers

BOBILib's feasibility checker (described in Section 5.2 of the BOBILib paper by Thürauf, Kleinert, Ljubić, Ralphs and Schmidt,[1] and distributed with the library as a script) accepts a solution file in three steps: (i) the HPR is feasible; (ii) the file's point passes the MIPLIB 2017 solution checker[2] (version 1.0.3) on the HPR with tolerance 10^{-4} for the linear rows, scaled by the size of the row (a relative test), and 10^{-4} for integrality, and the stated leader objective matches; (iii) the follower's problem with the leader's variables fixed to the file's values is solved to optimality (by Gurobi with feasibility tolerance 10^{-6}), giving φ_{check}, and the file's follower value φ_{sol} must satisfy

$$\frac{|\varphi_{\text{sol}} - \varphi_{\text{check}}|}{10^{-10} + |\varphi_{\text{check}}|} \leq 10^{-4}. \tag{A.2}$$

(The paper prints the denominator without the absolute value; the script uses $|\varphi_{\text{check}}|$. The script also contains an optimistic tie-break test, which its driver does not call.) Our verifier `bobi verify` checks (a) every row, bound and integrality requirement of the HPR at the file's point with absolute tolerance 10^{-6}; (b) the stated leader objective; (c) the follower's problem with the leader fixed, re-solved by an exact knapsack oracle when there is one follower row and by a HiGHS[3] MILP otherwise, against test (A.2), where differences of at most 10^{-9} count as equal. A MILP stopped at a time limit gives a verdict only when its incumbent is strictly better than the file's value by more than (A.2) allows ("not optimal") or its proven bound passes (A.2) ("optimal"); otherwise the verdict is "unproven". The follower problem contains only the follower's rows, as in BOBILib's checker; coupling rows constrain the leader. Our re-certification `bobi certify` (used for the points with coupling rows) adds the optimistic tie-break test, and both of its optimality tests use proven dual bounds and the relative tolerance 10^{-5}.

The *tight test* `bobi verify --tight` keeps (a) and (b) but replaces the relative test (A.2) in (c) by

$$\varphi_{\text{sol}} - \varphi \leq \max\left(10^{-6}, 10^{-9}|\varphi|\right) \quad \text{for every follower optimum } \varphi \text{ the solver's proof allows,} \tag{A.3}$$

with the follower solved to zero relative gap and absolute gap 10^{-9} (for an integer-valued follower objective the bound is rounded up). A file that passes BOBILib's test but not (A.3) uses BOBILib's tolerance on the follower's value; it is a benchmark-accepted point (Section A.4). All 1,478 solution files of the data package pass the tight test (`results/notes/tight_verify.md`). `bobi certify --tight` applies the same bound with proven dual bounds and also bounds the tie-break: the leader's objective within $10^{-6} + 10^{-9}|F|$ of the proven optimistic tie-break value.

"Optimal" in this appendix means: a bilevel feasible point in the sense of Definition A.1 (checked within the tolerances above) whose value equals a proven lower bound. How rigorous the point and the bound are is given by their certificate class (Section A.4): exact, solver-certified or benchmark-accepted.

A.3 Verification protocol

Every claim in the paper rests on one or more of the following checks.

1. **Solution files.** Every solution file in the data package is written in BOBILib's format and passes `bobi verify` (Section A.2): all rows, bounds and integrality requirements of the HPR within 10^{-6}; the stated leader objective; and the follower's problem with the leader fixed, re-solved by the exact knapsack oracle (one follower row) or by a HiGHS MILP built from the model (otherwise), accepted by BOBILib's relative test (A.2). This covers the 583 imported KIP files, the 16 subset-sum files, the 690 OR files (270 with one follower knapsack, 420 with several), the 64 general files, the 29 files with coupling rows, the 60 clique files and the 36 network files; all of them passed again when this supplement was assembled. All 1,478 files also pass the tight test (A.3) (`bobi verify --tight`), which holds the follower's value to $\max(10^{-6}, 10^{-9}|\varphi|)$ of its proven optimum instead of BOBILib's relative 10^{-4}; the 26 files they replace, kept for the record, include 25 benchmark-accepted points.

The points with coupling rows and 62 of the 64 without also pass `bobi certify --tight` (proven dual bounds, optimistic tie-break; Section E). `bobi verify` is separate from BOBILib's solvers but not from this work's code: it belongs to the same Rust crate as the solvers and shares their instance reader and knapsack oracle.

2. **Optimality proofs.** Each "optimal" status rests on a lower bound of one of these kinds: a published proof or published lower bound (cited; for 446 of the 528 published optima a verified solution file here has the published value, for the other 82 only the published value is used); the exact subset-sum search (Section B), a computational proof supported by formally verified lemmas; a theorem with a checkable certificate (Mantel, Brouwer and partitions for clique interdiction, Section D); exhaustive enumeration with checked witnesses (Section F); or the HPR bound, where the model comparison is exact but the bound is MIPLIB's published floating-point optimum (Section E). "No bilevel feasible point" rests on Theorem E.1 with an exact rational check (all 40 instances). Section A.4 gives the certificate class of each.

3. **Separate provers for the subset-sum proofs.** Two Python provers that share no code with the Rust search, one written with the search and one in a separate review session that read neither, parse the instance files themselves and re-prove all 16 lower bounds with only the direct rule and the per-node rule (Theorem B.19), in integer arithmetic; both compare themselves with brute force on random small instances. Two other designs of the search, built separately, prove `interKP-200-100-6-3`, and one of them all four instances of class 6 with instance number $I = 3$. The review session compared every search binary with exhaustive enumeration on 2,990 random small instances (no mismatch).

4. **Tests of the code against its specification.** Unit tests compare the subset-sum search with brute force over every deletion, the threshold tables with the bitset recursion, the node tables with the online game defined directly and with enumeration of every deletion, and the table code with a direct transcription of the Lean recursion thr (also with costs not ascending, as after restricted copies). The network DP is compared with brute force over all follower points on random forests, and the whole enumeration with a brute-force bilevel optimum.

5. **Separate reviews.** The clique, network and coupled-row results were each re-derived in a separate review session with its own instance reader and different solvers or code: two exact maximum-clique searches, HiGHS and a branch and bound written in C for the part-size cases of Proposition D.7; a C implementation of the tree DP, compared with ours at every attack of 0 to Φ nodes on the 40 `inter-fire` instances that are forests (9,344,902 attacks, each instance once: the 36 trees and 4 `rndgraph-25` graphs, Section F), and a brute force based on Lemma F.1; and a second reading of the coupled files with HiGHS's own MPS reader. The notes of each review and the corrections it led to are in the `README.md` of the folder concerned. All the software, the experiments and the review sessions were carried out with an AI coding agent directed by the author; the review sessions were fresh sessions of the same agent that wrote their own code (threat 5 below).

6. **Exact arithmetic where it is cheap.** The clique inequalities (Mantel, Brouwer's threshold, the size vectors of Proposition D.7) and the branch-and-bound certificates of the part-size cases (`scripts/check_clique_certificates.py`), the recession directions and the comparison of each HPR with its MIPLIB model are checked with integers or exact rationals. The subset-sum provers, the knapsack oracle, the network DP and the table of the merged-row online bound work in integer arithmetic. The clique checker passes on all 170 case certificates (134,368 tree nodes, about 3.1 s).

7. **Machine-checked lemmas.** The Lean project (Lean 4,[4] with Mathlib[5]) proves the lemmas listed in Section G. It contains no `sorry`, and `#print axioms` shows only Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`) for each main theorem.

A.4 Certificate classes

The results of this supplement are established with different degrees of rigour. We sort them into three classes and use the classes throughout.

Definition A.2 (Certificate classes)**.**

(i) An *exact certificate* is checked in exact rational or integer arithmetic, or consists of a mathematical proof together with an exhaustive computation in integer arithmetic. No tolerance enters it. It still relies on the code that carries out the computation, which is tested and, for its main lemmas, backed by machine-checked proofs, but is not itself verified.

(ii) A *solver-certified numerical result* rests on a floating-point LP, MILP or CP solver run with tolerances: a proven dual bound, an optimality or infeasibility verdict, or a point whose rows hold within 10^{-6} and whose follower value lies within solver precision of a solver's proven bound on the follower optimum at the point's leader values, that is, it passes the tight test (A.3). It is correct if the solver's numerics are; it is not an exact-arithmetic proof.

(iii) A *benchmark-accepted point* passes BOBILib's tolerance-based checker (Section A.2), and possibly `bobi verify`, only because of their tolerances: its follower value lies above the follower's optimum by more than solver precision (it fails the tight test (A.3)) but within the relative 10^{-4} of test (A.2), or its rows hold only within BOBILib's 10^{-4}. It need not be bilevel feasible in the sense of Definition A.1. BOBILib's own solution file of `general20-30-20-20-20-2` is the example (Remark E.7).

A result is in the weakest class of the steps it uses. Results taken from the literature (the published optima, the published lower bounds of the OR instances with several follower knapsacks, Brouwer's theorem) are used as published and not re-certified; where a verified file has a published optimal value, the file is classified with its family below. By the conventions of BOBILib and MIPLIB, a point and a lower bound that agree, both of class (ii), make a value "optimal"; the paper and this supplement use the word in that sense, and say so where a bound is not exact. The subset-sum results are computational proofs supported by formally verified lemmas (Section B), not formally verified solutions. The list below assigns each kind of result to its class; the column `certificate_class` of `instances.csv` (Section H.2) gives the class of each instance's result, and there a published optimum, used as published, counts as class (ii). Of the 878 instances with a result, 115 are class (i), 763 class (ii) and 0 class (iii).

Certificate class of each kind of result (Definition A.2).

Result	Class	Basis
16 subset-sum KIP optima (lower bounds; Section B)	(i)	computational proofs supported by formally verified lemmas: exhaustive integer search with Lean-proved pruning rules, repeated by two integer-only provers
Points with one follower knapsack: KIP files, the 16 subset-sum files, 270 OR points (Sections B, C)	(i)	integer data and 0/1 values, so no tolerance applies; follower optimum by the exact integer knapsack oracle
OR lower bounds, one follower knapsack, from the merged-row online bound (166 instances; 166 of the 191 improved bounds)	(i)	Theorem C.5(b): integer data with a proven rounding direction, integer DP

(continued)

Result	Class	Basis
OR lower bounds, one follower knapsack, from the LP bracket (104 instances; 25 improved bounds)	(ii)	HiGHS's dual bound (Theorem C.4), used only when finite and lowered by a safety margin before rounding up
OR points with several follower knapsacks (420)	(ii)	leader rows exact; follower optimum by HiGHS's dual bound (Proposition C.6)
OR upper bounds $U(\hat{x})$ (follower LP) in Table S5	(ii)	floating-point LP value
Clique: value at least 3 (Mantel), upper bounds 3 and 4 (partition files, 60 instances)	(i)	integer arithmetic, Lean (Theorem D.2, Proposition D.3); exact maximum-clique search for the follower
Clique: value at least 4 (30 instances at density 0.9)	(i)	Brouwer's published theorem (Theorem D.4) and the 170 part-size cases, each with a branch-and-bound certificate that the integer checker `scripts/check_clique_certificates.py` verifies (all pass); the solver verdicts on the same cases are class (ii)
Network: the 2 optima and the 34 trees re-derived (Section F)	(i)	every attack enumerated, 0/1 witnesses checked against the model, follower optimum at the minimiser by the DP (Proposition F.3) and the review's brute force
Network: 12 `rndgraph-25` values re-derived with HiGHS at every attack	(ii)	HiGHS as the follower (the review's brute force re-derives them exactly)
No bilevel feasible point, all 40 instances (Section E)	(i)	recession direction checked in rational arithmetic (Theorem E.1); for `supportcase42_0_100` and `supportcase42_50_50` a unit direction found in exact arithmetic
HPR equal to the MIPLIB model (Section E)	(i)	exact rational comparison of the two models
The 7 HPR-bound optima (Table S11)	(ii)	MIPLIB's published floating-point optima; optimal by accepted benchmark conventions, not by an exact-arithmetic proof (`noswot`, `ex10` and `neos-1445765` safe by integrality rounding given the published bound, `pigeon-10` likewise, its values being multiples of 1,000; `timtab1` agrees numerically)
Points without coupling rows (64 files, 17 of them re-derived under the strict tolerance)	(ii)	rows within 10^{-6}; follower optimum by HiGHS; all pass the tight test (A.3), and 62 also `bobi certify --tight`
Points with coupling rows (29 files, 9 of them re-derived)	(ii)	all 29 pass `bobi certify --tight`: proven dual bounds, follower within the tight test of its optimum, optimistic tie-break
25 of the 26 superseded files (`*/tolerance/`)	(iii)	fail the tight test and pass BOBILib's (follower about 10^{-7} relative above its optimum); kept for the record, not used for any result
Published optima (528)	—	used as published, not re-certified; counted as class (ii) in `instances.csv`
BOBILib's file of `general20-30-20-20-20-2`	(iii)	the example of Remark E.7; rejected by `bobi verify`
Lean lemmas (Section G)	—	machine-checked proofs about abstract models; they support class (i) results and are not results on instances

A.5 Threats to validity

1. **Trusted solvers.** The class (ii) results of Section A.4 come from MILP, LP or CP solvers: follower optimality for followers with several rows and for general instances (HiGHS), the dual bounds in the OR bracket (Theorem C.4) and the MIPLIB optima behind the HPR bounds. A solver bug would go unnoticed there. The 170 part-size verdicts behind $w_3(G) > k$ and the follower optimum at the minimising attack in the network proof are also solver results, but neither depends on a solver any more: each part-size case has a branch-and-bound certificate checked by the integer checker, and the network optimum is re-derived by the review's brute force.

2. **Floating point and tolerances.** Points of the general families are feasible within 10^{-6} and follower-optimal within the tight test (A.3), not exactly. The 17 files without coupling rows and 8 with coupling rows whose follower had used BOBILib's relative 10^{-4} (benchmark-accepted points) were replaced by points re-derived under the strict tolerance (Section E); no current file is benchmark-accepted. The MIPLIB optima behind the 7 HPR-bound optima are published floating-point values, so those optima hold by accepted benchmark conventions, not by an exact-arithmetic proof (Section E). All 40 ray certificates are checked in exact rational arithmetic. The merged-row online bound (Theorem C.5) computes its integer data in double precision; part (b) of the theorem shows that the rounding error cannot break the bound for the OR data, with margins below 10^{-10}. The LP bracket uses HiGHS's floating-point dual bound only when it is finite, lowered by $10^{-6}\max(1, |L|)$ before rounding up (Section C). The window rule of the subset-sum search uses a floating-point fractional bound; the 16 proofs are confirmed by the integer-only provers above, which do not use that rule.

3. **Scope of the Lean proofs.** Lean proves lemmas about abstract models, not the code. That the Rust loops compute the recursions, the range shortcuts and early exits of the table code, the bitset and cache code, and the absence of integer overflow are argued in doc comments and tested, not formalised. That the instance files have the stated parameters (n, m, k, budgets) is checked by scripts, not by Lean. Section G lists what is not formalised.

4. **Results taken from the literature.** Brouwer's theorem[6] was read in the scanned 1981 report; a refereed proof is Tyomkyn and Uzzell's,[7] and other refereed papers give the same statement (Section D). It is neither formalised nor reproved here. Published optima (Weninger and Fukasawa[8] for KIP, Fischetti et al.[9] for imkp, Lozano, Bergman and Cire[10] for OR, MIPLIB 2017 optima[2] for the HPR bounds) and the published lower bounds of the OR instances with several follower knapsacks are used as reported. For 446 of the 528 published optima a solution file at the published value is verified here; for 82 (52 imkp, 18 OR, 8 MIPLIB-derived, 4 generalized) only the published value is used. The published OR points include values of Lefebvre and Schmidt[11] that are not verified; they are taken only from runs that ended before the time limit, as in their own analysis (Sections C.5 and A.6).

5. **Common-mode errors.** The solvers, the verifier, the second prover and the review sessions were all written with the same AI coding agent directed by the same author. Fresh sessions with their own code and instance readers reduce, but do not remove, the risk of a shared misreading of the file format or a shared conceptual slip. The checks least exposed to it are the Lean proofs, which Lean's kernel checks, and short programs whose verdicts can be audited by hand: the exact rational checks of the recession directions, the exact HPR comparisons and the brute-force comparisons.

6. **Instance reading.** Every argument depends on reading the MPS and AUX files correctly. Each family's model is checked row by row against its expected pattern, and the key results were re-derived with separately written readers (two subset-sum provers, the exact ray and HPR scripts, the parsers of the review sessions).

7. **Running times.** All times were measured on a shared machine with four cores that was often oversubscribed; they are indicative only and do not affect any proof.

A.6 Lefebvre and Schmidt's run files

Lefebvre and Schmidt publish per-instance run files for their Dantzig–Wolfe reformulation and its heuristics on most of BOBILib.[11] The comparison of all their rows with this package is in `lefebvre_schmidt.md` and, with file and line numbers, in `lefebvre_schmidt_discrepancies.tsv` (both in `results/notes/`). Most rows agree with everything proven here, and none of the following concerns their reformulation itself.

- Rows that end at the time limit can carry the status "optimal" with a value that no bilevel feasible point has: 366 OR instances get the value 0, which published and elementary lower bounds rule out. It appears to be the value of the high point relaxation, kept as the incumbent when the bilevel-feasibility separation ran out of time. Their own analysis treats such rows as unsolved, so only their best-of-all-methods comparison (their Figure 6) is affected. This package does the same: their values count as published points only from runs that ended before the time limit (and, for their heuristic PADM, with status `Feasible`), and never as optima (`scripts/ls_mkip_points.py`; Table S6 lists the OR instances where a time-limit value would otherwise be the best published point).
- In the other direction, 90 instances with a verified point are reported infeasible, and two reported optima are worse than verified points; these runs enter their paper's counts of solved instances.
- Their heuristic point on `highschool1-aigio_0_100`, which their paper counts among its feasible points for open instances, cannot be bilevel feasible: the instance has none (an exact recession direction, Section E). The instance violates the boundedness assumption of their method.
- Their exact runs report `ex10_0_100` optimal at 100, which is correct (Section E); their paper does not state it, so it is counted as optimal by this work, at the MIPLIB bound, not as published.

B. KNAPSACK INTERDICTION PROOFS

This section proves the rules behind the exact search for subset-sum knapsack interdiction (KIP with $p = w$, Section A). The search answers a yes/no question: given an integer ℓ, does every leader set within budget leave the follower at least ℓ? A "yes" proves OPT $\geq \ell$; a "no" comes with a leader set whose follower value is below ℓ. Repeating with ℓ equal to the incumbent value proves optimality (Corollary B.25).

B.1 Monotonicity and the count formulation

Lemma B.1 (Monotonicity of interdiction)**.** *Let $\mathscr{F}$ be any family of subsets of N (the follower's feasible sets) and $\Phi_{\mathscr{F}}(X) = \max\{p(Y) : Y \in \mathscr{F},\ Y \cap X = \emptyset\}$ with $p \geq 0$ (a maximum over no set is 0). If $X \subseteq X'$ then $\Phi_{\mathscr{F}}(X') \leq \Phi_{\mathscr{F}}(X)$. If $\mathscr{F}$ is closed under taking subsets, then $\Phi_{\mathscr{F}}(X) = \max\{p(Y \setminus X) : Y \in \mathscr{F}\}$. The knapsack follower $\mathscr{F} = \{Y : \sum_{j \in Y} W_{\cdot j} \leq c\}$ is closed under taking subsets because $W \geq 0$; in particular $\Phi(X') \leq \Phi(X)$ for $X \subseteq X'$.*

Proof. Every $Y \in \mathscr{F}$ that avoids X' also avoids X, so the maximum for X' is over a subfamily. For the second claim, a set $Y \in \mathscr{F}$ avoiding X has $p(Y) = p(Y \setminus X)$, which gives "$\leq$"; conversely, for $Y \in \mathscr{F}$ the set $Y \setminus X$ lies in $\mathscr{F}$ and avoids X, which gives "$\geq$". A subset of a packing has no larger weight in any row because $W \geq 0$. □

Formalised: `Bobi/Knapsack.lean`, theorems `follower_antitone`, `bestAvoiding_antitone`, `bestAvoiding_eq_sup_sdiff`, `packings_downward_closed` and `K_eq_bestAvoiding`.

Lemma B.2 (Cheapest copies first)**.** *Consider KIP and a class $Q \subseteq N$ of identical items ($p_i = p_j$ and $w_i = w_j$ for $i, j \in Q$). For every leader set X there is a leader set X' with $X' \setminus Q = X \setminus Q$, $|X' \cap Q| = |X \cap Q|$, $\gamma_a \le \gamma_b$ for all $a \in X' \cap Q$ and $b \in Q \setminus X'$, $\gamma(X') \le \gamma(X)$ and $\Phi(X') = \Phi(X)$.*

Proof. First a single swap: if $i \in X \cap Q$, $j \in Q \setminus X$ and $\gamma_j \le \gamma_i$, let $X^s = (X \setminus \{i\}) \cup \{j\}$. The transposition of i and j preserves p and w and maps $N \setminus X$ onto $N \setminus X^s$, so it maps the packings avoiding X bijectively onto those avoiding X^s with the same profits; hence $\Phi(X^s) = \Phi(X)$, and $\gamma(X^s) = \gamma(X) - \gamma_i + \gamma_j \le \gamma(X)$. Now induct on the integer $\gamma(X)$. If some $i \in X \cap Q$ and $j \in Q \setminus X$ have $\gamma_j < \gamma_i$, the swap lowers the cost strictly, keeps $X \setminus Q$, $|X \cap Q|$ and Φ, and the induction hypothesis applies to X^s. Otherwise $X' = X$ works. □

Formalised: `Bobi/Knapsack.lean`, theorems `K_map_equiv`, `swap_block` and `cheapest_first`.

Definition B.3 (Class instance)**.** A *class instance* is a list of classes $1, \dots, L$; class i has a value $v_i \in \mathbf{Z}_{\ge 1}$ and n_i copies with deletion costs $\gamma_{i,1}, \dots, \gamma_{i,n_i} \in \mathbf{Z}_{\ge 0}$ (in a fixed order, not necessarily ascending). Let $P_i(r) = \sum_{h \le r} \gamma_{i,h}$ for $0 \le r \le n_i$. A *deletion* is a count vector $r = (r_1, \dots, r_L)$ with $0 \le r_i \le n_i$; its cost is $P(r) = \sum_i P_i(r_i)$ and its survivors form the multiset $R(r)$ with $n_i - r_i$ copies of v_i for each i. For a finite multiset M of nonnegative integers, $\mathrm{SS}(M)$ is the set of sums of its sub-multisets ($0 \in \mathrm{SS}(M)$). Given integers $0 \le lo \le hi$ and a budget $\beta \ge 0$, the instance is *safe* if every deletion with $P(r) \le \beta$ has $\mathrm{SS}(R(r)) \cap [lo, hi] \ne \emptyset$; a deletion within budget with $\mathrm{SS}(R(r)) \cap [lo, hi] = \emptyset$ is *blocking*.

Proposition B.4 (Reduction of subset-sum KIP)**.** *Consider subset-sum KIP with data γ, β, w, c (all nonnegative integers). Let $F = \{j : \gamma_j = 0\}$, $\mathscr{L} = \{j \notin F : 1 \le w_j \le c\}$ and $g = \gcd\{w_j : j \in \mathscr{L}\}$ ($g = 1$ if $\mathscr{L} = \emptyset$). Group $\mathscr{L}$ into classes by the value w_j/g, with each class's costs sorted ascending, and let $hi = \lfloor c/g \rfloor$. For a deletion r let $X(r)$ be F together with the r_i cheapest items of each class. Then*

1. *$\gamma(X(r)) = P(r)$ and $\Phi(X(r)) = g \cdot \max\big(\mathrm{SS}(R(r)) \cap [0, hi]\big)$;*
2. *for every leader set X with $\gamma(X) \le \beta$ there is a deletion r with $P(r) \le \gamma(X)$ and $\Phi(X(r)) \le \Phi(X)$;*
3. *for every integer $\ell \ge 0$ with $lo = \lceil \ell/g \rceil \le hi$: $\mathrm{OPT} \ge \ell$ if and only if the class instance is safe for the band $[lo, hi]$ and budget β; and r is blocking if and only if $P(r) \le \beta$ and $\Phi(X(r)) < \ell$.*

Proof. (1) Items of F cost nothing. A packing avoiding $X(r)$ contains no item of F; an item outside $F \cup \mathscr{L}$ has $w_j = 0$ (then $p_j = 0$) or $w_j > c$ (then it is in no packing), so dropping such items changes neither weight nor profit. Every weight in $\mathscr{L}$ is g times its class value, so a set $Y \subseteq \mathscr{L}$ has $w(Y) \le c$ iff its value sum is at most $\lfloor c/g \rfloor = hi$, and the value sums of packings of $\mathscr{L} \setminus X(r)$ are exactly $\mathrm{SS}(R(r))$. (2) Let $X_1 = X \cup F$: $\gamma(X_1) = \gamma(X)$ and $\Phi(X_1) \le \Phi(X)$ by Lemma B.1. Items outside $F \cup \mathscr{L}$ do not affect Φ (as in (1)) and blocking them costs something, so remove them from X_1 (cost does not increase, Φ unchanged). Apply Lemma B.2 to each class in turn. The result blocks in each class i some r_i copies, each no costlier than every unblocked copy, so its cost in class i is the sum of the r_i smallest costs, $P_i(r_i)$, and its follower value is that of $X(r)$. (3) Since $gs \ge \ell$ iff $s \ge lo$ for integers s, (1) gives: $\Phi(X(r)) \ge \ell$ iff $\mathrm{SS}(R(r)) \cap [lo, hi] \ne \emptyset$. If the instance is safe, (2) gives $\Phi(X) \ge \Phi(X(r)) \ge \ell$ for every X within budget. If a deletion r is blocking, $X(r)$ is within budget with $\Phi(X(r)) < \ell$, so $\mathrm{OPT} < \ell$. □

Not formalised as a whole; its two ingredients are Lemmas B.1 and B.2.

If $\lceil \ell/g \rceil > hi$, the question is trivial: the leader set $F = X(0)$ costs nothing and, by part (1), $\Phi(F) \le g \cdot hi < \ell$, so $\mathrm{OPT} < \ell$ and the deletion $r = 0$ is a witness. From now on we work with a class instance, a band $[lo, hi]$ with $lo \le hi$ and a budget β. All results below hold for arbitrary nonnegative costs in the count formulation; where ascending costs are needed (deletions of arbitrary copies) we say so.

B.2 Windows of reachable sums and the stall test

Lemma B.5 (Window extension)**.** *Let M be a multiset with $[a,b] \subseteq \mathrm{SS}(M)$, $a \le b$. If a value v satisfies $v \le b-a+1$ then $[a,b+v] \subseteq \mathrm{SS}(M \uplus \{v\})$. More generally, if values $v_1,\dots,v_q$ satisfy $v_h \le (b-a+1)+v_1+\cdots+v_{h-1}$ for every h, then $[a,\, b+\sum_h v_h] \subseteq \mathrm{SS}(M \uplus \{v_1,\dots,v_q\})$.*

Proof. A target $t \in [a,b]$ is a sum of M. A target $t \in (b,b+v]$ has $a \le t-v \le b$, because $t > b$ and $v \le b-a+1$; so t is v plus a sum of M. The general statement follows by induction on q: after adding $v_1,\dots,v_{h-1}$ the window is $[a,b+v_1+\cdots+v_{h-1}]$, of width $(b-a+1)+v_1+\cdots+v_{h-1} \ge v_h$. □

Formalised: `Bobi/SubsetSum.lean`, theorems `window_extend` and `window_extend_all`.

Lemma B.6 (Survivors that fit)**.** *Let $[a,b] \subseteq \mathrm{SS}(M)$ with $a \le b$ and $\theta = b-a+1$. Let $(v_1,\eta_1),\dots,(v_q,\eta_q)$ be a list of copies with values v_h and flags $\eta_h \in \{kept, deleted\}$. Write $\mu_h = \sum_{h'<h} v_{h'}$ for the mass before copy h and δ_h for the deleted mass before copy h. If $\delta_h + v_h \le \theta + \mu_h$ for every h, then $[a,\, b+\sigma] \subseteq \mathrm{SS}(M \uplus K)$, where K is the multiset of kept copies and σ its total.*

Proof. Let h be a kept copy. The kept mass before it is $\mu_h - \delta_h$, and the hypothesis says $v_h \le \theta + (\mu_h - \delta_h)$. So the kept copies, in list order, satisfy the hypothesis of Lemma B.5. □

Formalised: `Bobi/SubsetSum.lean`, theorems `fits_survivors` and `window_survivors` (`SubsetSum.perm`: subset sums do not depend on the order).

Lemma B.7 (Fractional deletion bound)**.** *Let copies h have values $v_h > 0$ and costs $\gamma_h \ge 0$, sorted so that $\gamma_1/v_1 \le \gamma_2/v_2 \le \cdots$. For $d > 0$ let $\Psi(d) = \min\{\sum_h \gamma_h z_h : \sum_h v_h z_h \ge d,\ 0 \le z \le 1\}$ ($+\infty$ if $\sum_h v_h < d$). The greedy solution, which fills copies in this order until mass d is reached, is optimal, and every set H of copies with $\sum_{h\in H} v_h \ge d$ has $\sum_{h\in H} \gamma_h \ge \Psi(d)$.*

Proof. The indicator vector of H is feasible, which proves the last claim. Let $m_h = v_h z_h$ and $\rho_h = \gamma_h/v_h$, so the cost is $\sum_h \rho_h m_h$ with $0 \le m_h \le v_h$ and $\sum_h m_h \ge d$. Let m^* be the greedy allocation, λ the ratio of the copy where it stops, and m any feasible allocation. Then $\sum_h \rho_h(m_h - m_h^*) = \sum_h (\rho_h - \lambda)(m_h - m_h^*) + \lambda \sum_h (m_h - m_h^*)$. Each term of the first sum is nonnegative ($m_h^* = v_h \ge m_h$ when $\rho_h < \lambda$, $m_h^* = 0 \le m_h$ when $\rho_h > \lambda$), and the last term is nonnegative because $\sum_h m_h^* = d \le \sum_h m_h$ and $\lambda \ge 0$. □

Not formalised (a standard linear programming fact).

The next rule is stated for a search node (Section B.8): the decided classes have left a multiset M of survivors, $S = \mathrm{SS}(M) \cap [0, hi]$, and the remaining (undecided) classes may be deleted with residual budget β'. A *completion* is a count vector for the undecided classes of cost at most β'.

Proposition B.8 (Window rule with stall test)**.** *Order the undecided classes by ascending value, $u_1,\dots,u_q$ (the proof works for any order; ascending value is the code's choice), and let $M_i = \sum_{i'<i} n_{u_{i'}} v_{u_{i'}}$ be the mass of the classes before u_i. Let Δ be the largest mass that a completion deletes and σ_0 the total undecided mass minus Δ. Let $[a,b] \subseteq S$ be a run of consecutive integers with $b < lo \le b + \sigma_0$ and width $\theta = b-a+1$, and put $D_i = M_i - v_{u_i} + \theta + 1$. Suppose that for every i: $D_i > 0$, and either $i = 1$ or the fractional bound of Lemma B.7 over the copies of $u_1,\dots,u_{i-1}$ satisfies $\Psi_i(D_i) > \beta'$. Then every completion leaves survivors R with $lo \in \mathrm{SS}(M \uplus R)$; the node is safe.*

Proof. Fix a completion and list the undecided copies class by class in the order $u_1,\dots,u_q$, flagging the deleted ones. Let e_i be the mass the completion deletes from $u_1,\dots,u_{i-1}$. Those deleted copies form a set of total cost at most β', so $e_i \ge D_i$ would contradict $\Psi_i(D_i) > \beta'$ by Lemma B.7; for $i = 1$, $e_1 = 0 < D_1$. Hence $e_i \le D_i - 1$. At the copy number t ($t = 0, 1, \dots$) of class u_i, the mass before it is $M_i + t v_{u_i}$ and the deleted mass before it is at most $e_i + t v_{u_i}$, so deleted mass plus v_{u_i} is at most $D_i - 1 + (t+1)v_{u_i} = \theta + M_i + t v_{u_i}$. Lemma B.6 gives $[a, b+\sigma] \subseteq \mathrm{SS}(M \uplus R)$ with $\sigma \ge \sigma_0$ the surviving mass, and $a \le b < lo \le b + \sigma_0$ puts lo in that window. □

The per-copy hypothesis is formalised (`Bobi/SubsetSum.lean`, theorem `window_survivors`); the reduction to one test per class and the use of Lemma B.7 are argued here and in the doc comment of the code, not formalised. Δ is computed exactly by a bounded-knapsack recursion over the undecided classes.

B.3 The online relaxation

The online relaxation of Weninger and Fukasawa[8,12] (Theorem 2 in Section 3 of their preprint) applies to general KIP. Fix a list $o = (j_1, \dots, j_L)$ of distinct items, any subset of N in any order. The *online game* on o has $2L$ rounds: at item j_h the leader first decides whether to block it (allowed if its cost fits the remaining budget); if the leader does not block it, the follower decides whether to take it (allowed if its weight fits the remaining capacity). The follower collects the profits of the items it takes. Both players see all earlier moves. The value of this finite game of perfect information is given by backward induction: $\Omega(\emptyset, \beta', \kappa) = 0$ and, for $o = (j, o')$,

$$\Omega(o, \beta', \kappa) = \begin{cases} \min\{\Omega(o', \beta' - \gamma_j, \kappa),\ T\} & \text{if } \gamma_j \le \beta', \\ T & \text{otherwise,} \end{cases}$$

$$T = \begin{cases} \max\{\Omega(o', \beta', \kappa),\ \Omega(o', \beta', \kappa - w_j) + p_j\} & \text{if } w_j \le \kappa, \\ \Omega(o', \beta', \kappa) & \text{otherwise.} \end{cases}$$

Theorem B.9 (Online lower bound, any play order)**.** *For every list o of distinct items and every $X \subseteq N$ with $\gamma(X) \le \beta$, $\Omega(o, \beta, c) \le K(o \setminus X) \le \Phi(X)$, where $K(T)$ is the best packing of the items of T with capacity c. Hence $\Omega(o, \beta, c) \le \mathrm{OPT}$.*

Proof. In the online game the leader may adapt to the follower's earlier moves; the real game corresponds to the leader strategy "block exactly the items of X", which does not adapt. Formally, we prove by induction on the length of o, for all $\beta', \kappa \ge 0$ and all $X \subseteq o$ with $\gamma(X) \le \beta'$, that $\Omega(o, \beta', \kappa) \le K_\kappa(o \setminus X)$ (best packing with capacity κ). For $o = \emptyset$ both sides are 0. Let $o = (j, o')$. If $j \in X$, then $\gamma_j \le \gamma(X) \le \beta'$, so the blocking branch is offered and $\Omega(o, \beta', \kappa) \le \Omega(o', \beta' - \gamma_j, \kappa) \le K_\kappa(o' \setminus X) = K_\kappa(o \setminus X)$ by induction applied to $X \setminus \{j\}$. If $j \notin X$, then $X \subseteq o'$ and $\Omega(o, \beta', \kappa) \le T$. By induction $\Omega(o', \beta', \kappa) \le K_\kappa(o' \setminus X)$ and, if $w_j \le \kappa$, $\Omega(o', \beta', \kappa - w_j) + p_j \le K_{\kappa - w_j}(o' \setminus X) + p_j \le K_\kappa((o' \setminus X) \cup \{j\})$ (add j to an optimal packing). So in both cases of T we get $T \le K_\kappa((o' \setminus X) \cup \{j\}) = K_\kappa(o \setminus X)$. For an arbitrary X within budget apply this to $X \cap o$ (its cost is no larger and $o \setminus (X \cap o) = o \setminus X$). Finally $K(o \setminus X) \le K(N \setminus X) = \Phi(X)$ because more items can only help. □

Formalised: `Bobi/Online.lean`, theorems `online_le_K`, `online_le_K_any` and `online_lower_bound`.

Remark B.10. The bound need not be tight. In Weninger and Fukasawa's Table 1 (profits $(1,1,2)$, leader costs $(1,1,2)$, follower weights $(1,1,3)$, $\beta = 2$, $c = 3$) the online game in the play order $(1,2,3)$ is worth 1 and the bilevel optimum is 2 (Lean: `table1_online`, `table1_opt`, `table1_block`, checked by evaluation).

B.4 Robust tables with an end set

For subset-sum instances the same idea is applied class by class, in the count formulation. Let $Q = (q_1, \dots, q_L)$ be a list of classes (the *play order*) and $A \subseteq \mathbf{Z}_{\ge 0}$ an *end set*. The game $\Gamma_A(Q, \beta', t)$ starts with residual t and budget β'; at class q the leader announces $r \le n_q$ with $P_q(r)$ at most the remaining budget, then the follower takes $j \le n_q - r$ copies with jv_q at most the current residual, which decreases by jv_q; the follower wins if the final residual lies in A.

Definition B.11 (Robust tables)**.** $G_A(\emptyset,\beta') = A$ and, for $Q = (q, Q')$,

$$t \in G_A(Q,\beta') \iff \forall r \le n_q \text{ with } P_q(r) \le \beta' \;\; \exists j \le n_q - r : \;\; jv_q \le t \text{ and } t - jv_q \in G_A(Q', \beta' - P_q(r)).$$

So $G_A(Q,\beta')$ is the set of residuals from which the follower wins $\Gamma_A(Q,\beta',\cdot)$. The robust table is $H = G_{\{0\}}$ and the band table of width $w \ge 1$ is $G_w = G_{[0,w-1]}$.

Theorem B.12 (Soundness of the tables)**.** *If $t \in G_A(Q,\beta')$ and r is a deletion of the classes of Q with $P(r) \le \beta'$, then there are $s \in \mathrm{SS}(R(r))$ and $a \in A$ with $s + a = t$. In particular $t \in H(Q,\beta')$ gives $t \in \mathrm{SS}(R(r))$, and $t \in G_w(Q,\beta')$ gives some $s \in \mathrm{SS}(R(r))$ with $s \le t < s + w$. If each class's costs are ascending, the same holds for a deletion of arbitrary copies of total cost at most β'.*

Proof. Induction on L. For $L = 0$ take $s = 0$, $a = t$. For $Q = (q, Q')$: $P_q(r_1) \le P(r) \le \beta'$, so the definition gives $j \le n_q - r_1$ with $t - jv_q \in G_A(Q', \beta' - P_q(r_1))$, and the rest of r costs at most $\beta' - P_q(r_1)$. By induction $t - jv_q = s' + a$ with s' a subset sum of the survivors of Q'; then $s = jv_q + s'$ uses j of the $n_q - r_1$ survivors of q. For deletions of arbitrary copies with ascending costs, deleting r_i copies of class i costs at least $P_i(r_i)$, and the survivors are again $n_i - r_i$ copies of v_i. □

Formalised: `Bobi/EndTable.lean`, theorems `GA_sound`, `GA_sound_any`, `G_iff_GA`, `H_iff_GA`, `G_sound_of_GA`, `H_sound_of_GA`; `Bobi/RobustTable.lean`, `H_sound`, `H_sound_any`, `take_sum_le`, `counts_of_deletion`; `Bobi/BandTable.lean`, `G_sound`, `G_sound_any`, `G_one`, `G_sound_one`.

Proposition B.13 (Any play order)**.** *Let Q be the undecided classes in the search order and $\pi(Q)$ any permutation of them. If $t \in G_A(\pi(Q),\beta')$, the conclusion of Theorem B.12 holds for every deletion r of Q with $P(r) \le \beta'$.*

Proof. Permuting the pairs (class, count) permutes the survivors and leaves the cost unchanged, and subset sums depend only on the multiset. Apply Theorem B.12 to the permuted deletion. □

Formalised: `Bobi/EndTable.lean`, theorems `counts_perm`, `remaining_eq_flatten`, `totalCost_eq_sum`, `remaining_perm`, `totalCost_perm`, `GA_sound_perm` and `GA_sound_perm_any`.

Corollary B.14 (Root-table rules)**.** *Let M be the decided survivors and Q the undecided classes (any play order). (i) If $s \in \mathrm{SS}(M)$ and $t \in H(Q,\beta')$, every completion leaves $s + t \in \mathrm{SS}(M \uplus R)$. (ii) If $[a,b] \subseteq \mathrm{SS}(M)$, $t \in G_w(Q,\beta')$, $a + t \le \tau$ and $\tau + w \le b + t + 1$, every completion leaves $\tau \in \mathrm{SS}(M \uplus R)$. The search prunes when $s + t$, respectively τ, lies in $[lo, hi]$.*

Proof. (i) Theorem B.12 gives $t \in \mathrm{SS}(R)$. (ii) It gives $s' \in \mathrm{SS}(R)$ with $s' \le t < s' + w$; then $a \le \tau - t \le \tau - s'$ and $\tau - s' \le \tau - t + w - 1 \le b$, so $\tau - s' \in \mathrm{SS}(M)$. □

Formalised: `Bobi/RobustTable.lean`, theorem `covers_sound`; `Bobi/BandTable.lean`, `band_covers_sound`, `covers_sound_of_band`; `Bobi/Threshold.lean`, `covers_threshold_sound`, `covers_threshold_sound_perm`, `band_covers_threshold_sound`, `band_covers_threshold_sound_perm` (the same on stored thresholds).

Lemma B.15 (Antitone in the budget)**.** *If $\beta'' \le \beta'$ then $G_A(Q,\beta') \subseteq G_A(Q,\beta'')$. If $A \subseteq A'$ then $G_A(Q,\beta') \subseteq G_{A'}(Q,\beta')$.*

Proof. Induction on L; the base case does not depend on the budget. A deletion r with $P_q(r) \le \beta''$ also has $P_q(r) \le \beta'$; the answer j for budget β' leads to $G_A(Q', \beta' - P_q(r)) \subseteq G_A(Q', \beta'' - P_q(r))$ by induction. The second claim is the same induction with a larger base set. □

Formalised: `Bobi/EndTable.lean`, theorems `GA_antitone`, `G_antitone`, `H_antitone` and `GA_mono`.

B.5 Threshold form

By Lemma B.15, for each residual t the budgets at which $t \in G_A(Q,\cdot)$ form an initial segment, so one number per t describes the table. Fix a cap $\mathrm{cap} \geq 0$ (the root budget β).

Definition B.16 (Threshold)**.** $\mathrm{thr}(\emptyset,t) = \mathrm{cap}$ if $t \in A$ and -1 otherwise; for $Q = (q,Q')$ with value v and n copies,

$$\mathrm{thr}(Q,t) = \min\Big(\mathrm{cap},\ \min_{\substack{r\leq n\\ P_q(r)\leq \mathrm{cap}}} \Big[P_q(r) + \max\Big(-1,\ \max_{\substack{j\leq n-r\\ jv\leq t}} \mathrm{thr}(Q',t-jv)\Big)\Big]\Big).$$

Theorem B.17 (Threshold form)**.** *For every integer $B \geq 0$: $B \leq \mathrm{thr}(Q,t)$ if and only if $B \leq \mathrm{cap}$ and $t \in G_A(Q,B)$. Consequently $-1 \leq \mathrm{thr}(Q,t) \leq \mathrm{cap}$; $\mathrm{thr}(Q,t) = -1$ iff $t \notin G_A(Q,0)$; and otherwise $\mathrm{thr}(Q,t)$ is the largest $B \leq \mathrm{cap}$ with $t \in G_A(Q,B)$.*

Proof. Induction on L. For $L = 0$: if $t \in A$ the claim reads $B \leq \mathrm{cap}$; if $t \notin A$ both sides are false since $B \geq 0 > -1$. For $Q = (q,Q')$ write $m_r(t)$ for the inner maximum (at least -1). Then $B \leq \mathrm{thr}(Q,t)$ iff $B \leq \mathrm{cap}$ and $B \leq P_q(r) + m_r(t)$ for all r with $P_q(r) \leq \mathrm{cap}$. Assume $B \leq \mathrm{cap}$. For r with $P_q(r) > B$ the inequality holds automatically, as $B \leq P_q(r) - 1$. For r with $P_q(r) \leq B$ (these have $P_q(r) \leq \mathrm{cap}$): $B \leq P_q(r) + m_r(t)$ iff $0 \leq B - P_q(r) \leq m_r(t)$ iff some $j \leq n-r$ with $jv \leq t$ has $B - P_q(r) \leq \mathrm{thr}(Q',t-jv)$ iff, by induction (note $B - P_q(r) \leq \mathrm{cap}$), $t - jv \in G_A(Q',B-P_q(r))$. This is the definition of $t \in G_A(Q,B)$. The consequences follow by taking $B = 0$ and $B = \mathrm{thr}(Q,t)$. □

Formalised: `Bobi/Threshold.lean`, theorems `thr_iff`, `GA_iff_thr`, `thr_bounds`, `thr_eq_neg_one_iff`, `thr_greatest`, `G_iff_thr` and `H_iff_thr` (definition `thr`).

Remark B.18 (What the code computes). The recursion at residual t reads only residuals $\leq t$, so arrays indexed $0,\ldots,hi$ hold the exact thresholds of those residuals. With nonnegative costs P_q is nondecreasing, so stopping at the first r with $P_q(r) > \mathrm{cap}$ implements the filter. That the code's loops compute thr (including the shortcuts that compute only residuals in a range that can matter, and an early exit once every computed threshold is below the budget) is argued in the code's doc comments and tested against a direct transcription of Definition B.16 and against brute force; it is not formalised.

B.6 The per-node rule

At a search node the decided survivors M reach the sums $S = \mathrm{SS}(M) \cap [0,hi]$. The follower wins with any total in $[lo,hi]$, so it need not fix its target in advance. The node's end set is

$$A_{\mathrm{node}} = \{a \in [0,hi] : \exists s \in S,\ s \leq a \leq s + (hi - lo)\},$$

that is, $(S + [0,hi-lo]) \cap [0,hi]$, and the node table asks whether $hi \in G_{A_{\mathrm{node}}}(Q,\beta')$.

Theorem B.19 (Per-node rule)**.** *Let $S \subseteq \mathrm{SS}(M)$, $lo \leq hi$, A any set whose elements a each have some $s \in S$ with $s \leq a \leq s + (hi - lo)$, and $\pi(Q)$ any permutation of the undecided classes Q. If $hi \in G_A(\pi(Q),\beta')$, then every completion (a deletion r of Q with $P(r) \leq \beta'$, costs in any order) leaves some $u \in \mathrm{SS}(M \uplus R(r))$ with $lo \leq u \leq hi$. In threshold form: if $\beta' \leq \mathrm{cap}$ and $\beta' \leq \mathrm{thr}(\pi(Q),hi)$ computed with end set A_{node}, the node is safe at every budget up to $\mathrm{thr}(\pi(Q),hi)$.*

Proof. Proposition B.13 gives $x \in \mathrm{SS}(R(r))$ and $a \in A$ with $x + a = hi$. Pick $s \in S$ with $s \leq a \leq s + (hi - lo)$. Then $u = s + x$ is a subset sum of $M \uplus R(r)$ and $u = s + hi - a \in [hi - (hi - lo), hi] = [lo,hi]$. The threshold form follows from Theorem B.17 with $B = \beta'$ (and for smaller budgets, since $B \leq \mathrm{thr}$ is the same condition). The hypothesis $lo \leq hi$ is needed: with $lo > hi$ there is no target, and the code returns -1. □

Formalised: `Bobi/EndTable.lean`, theorems `node_rule`, `node_table_sound`, `node_table_sound_perm` and `node_table_sound_any` (definition `nodeEnd`); `Bobi/Threshold.lean`, `node_table_threshold_sound` and `node_table_threshold_sound_counts` (the form the search uses: count formulation, any play order, any cost order).

Remark B.20 (Strength). The per-node rule contains the rule of H in the same play order: if $s \in S$, $s \le \tau$, $\tau \in [lo, hi]$ and $\tau - s \in H(Q, \beta')$, then $hi \in G_{A_{\text{node}}}(Q, \beta')$ (shift the follower's strategy up by $hi - \tau + s$). Formalised: `Bobi/EndTable.lean`, theorem `node_table_of_H`. The analogous containment of the band rules is argued in the code's doc comment, not formalised; soundness does not need either. The play order matters only for strength; the search plays the classes in ascending order of deletion cost per unit of mass.

B.7 Restricted copies

Proposition B.21 (Restricted follower)**.** *For $k \ge 1$ let $I^{(k)}$ be the class instance in which every class i with $n_i > k$ is replaced by a class with the same value, k copies and costs $(\gamma_{i,1} + \cdots + \gamma_{i,n_i-k+1},\ \gamma_{i,n_i-k+2}, \ldots, \gamma_{i,n_i})$. If $I^{(k)}$ is safe (same band and budget), then so is the original instance.*

Proof. Map a deletion r of the original to r' with $r'_i = \max(0, r_i - (n_i - k))$ for the restricted classes and $r'_i = r_i$ otherwise. If $r'_i = 0$ its cost is $0 \le P_i(r_i)$; otherwise it is $\gamma_{i,1} + \cdots + \gamma_{i,n_i-k+r'_i} = P_i(r_i)$. So $P'(r') \le P(r)$. The restricted survivors number $k - r'_i = \min(k, n_i - r_i) \le n_i - r_i$, so $R'(r') \subseteq R(r)$ as multisets and $\mathrm{SS}(R'(r')) \subseteq \mathrm{SS}(R(r))$. Safety of $I^{(k)}$ applied to r' gives a sum in $[lo, hi]$ for r. □

Not formalised. The restricted costs are not ascending, which is why the count-formulation results above make no assumption on the order of costs. (The option is off by default; no run recorded in the repository uses it.)

B.8 Why the search is exact

The search fixes an order of the classes (any order). A node at depth k has decided counts $r_1, \ldots, r_k$ with residual budget $\beta' = \beta - \sum_{i \le k} P_i(r_i) \ge 0$, decided survivors M_k and $S_k = \mathrm{SS}(M_k) \cap [0, hi]$. The node is *safe* if every completion $(r_{k+1}, \ldots, r_L)$ with $\sum_{i>k} P_i(r_i) \le \beta'$ has $\mathrm{SS}(M_k \uplus R_{>k}) \cap [lo, hi] \ne \emptyset$. The root ($k = 0$, $S_0 = \{0\}$) is safe iff the class instance is safe.

Lemma B.22 (Node state)**.** *Whether a node is safe depends only on (k, S_k, β'), and a node safe at β' is safe at every $\beta'' \le \beta'$.*

Proof. $\mathrm{SS}(M \uplus R) = \mathrm{SS}(M) + \mathrm{SS}(R)$, and all values are nonnegative, so a sum in $[lo, hi]$ has its $\mathrm{SS}(M)$ part in $[0, hi]$. Hence the condition reads $(S_k + \mathrm{SS}(R_{>k})) \cap [lo, hi] \ne \emptyset$, and the completions depend only on k and β', with fewer completions for smaller budgets. □

Theorem B.23 (Sound pruning)**.** *Each of the following implies that the node (k, S_k, β') is safe.*

1. *Direct: $S_k \cap [lo, hi] \ne \emptyset$.*
2. *Guaranteed copies: with $d_i = \max\{r \le n_i : P_i(r) \le \beta'\}$, the multiset U with $n_i - d_i$ copies of v_i for each undecided i has $(S_k + \mathrm{SS}(U)) \cap [lo, hi] \ne \emptyset$.*
3. *Window: the hypothesis of Proposition B.8 holds for a run of S_k.*
4. *Root tables: Corollary B.14 gives a target in $[lo, hi]$, for a table built once at the root (any play order).*
5. *Node table: $\beta' \le \mathrm{thr}(\pi(Q), hi)$ with end set A_{node} (Theorem B.19).*

6. *Cache: an entry $(k, S_k) \mapsto \beta^*$ with $\beta' \le \beta^*$, where an entry is stored only after (k, S_k) was shown safe at β^* by one of the rules or by exhausting its children; rule 5 stores $\beta^* = \mathrm{thr}(\pi(Q), hi)$.*

Proof. (1) The decided survivors are present in every completion. (2) Costs are nonnegative, so a completion within β' has $P_i(r_i) \le \beta'$ and, P_i being nondecreasing, $r_i \le d_i$; at least $n_i - d_i$ copies of v_i survive, and a subset sum of a sub-multiset is a subset sum of the multiset. (3)–(5) are Proposition B.8, Corollary B.14 and Theorem B.19. (6) is Lemma B.22 together with Theorem B.19, which holds at every budget up to the stored threshold. □

Not formalised as a whole: rules (3)–(5) are the results cited in the proof (rule (3) only in part, see Proposition B.8), and rules (1), (2) and (6) and Lemma B.22 are argued here.

Theorem B.24 (Exactness)**.** *The search visits the root; at a node it returns "safe" if a rule of Theorem B.23 applies or if all children $r_{k+1} \in \{r \le n_{k+1} : P_{k+1}(r) \le \beta'\}$ return "safe"; at a leaf ($k = L$) to which no rule applies it returns "blocking" with the counts on the path. If the search ends without reaching its node limit, then "safe" at the root means the class instance is safe, and "blocking" returns a blocking deletion.*

Proof. By induction from the leaves, every node that returns "safe" is safe: for the rules this is Theorem B.23; for a node whose children are all safe, every completion within β' starts with some r_{k+1} with $P_{k+1}(r_{k+1}) \le \beta'$ (costs are nonnegative), and the rest of it is a completion of that child within $\beta' - P_{k+1}(r_{k+1})$. A leaf returning "blocking" has $S_L \cap [lo, hi] = \emptyset$ (rule 1 did not apply), and $M_L = R(r)$ for the path's deletion r, whose cost is $\beta - \beta' \le \beta$; so r is blocking. A node returns "blocking" only when a child does. □

Not formalised (the search code; see Section G). The rules it relies on are formalised as listed above, except the fractional bound of Lemma B.7.

Corollary B.25 (Exact solve)**.** *Start from any leader set X_0 within budget with $u_0 = \Phi(X_0)$ (the code uses Weninger and Fukasawa's greedy leader). For $h = 0, 1, \ldots$ run the search with $\ell = u_h$: if it returns "safe", $\mathrm{OPT} = u_h$; if it returns a blocking deletion r, set $X_{h+1} = X(r)$ and $u_{h+1} = \Phi(X(r)) < u_h$. If no search hits its node limit, this ends after at most $u_0 + 1$ searches with a proof of optimality and an optimal leader set.*

Proof. By Proposition B.4(3) and Theorem B.24, "safe" proves $\mathrm{OPT} \ge u_h = \Phi(X_h) \ge \mathrm{OPT}$, and "blocking" yields $X(r)$ within budget with $\Phi(X(r)) < u_h$. (If $\lceil u_h/g \rceil > hi$, no search is needed: $r = 0$ is blocking, as noted after Proposition B.4.) The values u_h are nonnegative integers that strictly decrease. □

For each of the 16 instances that were open in the literature the loop ended with "safe" at BOBILib's own value, so BOBILib's point was already optimal and the proof of optimality is new. These proofs are computational proofs supported by formally verified lemmas, not formally verified solutions: Lean proves the pruning rules (Section G), and the search that applies them, like the two integer-only provers that repeat it (Section A.3), is an exhaustive computation in integer arithmetic whose code is tested but not verified. In the terms of Section A.4 they are exact certificates (class (i)).

C. MULTIDIMENSIONAL KNAPSACK INTERDICTION PROOFS

We use the packing interdiction notation of Section A: leader rows $Gx \le g$, follower rows $Wy \le c$ with m rows, $p, G, g, W, c \ge 0$ integral, and $\Phi(x)$ the follower's optimum. Let

$$U(x) = \max\{p^\top y : Wy \le c,\ 0 \le y \le \mathbf{1} - x\}$$

be the value of the follower's linear programming (LP) relaxation, and $p_{\max} = \max_j p_j$.

C.1 The dualised follower LP

Lemma C.1 (Dual form of U)**.** *For $x \in \{0,1\}^n$,*

$$U(x) = \min_{\lambda \in \mathbf{R}^m_{\geq 0}} D(x,\lambda), \qquad D(x,\lambda) = c^\top \lambda + \sum_{j=1}^{n} \max\bigl(0,\, p_j - (W^\top \lambda)_j\bigr)(1-x_j),$$

and the minimum is attained by some λ with $0 \leq \lambda_k \leq p_{\max}/w_k^{\min}$ for every row k, where $w_k^{\min}$ is the smallest positive entry of row k of W ($w_k^{\min} = 1$ if the row is zero).

Proof. The LP is feasible ($y = 0$, since $c \geq 0$) and bounded ($0 \leq y \leq \mathbf{1}$), so by LP duality $U(x) = \min\{c^\top \lambda + \sum_j (1-x_j)\nu_j : W^\top \lambda + \nu \geq p,\ \lambda, \nu \geq 0\}$, and the minimum is attained. For fixed λ the best feasible ν_j with $x_j = 0$ is $\max(0, p_j - (W^\top \lambda)_j)$; for $x_j = 1$ the term has coefficient 0. This gives $U(x) = \min_\lambda D(x,\lambda)$. For the box: if $\lambda_k > p_{\max}/w_k^{\min}$, lower λ_k to $p_{\max}/w_k^{\min}$. Every item j with $W_{kj} > 0$ still has $(W^\top \lambda)_j \geq \lambda_k W_{kj} \geq p_{\max} \geq p_j$, so its term stays 0; items with $W_{kj} = 0$ are unaffected; and $c_k \lambda_k$ does not increase because $c_k \geq 0$. So D does not increase. □

Not formalised.

Proposition C.2 (Exact linearisation)**.** *Let $U^* = \min\{U(x) : Gx \leq g,\ x \in \{0,1\}^n\}$. Then U^* is the optimal value of the MILP*

$$\begin{aligned} \min \quad & c^\top \lambda + \sum_{j:\, p_j > 0} v_j \\ s.t. \quad & v_j \geq p_j - (W^\top \lambda)_j - p_j x_j, \quad v_j \geq 0 \quad (p_j > 0), \\ & 0 \leq \lambda_k \leq p_{\max}/w_k^{\min} \quad (k = 1, \ldots, m), \qquad Gx \leq g, \quad x \in \{0,1\}^n. \end{aligned}$$

Proof. For $x_j \in \{0,1\}$ and $(W^\top \lambda)_j \geq 0$,

$$\max\bigl(0, p_j - (W^\top \lambda)_j\bigr)(1-x_j) = \max\bigl(0,\, p_j - (W^\top \lambda)_j - p_j x_j\bigr) :$$

both sides equal $\max(0, p_j - (W^\top \lambda)_j)$ when $x_j = 0$, and both are 0 when $x_j = 1$. For fixed (x, λ) the smallest feasible v_j is this maximum, and items with $p_j = 0$ contribute 0. So the MILP's value is the minimum of $D(x,\lambda)$ over leader-feasible x and λ in the box, which is $\min_x U(x) = U^*$ by Lemma C.1. □

Not formalised.

C.2 The integrality-gap bracket

Lemma C.3 (Integrality gap of a packing problem with m rows)**.** *For every $x \in \{0,1\}^n$, with $p_{\max}(x) = \max\{p_j : x_j = 0\}$ (0 if there is no such j),*

$$U(x) - m\, p_{\max}(x) \ \leq\ \Phi(x) \ \leq\ U(x).$$

Proof. $\Phi(x) \leq U(x)$ because the LP is a relaxation. The LP's feasible region $P = \{y : Wy \leq c,\ 0 \leq y_j \leq 1 - x_j\}$ is a nonempty polytope, so the LP has an optimal vertex y^*. A vertex of $P \subseteq \mathbf{R}^n$ satisfies n linearly independent constraints of P with equality. Let $F = \{j : 0 < y_j^* < 1\}$; these coordinates have $x_j = 0$ and neither bound tight. The bound constraints of coordinate j are $y_j \geq 0$ and $y_j \leq 1 - x_j$, with normals $\pm e_j$, so a linearly independent family of tight constraints contains at most one bound constraint per coordinate outside F and none for coordinates in F. Hence at least $|F|$ of the n independent tight constraints are rows of W, and $|F| \leq m$. Let $\tilde{y}_j = 1$ if $y_j^* = 1$ and $\tilde{y}_j = 0$ otherwise. Then $\tilde{y} \leq y^*$, so $W\tilde{y} \leq Wy^* \leq c$ (as $W \geq 0$) and $\tilde{y} \leq \mathbf{1} - x$; $\tilde{y}$ is a binary follower response, and $\Phi(x) \geq p^\top \tilde{y} = U(x) - \sum_{j \in F} p_j y_j^* \geq U(x) - m\, p_{\max}(x)$. □

Not formalised.

Theorem C.4 (Bracket)**.** *Let $L \le U^*$ be any lower bound on U^* (for example a MILP solver's proven dual bound for the MILP of Proposition C.2) and let $\hat{x}$ be any leader-feasible choice. Then*

$$\lceil L - m\,p_{\max} \rceil \;\le\; \mathrm{OPT} \;\le\; \Phi(\hat{x}) \;\le\; U(\hat{x}).$$

Proof. For every leader-feasible x, Lemma C.3 gives $\Phi(x) \ge U(x) - m\,p_{\max}(x) \ge U^* - m\,p_{\max} \ge L - m\,p_{\max}$, so $\mathrm{OPT} \ge L - m\,p_{\max}$, and OPT is an integer because p is. The upper bounds hold because the leader may play $\hat{x}$ and the game is zero-sum. □

Not formalised.

For the OR instances with one follower knapsack (variant -100, $m = 1$) the lower bound is $\lceil L - p_{\max} - 10^{-6}\max(1,|L|) \rceil$, where L is HiGHS's proven dual bound for the MILP of Proposition C.2. The code (`mkip::bracket_lower_bound`) uses L only when it is finite; otherwise there is no bracket bound (the solver's incumbent value is an upper bound on U^*, not a lower one). The margin $10^{-6}\max(1,|L|)$ keeps the floating-point error of L from rounding the bound up past an integer: a dual bound such as 13,322.000000000024 gives $13{,}322 - p_{\max}$, not one more. A unit test checks the bound by brute force on random instances. The bracket's lower bound is solver-certified (class (ii), Section A.4): it rests on HiGHS's dual bound within that margin. It is the reported lower bound for 104 of the 270 instances, all re-run with this code and a 120 s limit: every run had a finite dual bound, and 34 of the MILPs were solved to optimality. For the other 166 the merged-row online bound below is at least as strong. With several follower rows the width $m\,p_{\max}$ makes the lower bound weak, and only the upper bounds are used.

C.3 The merged-row online bound

Theorem B.9 needs one leader row and one follower row. For one follower row ($m = 1$, weights $w_j = W_{1j}$, capacity $c = c_1$) and several leader rows, `mkip::omega_root` merges the leader rows into one surrogate row and rescales both rows to small integers. Its inputs are multipliers $\mu \in \mathbf{R}^{m_u}_{\ge 0}$ and scales $s_u, s_l > 0$, all double-precision numbers. For these numbers let

$$a_j = s_u \sum_{i=1}^{m_u} \mu_i G_{ij} \quad (j \in N), \qquad b = s_u \sum_{i=1}^{m_u} \mu_i g_i$$

be the exact real values. The code computes them in IEEE double precision (rounding to nearest): it forms the products $\mu_i G_{ij}$, adds them in the order of the rows and multiplies the sum by s_u; we write $\hat{a}_j$ and $\hat{b}$ for the computed values, and $\widehat{s_l w_j}$, $\widehat{s_l c}$ for the computed products of the follower data with s_l. It then sets

$$\gamma'_j = \lfloor \hat{a}_j \rfloor, \qquad \beta' = \lceil \hat{b} \rceil, \qquad w'_j = \lceil \widehat{s_l w_j} \rceil, \qquad c' = \lfloor \widehat{s_l c} \rfloor,$$

which are nonnegative integers for every $\mu \ge 0$ and $s_u, s_l > 0$, since all the data are nonnegative: leader costs and follower capacity are rounded down, the leader budget and follower weights up. Finally it evaluates the online game of Theorem B.9 with profits p, leader costs γ', budget β', follower weights w' and capacity c' on the list o of the items with $p_j > 0$ in decreasing order of p_j / w_j (ties: larger profit first), by the backward recursion of Section B over a table of $(\beta' + 1)(c' + 1)$ entries per item in 64-bit integer arithmetic. Let $\Omega' = \Omega(o, \beta', c')$ be its value. Below, $u = 2^{-53}$ is the unit roundoff of double precision and $\varepsilon_k = (1+u)^k - 1$ (so $\varepsilon_k \le ku/(1-ku)$).

Theorem C.5 (Merged-row online bound)**.** *Let p, G, g, w and c be nonnegative integers, and $\mu \ge 0$, $s_u, s_l > 0$.*

(a) If the four rounded quantities are computed exactly ($\hat{a}_j = a_j$, $\hat{b} = b$, $\widehat{s_l w_j} = s_l w_j$, $\widehat{s_l c} = s_l c$), then $\Omega' \le \mathrm{OPT}$.

(b) If every entry of G, g, w and c is below 2^{53} (so it is a double), no product underflows, and

$$2\,\varepsilon_{m_u+1}\,b < 1 \qquad \textit{and} \qquad 2uc < 1-u,$$

then $\Omega' \le \mathrm{OPT}$ *for the data computed in double precision.*

Proof. Two properties suffice. Suppose the integer data satisfy

(L) $\gamma'(X) \le \beta'$ for every leader-feasible X (that is, $G\mathbf{1}_X \le g$), and

(F) $w(Y) \le c$ for every $Y \subseteq N$ with $w'(Y) \le c'$.

By (F) every packing for (w', c') is a packing for (w, c), so the best packings satisfy $K'(T) \le K(T)$ for every $T \subseteq N$. Let X be leader-feasible. By (L), X fits the merged budget, so Theorem B.9 for the data $(p, \gamma', \beta', w', c')$ gives $\Omega' \le K'(o \setminus X) \le K(o \setminus X) \le K(N \setminus X) = \Phi(X)$. Minimising over X gives $\Omega' \le \mathrm{OPT}$. Both properties use

$$a(X) = s_u \sum_i \mu_i (G\mathbf{1}_X)_i \le s_u \sum_i \mu_i g_i = b \qquad \text{for leader-feasible } X, \tag{C.1}$$

which holds because $\mu \ge 0$ and $s_u > 0$, and the fact that $w(Y)$ and c are integers.

(a) By (C.1), $\gamma'(X) = \sum_{j \in X} \lfloor a_j \rfloor \le a(X) \le b \le \lceil b \rceil = \beta'$, which is (L). If $w'(Y) \le c'$, then $s_l w(Y) \le \sum_{j \in Y} \lceil s_l w_j \rceil = w'(Y) \le c' = \lfloor s_l c \rfloor \le s_l c$, so $w(Y) \le c$, which is (F).

(b) Each operation on doubles returns the exact result times a factor in $[1-u, 1+u]$, and every quantity involved is nonnegative. A term $\mu_i G_{ij}$ of a_j passes through at most $m_u + 1$ roundings (its product, at most $m_u - 1$ additions, the final product with s_u; adding the first term to zero is exact), so $(1-\varepsilon_{m_u+1}) a_j \le \hat{a}_j \le (1+\varepsilon_{m_u+1}) a_j$, using $1-(1-u)^k \le ku \le \varepsilon_k$; the same holds for $\hat{b}$ and b. The follower's products have one rounding each: $\widehat{s_l w_j} \ge (1-u) s_l w_j$ and $\widehat{s_l c} \le (1+u) s_l c$. For (L), let X be leader-feasible. By (C.1), $\gamma'(X) \le \sum_{j \in X} \hat{a}_j \le (1+\varepsilon_{m_u+1})\, a(X) \le (1+\varepsilon_{m_u+1})\, b$, and $\beta' \ge \hat{b} \ge (1-\varepsilon_{m_u+1})\, b$. If $\gamma'(X) \ge \beta' + 1$, these inequalities give $2\varepsilon_{m_u+1} b \ge 1$, contrary to the hypothesis; as both sides are integers, $\gamma'(X) \le \beta'$. For (F), if $w'(Y) \le c'$ then $(1-u)\, s_l w(Y) \le \sum_{j \in Y} \widehat{s_l w_j} \le w'(Y) \le c' \le \widehat{s_l c} \le (1+u)\, s_l c$, so $w(Y) \le c + 2uc/(1-u) < c + 1$, and $w(Y) \le c$ because both are integers. □

The online bound itself is formalised (Theorem B.9); properties (L) and (F), the rounding analysis and the code are not.

The proof needs only the directions of the roundings, not the size of the scales: lowering leader costs and raising the leader's budget enlarge the leader's options in the relaxed game, raising follower weights and lowering its capacity shrink the follower's options, and both moves can only lower the value of the game. In exact arithmetic $\beta' = \lfloor b \rfloor$ would also give (L), because $\gamma'(X)$ is an integer; rounding the budget up leaves the margin that part (b) uses when b is computed as a number just below an integer.

What the code chooses affects only the strength of the bound. `bobi bounds` starts from $\mu_i = 1/g_i$ and improves μ by coordinate ascent (`mkip::tune_mu`: each μ_i in turn is multiplied by 0, 0.5, 0.8, 1.25 or 2 while the bound on a coarse table improves; at most two sweeps). With its default of $4 \cdot 10^6$ table entries per item, `mkip::omega_scales` gives $s_u = 1000/\hat{\sigma}$, where $\hat{\sigma}$ is the computed value of $\sum_i \mu_i g_i$, and $s_l = 4000/c$; so b differs from 1000 by less than 10^{-11}, $\beta' \in \{1000, 1001\}$ and $c' \in \{3999, 4000\}$. In the 270 OR instances with one follower knapsack all data are integers (read from the instance files: the largest coefficient is 1,366 and the largest right-hand side 200,032), $m_u \le 29$ and no nonzero product comes near the underflow range, so $2\varepsilon_{m_u+1} b < 10^{-11}$ and $2uc < 10^{-10}$: the hypotheses of part (b) hold with a wide margin, and every merged-row bound reported for them is a valid lower bound in exact arithmetic: a mathematical proof plus an integer computation, so an exact certificate (class (i), Section A.4). It gives 166 of the 191 lower bounds that improve on the published ones; the other 25 come from the LP bracket.

C.4 Certified points

Proposition C.6 (Certified point)**.** *Let $\hat{x}$ satisfy $G\hat{x} \leq g$, and let $\hat{y}$ be a follower response at $\hat{x}$ proven optimal (by the exact knapsack oracle when $m = 1$, or by a MILP solver whose dual bound meets $p^\top \hat{y}$). Then $(\hat{x}, \hat{y})$ is bilevel feasible with value $\Phi(\hat{x})$, so* OPT $\leq \Phi(\hat{x})$*. Without an exact follower solve,* OPT $\leq \lfloor U(\hat{x}) \rfloor$ *still holds.*

Proof. $\hat{y} \in S(\hat{x})$ by definition; the game is zero-sum, so every element of $S(\hat{x})$ gives the leader the value $\Phi(\hat{x})$. The last claim is Lemma C.3 and integrality of Φ. □

Not formalised.

For several follower knapsacks the leader choice $\hat{x}$ is the minimiser found for the MILP of Proposition C.2; when HiGHS proves the follower's optimum at $\hat{x}$, the point is certified. For one follower knapsack, a first-improvement local search on the leader's choice (add an item the follower uses, or swap a blocked item for one it uses, keeping every budget row) evaluates each candidate with the exact knapsack oracle, so every incumbent it reports is a certified point. Every point reported is checked again from its file by `bobi verify` (Section A.3). The data are integers and the points are 0/1 vectors, so the leader's rows hold exactly. With one follower knapsack the follower's optimum comes from the exact knapsack oracle, which works in integer arithmetic, and the 270 points are exact certificates (class (i), Section A.4); with several, it rests on HiGHS's dual bound, and the 420 points are solver-certified (class (ii)). The follower LP bound $U(\hat{x})$, an upper bound in the gaps of Table S5, is a floating-point LP value (class (ii)).

C.5 Published bounds of the OR set

The paper compares this work's bounds with the published ones. The published lower bounds of the OR instances are Lozano, Bergman and Cire's;[10] the best published point of an instance is the best of theirs (with a solution file in their repository[13]), Lefebvre and Schmidt's[11] and BOBILib's. Lefebvre and Schmidt's values count only from runs that ended before the time limit, the rule of their own analysis (Section A.6); for 21 OR instances a value from a run stopped at the limit would otherwise be the best published point, and for 10 of them it lies below Lozano, Bergman and Cire's lower bound (Table S6). The published points are not verified here.

For one instance a published lower bound exceeds the best published point. `OR10x100-0.75_4.dat-0` is closed in the published data. Lozano et al. publish a point of value 8,813 and a lower bound of 8,814 from runs they flag as numerically inconsistent; their repository also records a branch-and-cut bound of 8,813, which with the point proves the value optimal. The instance is therefore counted among the published optima. This work's verified solution file at 8,813 (checked also with SCIP) confirms that the bound 8,814 is wrong.

D. CLIQUE INTERDICTION PROOFS

D.1 The model

BOBILib's instance `normalClique`*n*`-`*d*`-`*i* has a graph $G = (V, E)$ with $n = |V|$, $m = |E|$, and budget $k = \lfloor m/4 \rfloor$. All variables are binary. The leader has w_e for $e \in E$ and the single row $\sum_e w_e \leq k$. The follower has x_v ($v \in V$) and y_e ($e \in E$) and the rows

$$x_u + x_v \leq 1 \ \ (uv \notin E), \qquad x_u + x_v - y_{uv} \leq 1 \ \ (uv \in E), \qquad w_{uv} + y_{uv} \leq 1 \ \ (uv \in E).$$

The leader minimises $\sum_v x_v$ and the follower minimises $-\sum_v x_v$. (Every row, coefficient and objective of the 60 files was checked against this pattern, once by the instance reader of the clique scripts and once by the parser of a separate review session, which classifies rows by their coefficients.)

Lemma D.1 (Follower's problem)**.** *Fix the leader's choice and let $D = \{e : w_e = 1\}$. A vector $x \in \{0,1\}^V$ extends to a feasible follower point if and only if $\{v : x_v = 1\}$ is a clique of $G - D$. Hence the follower's optimum is $\omega(G-D)$, the game is zero-sum, and the bilevel optimum is* $\min\{\omega(G-D) : D \subseteq E,\ |D| \le k\}$.

Proof. Let x be part of a feasible point and $x_u = x_v = 1$. The first family of rows forces $uv \in E$. If $uv \in D$ then $w_{uv} = 1$ forces $y_{uv} = 0$, and the second family gives $x_u + x_v \le 1$, a contradiction. So $uv \in E \setminus D$. Conversely, for a clique of $G - D$ set $y_{uv} = 1 - w_{uv}$: the third family holds with equality; the second holds for $uv \notin D$ because $y_{uv} = 1$, and for $uv \in D$ because a clique of $G - D$ does not contain both u and v; the first holds because a clique contains no non-edge of G. The leader's objective is minus the follower's. □

Not formalised (the Lean file states the bounds directly in terms of $\omega(G-D)$).

D.2 Lower bound 3: Mantel's theorem

Theorem D.2 (Mantel 1907, deletion form)**.** *A graph on n vertices with more than $\lfloor n^2/4 \rfloor$ edges contains a triangle. Consequently, if $m - k > \lfloor n^2/4 \rfloor$, then $G - D$ contains a triangle for every set D of at most k edges, and the bilevel optimum is at least* 3.

Proof. Mantel's theorem[14] is Turán's theorem[15] for K_3. After deleting at most k edges at least $m - k$ edges remain, and $m - k > \lfloor n^2/4 \rfloor$. □

Formalised: `Bobi/Clique.lean`, theorems `mantel`, `mantel'` and `triangle_survives`; the instance arithmetic in `mantel_40`, `mantel_50`, `mantel_60`, `clique40`, `clique50` and `clique60`.

All 30 density-0.7 instances have $(n,m,k) \in \{(40,546,136),(50,857,214),(60,1239,309)\}$, so $m - k = 410,643,930$ exceeds $\lfloor n^2/4 \rfloor = 400,625,900$. The density-0.9 instances satisfy the same inequality.

D.3 Upper bounds: partitions

Proposition D.3 (Partition upper bound)**.** *Let $f : V \to \{1,\dots,r\}$ and let D contain every edge of G whose ends have the same colour. Then $G - D$ has no clique on $r+1$ vertices. If at most k edges of G lie inside the colour classes, deleting exactly those edges is a leader choice, and the bilevel optimum is at most r.*

Proof. f is a proper colouring of $G - D$, and a clique on $r+1$ vertices needs $r+1$ colours. □

Formalised: `Bobi/Clique.lean`, theorem `cliqueFree_of_partition`.

For a partition $\mathscr{P}$ of V into r parts (parts may be empty) let $\mathrm{in}_G(\mathscr{P})$ be the number of edges of G inside parts, and let $w_r(G) = \min_{\mathscr{P}} \mathrm{in}_G(\mathscr{P})$. Each solution file deletes the inside edges of a 3- or 4-partition found by tabu search and states a maximum clique of the remaining graph, computed exactly (the review re-checks it with two exact clique searches[16–18]); the files pass the checks of Section A.3.

D.4 Lower bound 4: Brouwer's theorem

Let $t_r(n)$ be the number of edges of the balanced complete r-partite graph on n vertices (Turán's number), and $T(n,r) = \binom{n}{2} - t_r(n)$, the number of edges of the disjoint union of r cliques of balanced sizes.

Theorem D.4 (Brouwer 1981, Theorem 2;[6] quoted, with his G renamed H and his $m := \lfloor n/t \rfloor$ written out)**.** *Let H be a graph on n points such that $\alpha(H) \le t$, and $n > 2t$. If H has no more than $T(n,t) + \lfloor n/t \rfloor - 2$ edges, then $V(H)$ can be written as a union of t cliques. Moreover, this bound is best possible.*

Here $\alpha(H)$ is the independence number. The theorem is restated by Tyomkyn and Uzzell[7] (their equation (1) and Theorem 1.1), who credit Brouwer with the first explicit proof, and, in the complemented form below for $t \ge 2$, by Korándi, Roberts and Scott[19] (their Theorem 1.1). Brouwer's report is not refereed, but the statement is: Tyomkyn and Uzzell give a new proof of it in their Section 2, Hanson and Toft and Kang

and Pikhurko found it again,[20,21] and Amin, Faudree, Gould and Sidorowicz[22] determine the largest number of edges of a maximal K_p-free graph that is not $(p-1)$-partite, which gives the same threshold (a K_p-free graph that is not $(p-1)$-partite stays so when edges are added until it is maximal). We use the theorem in complemented form.

Corollary D.5 (Complement form)**.** *Let $t \geq 1$ and $n \geq 2t+1$. Every K_{t+1}-free graph F on n vertices with $e(F) \geq t_t(n) - \lfloor n/t \rfloor + 2$ is t-partite (its vertex set is the union of t independent sets, some possibly empty).*

Proof. Apply Theorem D.4 to the complement $H = \overline{F}$: $\alpha(H) = \omega(F) \leq t$, $n > 2t$, and $e(H) = \binom{n}{2} - e(F) \leq \binom{n}{2} - t_t(n) + \lfloor n/t \rfloor - 2 = T(n,t) + \lfloor n/t \rfloor - 2$. So V is a union of t cliques of H, that is, of t independent sets of F. Assigning each vertex to one of the sets containing it gives a partition into independent sets. □

Not formalised: Brouwer's theorem is not in Mathlib, and it is neither proved nor assumed in the Lean files. The statement above was read in the scanned report, and the translation was checked by computer (for $t = 3$: $T(n,3) + t_3(n) = \binom{n}{2}$).

Theorem D.6 (Density 0.9: value at least 4)**.** *Let $n \geq 7$, $m - k \geq t_3(n) - \lfloor n/3 \rfloor + 2$ and $w_3(G) > k$. Then $G - D$ contains a K_4 for every $D \subseteq E$ with $|D| \leq k$, so the bilevel optimum is at least 4.*

Proof. Suppose $|D| \leq k$ and $G - D$ is K_4-free. It has $m - |D| \geq m - k$ edges, so by Corollary D.5 with $t = 3$ ($n \geq 7 = 2t+1$) it is 3-partite with some partition $\mathscr{P}$. No edge of $G - D$ lies inside a part, so every edge of G inside a part lies in D, and $|D| \geq \mathrm{in}_G(\mathscr{P}) \geq w_3(G) > k$, a contradiction. □

Not formalised (it rests on Theorem D.4).

For the density-0.9 instances $(n,m,k) \in \{(40,702,175),(50,1102,275),(60,1593,398)\}$, so $m - k = 527, 827, 1195$ and $t_3(n) - \lfloor n/3 \rfloor + 2 = 522, 819, 1182$ ($t_3(n) = 533, 833, 1200$): the hypothesis holds with margins 5, 8 and 13, while $m - k < t_3(n)$, so Turán's theorem alone does not give a K_4. It remains to show $w_3(G) > k$.

D.5 Reducing $w_3(G) > k$ to finitely many part-size cases

Let $\overline{E}$ be the edge set of the complement $\overline{G}$, so $|\overline{E}| = \binom{n}{2} - m$, and for a partition $\mathscr{P}$ let $\mathrm{cut}_{\overline{G}}(\mathscr{P})$ be the number of edges of $\overline{G}$ joining different parts.

Proposition D.7 (Part-size reduction)**.** *For a partition $\mathscr{P} = (P_1, P_2, P_3)$ of V with part sizes s_1, s_2, s_3,*

$$\mathrm{in}_G(\mathscr{P}) = \sum_{i=1}^{3} \binom{s_i}{2} - |\overline{E}| + \mathrm{cut}_{\overline{G}}(\mathscr{P}).$$

Hence, with $b(s) = k + |\overline{E}| - \sum_i \binom{s_i}{2}$, $w_3(G) > k$ holds if and only if, for every size vector $s_1 \geq s_2 \geq s_3 \geq 0$ with $s_1 + s_2 + s_3 = n$ and $b(s) \geq 0$, no partition with these part sizes has $\mathrm{cut}_{\overline{G}}(\mathscr{P}) \leq b(s)$.

Proof. Each of the $\sum_i \binom{s_i}{2}$ pairs inside a part is an edge of G or of $\overline{G}$, and the edges of $\overline{G}$ inside parts number $|\overline{E}| - \mathrm{cut}_{\overline{G}}(\mathscr{P})$. So $\mathrm{in}_G(\mathscr{P}) \leq k$ iff $\mathrm{cut}_{\overline{G}}(\mathscr{P}) \leq b(s)$. Relabelling the parts, every partition has sorted sizes, and since $\mathrm{cut}_{\overline{G}} \geq 0$, size vectors with $b(s) < 0$ admit no partition with $\mathrm{in}_G \leq k$. □

Not formalised (checked in exact integer arithmetic by script, and on random partitions of every instance).

Since n, m and k are the same for all ten graphs of a group, so are the size vectors with $b(s) \geq 0$:

n	$\lvert\overline{E}\rvert$	size vectors s with $b(s)$ in brackets
40	78	(14,13,13) [6], (14,14,12) [5], (15,13,12) [4], (15,14,11) [2], (16,12,12) [1], (16,13,11) [0]
50	123	(17,17,16) [6], (18,16,16) [5], (18,17,15) [4], (19,16,15) [2], (18,18,14) [1], (19,17,14) [0]
60	177	(20,20,20) [5], (21,20,19) [4], (21,21,18) [2], (22,19,19) [2], (22,20,18) [1]

That is $10 \cdot 6 + 10 \cdot 6 + 10 \cdot 5 = 170$ cases (instance, size vector), each asking for a partition of V with prescribed part sizes and at most $b(s) \leq 6$ edges of the sparse complement between parts. All 170 were decided infeasible by five methods: CP-SAT[23] and SCIP[24] on two different models, an exact solver-free depth-first search (tested against brute force), and, in a separate review session with its own code, HiGHS on a third model with an explicit budget row and a separate exact branch and bound written in C (tested against brute force on 450 random graphs and on planted instances). The three solver verdicts are solver-certified (class (ii), Section A.4). For an exact certificate, each case also has a branch-and-bound tree (`results/clique/certificates/`), checked by the integer checker `scripts/check_clique_certificates.py`: it reads the instance files itself, re-derives the graph, n, m and k from the rows, checks the hypothesis $m - k \geq t_3(n) - \lfloor n/3 \rfloor + 2$, enumerates the size vectors with $b(s) \geq 0$, and verifies at every leaf of each tree a lower bound on $\mathrm{cut}_{\overline{G}}$ above $b(s)$, using only Python's integers and no project code (174 lines of code). It passes on all 170 cases of the 30 instances (`check.log`): the trees have 134,368 nodes and 89,132 leaves in all, the largest 9,888 nodes, and the check takes about 3.1 s. With Theorem D.6 and Proposition D.3 (4-partitions with at most k inside edges), all 30 density-0.9 instances have value exactly 4: an exact certificate (class (i)) resting on Brouwer's published theorem, which is neither formalised nor reproved here. The case computations are not formalised proofs.

D.6 The three open instances

For normalClique60-70-2, -3 and -5 ($n = 60$, $m = 1239$, $k = 309$) Mantel gives value ≥ 3 and a 4-partition gives value ≤ 4. Let $\tau_4(G)$ be the fewest edges meeting every K_4 of G. By Lemma D.1 the value is 3 if and only if $\tau_4(G) \leq k$. Deleting the inside edges of a 3-partition destroys every K_4 (two of any four vertices share a part), so $\tau_4(G) \leq w_3(G)$, and $w_3(G) \leq k$ would give value 3; the best 3-partitions found have 311, 310 and 310 inside edges, against $k = 309$.

The converse argument of Theorem D.6 is not available here. It would need every K_4-free subgraph with at least $m - k = 930$ edges to be 3-partite, but $930 < t_3(60) - 20 + 2 = 1182$, so Corollary D.5 does not apply, and its conclusion is false at this edge count: the disjoint union of the wheel W_5 (a 5-cycle plus a vertex joined to all of it; 6 vertices, 10 edges, K_4-free, chromatic number 4) and any 3-partite graph with 920 edges on the other 54 vertices ($920 \leq t_3(54) = 972$) is a K_4-free graph on 60 vertices with 930 edges that is not 3-partite. So a proof of value 4 must bound $\tau_4(G)$ itself from below. Linear programming bounds on τ_4 (rows for every K_4, for maximal cliques and for vertex sets whose induced subgraph must lose a Turán excess of edges) stop near 296, below the 310 needed, and searches for a transversal with at most 309 edges that are not restricted to 3-partite remainders stop at the best 3-partition found. The three instances stay open with $3 \leq \text{value} \leq 4$.

E. GENERAL INSTANCES: RAYS, HPR BOUND AND PIPELINES

Notation is that of (A.1). In this section a follower row is written $\ell_i \leq C_i x + D_i y \leq h_i$ with $\ell_i \in \mathbf{R} \cup \{-\infty\}$ and $h_i \in \mathbf{R} \cup \{+\infty\}$, and the follower's variable bounds as $\ell^y_j \leq y_j \leq h^y_j$, because the direction test below depends on which sides are finite.

E.1 Unbounded followers: no bilevel feasible point

Theorem E.1 (Recession direction)**.** *Let $r \in \mathbf{Q}^{n_y}$ satisfy, for every follower row i, $D_i r \geq 0$ if $\ell_i > -\infty$ and $D_i r \leq 0$ if $h_i < +\infty$; for every follower variable j, $r_j \geq 0$ if $\ell^y_j > -\infty$ and $r_j \leq 0$ if $h^y_j < +\infty$; and $f^\top r < 0$. Then for every leader choice x (real or integral) the follower's problem $P(x)$ is infeasible or unbounded, so $S(x) = \emptyset$ and the bilevel problem has no feasible point.*

Proof. Let ν be a positive integer such that $\nu r_j \in \mathbf{Z}$ for every $j \in I_l$ (r is rational). Suppose y is feasible for $P(x)$. For every integer $t \geq 0$, $y + t\nu r$ satisfies every follower row, since $C_i x$ is a constant and $D_i(y +$

$t\nu r$) moves only in the permitted direction; it satisfies every bound for the same reason; and its integer-constrained entries stay integral. Its objective $f^\top y + t\nu f^\top r$ tends to $-\infty$. So $P(x)$ has no optimal solution whenever it is feasible. □

Not formalised.

Remark E.2 (Meyer's theorem). The theorem only needs an explicit rational direction. It is the case we need of a classical result of Meyer:[25] with rational data, the convex hull of the feasible set of a mixed-integer program is a rational polyhedron with the same recession cone as the LP relaxation, so a feasible mixed-integer program whose LP relaxation is unbounded is itself unbounded. Rationality matters: without it, a direction of the LP relaxation need not be usable by integer points. The directions here are found by an LP solver in floating point, written as decimals (which are rationals) and checked exactly.

The exact check. For each certificate, a script that reads the instance files independently of the solver code converts every entry of r to an exact rational and checks the conditions of Theorem E.1 in rational arithmetic, counting only follower variables in each follower row (leader variables enter those rows as constants). When rounding leaves tiny residuals, the direction is first projected onto the exact null space of its tight rows. This succeeds for all open instances but `supportcase42_0_100` and `supportcase42_50_50`, whose floating-point directions (support 9,733) keep residuals of about 10^{-12} (relative). For these two a second script (`scripts/exact_ray.py`) first tries every unit direction $r = \pm e_j$ in exact arithmetic and finds one of support one: the continuous follower variable `C9734`, with bounds $[0,\infty)$ and follower objective coefficient -1, occurs in a single follower row, a $\leq$ row with coefficient -1, so $r = e_{\texttt{C9734}}$ meets every condition of Theorem E.1 with $f^\top r = -1$ (here even without Meyer's theorem: from any feasible follower point, increasing `C9734` stays feasible and lowers the follower's objective without bound). `scripts/verify_ray.py` checks these two directions like the others. All 40 certificates of the 40 open instances are therefore exact (class (i), Section A.4).

E.2 The HPR bound and the MIPLIB optimum

Lemma E.3 (HPR bound)**.** *Every bilevel feasible point is feasible for the HPR, so* HPR $\leq$ OPT. *If a bilevel feasible point has value* HPR*, it is optimal.*

Proof. A bilevel feasible point satisfies $Ax + By \geq a$, the integrality conditions, and, as $y \in S(x)$, also $Cx + Dy \geq b$. The HPR minimises the same objective over a superset. □

Not formalised.

BOBILib builds its general instances from a MIPLIB problem by assigning variables and rows to the two levels and keeping the problem's objective as the leader's. If nothing else changes, dropping the condition $y \in S(x)$ gives back the MIPLIB problem: the HPR *is* the MIPLIB model, and its published optimum[2] z^* is a lower bound on OPT by Lemma E.3. This is not assumed: a script reads the bilevel instance and the MIPLIB 2017 model and checks in exact rational arithmetic that they have the same variables, rows, coefficients, sides, bounds, integrality requirements and objective (up to the objective's sense). For the 7 instances where it holds and a verified bilevel feasible point attains the bound (`pigeon-10_0_100`, `pigeon-10_50_50`, `timtab1_0_100`, `noswot_0_100`, `noswot_50_50`, `ex10_0_100`, `neos-1445765_0_100`; Table S11), the point is optimal by accepted benchmark conventions. The values z^* are the optima in MIPLIB 2017's solution file. They were computed by MILP solvers in floating point and are solver-certified numerical results (class (ii), Section A.4), not exact-arithmetic proofs, so these optima are no more rigorous than the published values:

- `noswot`: $z^* = -41.00000885$. The objective has integer coefficients on integer variables only, so the HPR optimum is an integer, and a valid lower bound L gives the lower bound $\lceil L \rceil$. The bound

$\lceil z^* \rceil = -41$ therefore holds unless the published optimum overstates the HPR optimum by 0.99999 or more (the HPR optimum would have to be -42 or less); BOBILib's own points attain it.

- `ex10`: $z^* = 100$; the objective is the all-ones vector on binary variables, so the same integrality argument applies, and the point of `bobi general` attains it. Lefebvre and Schmidt's run files[11] report the same optimum for `ex10_0_100` (an exact MILP run that ended well before its time limit); their paper does not state it, so it is not counted as a published optimum (`results/notes/lefebvre_schmidt.md`).

- `neos-1445765`: $z^* = -17{,}783$; the objective has integer coefficients on integer variables only, so the same argument applies, and the point of `bobi general` attains it. Integrality rounding makes these three, and `pigeon-10` below, the safest of the HPR-bound optima.

- `pigeon-10`: $z^* = -9{,}000$, attained by both points. The objective is $-1{,}000$ on ten integer variables and zero elsewhere, so every value is a multiple of 1,000, and the bound holds unless the published optimum overstates the HPR optimum by 1,000 or more.

- `timtab1`: $z^* = 764{,}771.99999978$ (printed as 764,772 on the MIPLIB web pages); the point attains it with row violations below 10^{-12}, so the agreement of point and bound is numerical.

"Optimal" for the HPR-bound optima therefore means equal to the published optimum to its printed precision (after rounding up for `noswot`), as BOBILib and MIPLIB use the word; no exact-arithmetic proof of the bound is claimed. The model comparison is exact; the points are class (ii) (Section A.4).

E.3 Feasible points without coupling rows

Proposition E.4 (HPR, follower, tie-break)**.** *Assume there are no coupling rows ($B = 0$). Let $(\hat{x}, \hat{y})$ be feasible for the HPR, and suppose $P(\hat{x})$ has an optimal solution, with value $\varphi(\hat{x})$. Then the tie-break problem at $\hat{x}$ (the optimistic choice among the follower's optima, Section A) is feasible, and for any of its feasible points y^*, the point $(\hat{x}, y^*)$ is bilevel feasible. If y^* is optimal for the tie-break problem, $(\hat{x}, y^*)$ is the best bilevel feasible point with leader part $\hat{x}$.*

Proof. $A\hat{x} \geq a$ holds because $(\hat{x}, \hat{y})$ is HPR feasible and $B = 0$, and $\hat{x}$ satisfies its integrality conditions. Any optimal solution of $P(\hat{x})$ is feasible for the tie-break problem, whose rows are those of $P(\hat{x})$, $f^\top y \leq \varphi(\hat{x})$, and $A\hat{x} + By \geq a$, which does not involve y. A feasible point y^* of it is feasible for $P(\hat{x})$ with $f^\top y^* \leq \varphi(\hat{x})$, so $y^* \in S(\hat{x})$. The bilevel feasible points with leader part $\hat{x}$ are exactly the pairs $(\hat{x}, y)$ with $y \in S(\hat{x})$, the feasible set of the tie-break problem. □

Not formalised. The existence of an optimal solution of $P(\hat{x})$ is what the follower solve establishes; the HPR point $\hat{y}$ shows that $P(\hat{x})$ is feasible. The computations accept $f^\top y \leq \varphi(\hat{x}) + 10^{-7}(1 + |\varphi(\hat{x})|)$ in the tie-break, well inside BOBILib's relative tolerance (A.2). The pipeline handles continuous linking variables, which neither of BOBILib's two solvers accepts.

The tie-break used this slack. All solution files were therefore checked again with `bobi verify --tight` (Section A.2), which holds the follower's value to $\max(10^{-6}, 10^{-9}|\varphi|)$ of its proven optimum. Of the original files without coupling rows, 17 failed it: their follower values lay 5×10^{-8} to 10^{-7} (relative) above the optimum, within BOBILib's tolerance, so they were benchmark-accepted points (class (iii), Section A.4). Without coupling rows every optimal follower response is feasible for the leader's rows (Proposition E.4), so the same leader values admit an exactly optimal response. `bobi coupled-tighten` computed one: it solves $P(\hat{x})$ at the old file's leader values with zero relative gap (absolute gap and feasibility tolerance 10^{-9}), giving φ^*, and then the tie-break with $f^\top y \leq \varphi^* + 10^{-9}$ and every leader variable fixed. The new files replace the old ones (kept in `results/general/tolerance/`); each new leader value is worse than

the old one by the old follower slack, as it must be, since the old leader used that slack. All 64 files now pass the tight test; 2 of them keep a slack above 10^{-8} relative but below the test's absolute 10^{-6}, and every other file's follower lies at most 1.3×10^{-9} above the proven bound. `bobi certify --tight`, a re-certification from scratch with proven bounds, accepts 62 of the 64; for the other 2 (`neos-3627168-kasai_0_100` and `neos-3754480-nidda_0_100`) the follower's problem at the files' leader values is infeasible at its feasibility tolerance of 10^{-9}, so it gives no follower bound; at HiGHS's default tolerance the follower is optimal and the rows hold within 10^{-6}. An independent check with HiGHS through its Python interface and its own reader of the files confirmed 13 of the 17 re-derived files (for the others it did not finish within 600 s). All 64 files, among them the 4 optimal points of Table S11, are solver-certified points (class (ii)).

E.4 Coupling rows

With coupling rows ($B \neq 0$) the follower's optima at the HPR's leader values may all violate them. Recall that $P(x)$, $\varphi(x)$ and $S(x)$ depend on x only through the linking variables x_L; write $\varphi(x_L)$. For fixed values $\bar{x}_L$ define the *best response*

$$\mathrm{BR}(\bar{x}_L) = \min\{c^\top x + d^\top y :\ x_L = \bar{x}_L,\ Ax + By \geq a,\ Cx + Dy \geq b,\ f^\top y \leq \varphi(\bar{x}_L),\ \text{integrality}\},$$

in which the other leader variables are free.

Proposition E.5 (Best response and blocked leader choices)**.** *Suppose $P(\bar{x}_L)$ has an optimal solution. Then the feasible points of $\mathrm{BR}(\bar{x}_L)$ are exactly the bilevel feasible points with linking part $\bar{x}_L$, and $\mathrm{BR}(\bar{x}_L)$ is the best value among them. If $\mathrm{BR}(\bar{x}_L)$ is infeasible, no bilevel feasible point has linking part $\bar{x}_L$ (the choice $\bar{x}_L$ is* blocked*). Consequently $\mathrm{OPT} = \inf_{\bar{x}_L} \mathrm{BR}(\bar{x}_L)$ over linking values for which $P(\bar{x}_L)$ has an optimal solution.*

Proof. (x, y) is bilevel feasible iff it satisfies the upper-level rows and integrality and $y \in S(x)$, and $y \in S(x)$ iff y is feasible for $P(x_L)$ and $f^\top y \leq \varphi(x_L)$. These are the constraints of $\mathrm{BR}(x_L)$. If $P(\bar{x}_L)$ has no optimal solution, $S(x) = \emptyset$ for every x with these linking values. □

Not formalised.

Lemma E.6 (Value-function cuts[26,27])**.** *Let $\hat{y}$ be feasible for $P(\hat{x}_L)$ with $f^\top \hat{y} = \hat{f}$. Every bilevel feasible (x, y) satisfies: $\hat{y}$ is infeasible for $P(x_L)$, or $f^\top y \leq \hat{f}$.*

Proof. If $\hat{y}$ is feasible for $P(x_L)$ then $\varphi(x_L) \leq \hat{f}$, and $y \in S(x)$ gives $f^\top y = \varphi(x_L)$. □

Not formalised.

The search for an unblocked linking choice tries, in turn within a time budget: the HPR's own x_L; best-response dynamics; the HPR with objective $c^\top x + d^\top y + \rho f^\top y$ for increasing ρ; a penalty alternating direction method after Kleinert and Schmidt;[28] moves of continuous linking variables towards the interior; the HPR with the cuts of Lemma E.6 (modelled with binary indicators, exact when every follower row's linking part takes integer values); and no-good cuts on integer linking variables. These are heuristics: none of them is needed for correctness, and the "blocked" verdicts (solver tolerances included) only steer the search; no instance is claimed to have no feasible point on that basis. What is certified is each returned point, re-checked from scratch: all rows, bounds and integrality within 10^{-6}; a fresh follower solve whose proven dual bound puts the point's follower value within relative 10^{-5} of the optimum; and the optimistic tie-break at the point's leader values within relative 10^{-5} by a proven bound. Each file then passes `bobi verify`.

In 9 of the files with coupling rows the follower's value had been about 10^{-7} (relative) above its optimum, which BOBILib's test (A.2) accepts; 8 of these files failed the tight test (benchmark-accepted points, class (iii)). All 9 were re-derived as above. For seven of them every leader value was kept. For `pg_50_50`

and `neos-2978193-inde_50_50` the coupling rows force the follower about 10^{-7} above its optimum at the old files' leader values, so only the linking variables were kept and the other leader variables re-optimised in the same tie-break, which is $\mathrm{BR}(\bar{x}_L)$ of Proposition E.5 with the follower's value capped at $\varphi^* + 10^{-9}$. All 29 files with coupling rows now pass `bobi verify --tight` and `bobi certify --tight` (the follower within 10^{-6} of the proven bound on its optimum, the leader's objective within $10^{-6} + 10^{-9}|F|$ of the proven tie-break bound), and the independent HiGHS check confirmed the 9 new ones: they are solver-certified points (class (ii)). The superseded files are kept in `results/general-coupled/tolerance/`.

Few binary linking variables. If the linking variables are binary and few, Proposition E.5 gives the exact optimum by enumerating all $\bar{x}_L$, solving $P(\bar{x}_L)$ and $\mathrm{BR}(\bar{x}_L)$ for each. This reproduces the published optimum 299,577 of Lozano, Bergman and Cire[10] for `general30-30-10-20-20-8` (10 linking bits); with 20 bits it is too slow.

E.5 A tolerance observation

Remark E.7 (`general20-30-20-20-20-2`). Test (A.2) is relative. In the generalized interdiction instance `general20-30-20-20-20-2` the follower's objective values are near 10^8, so a response up to about 10^4 units from the follower's optimum passes. In BOBILib's solution file for this instance the follower's value is 106,899,984.14 against the follower's optimum 106,899,953 at the same leader choice, 31 units above it (relative error about 2.9×10^{-7}), and rows are violated by up to 3.3×10^{-4} (our verifier's record is in `results/notes/`). The row violations pass the 10^{-4} tolerance of the MIPLIB checker inside BOBILib's checker (Section A.2) but not the 10^{-6} of `bobi verify`, and the stated objective 100,056 differs from the 100,056.207 of the point, so `bobi verify` rejects the file. Such a point is not bilevel feasible in the sense of Definition A.1: with an exactly optimal follower response the leader's value at the same leader choice can differ. A value accepted only under a relative tolerance is therefore not proven attainable with an exactly optimal follower response: the file is the example of a benchmark-accepted point (class (iii), Section A.4).

F. NETWORK INTERDICTION

F.1 The model as encoded in the files

BOBILib's `inter-fire` instances (from Baggio, Carvalho, Lodi and Tramontani,[29] with the vaccinated nodes already removed) have a graph (V, E) and three binary variables per node v: y_v (the leader attacks v), x_v (the follower protects v) and a_v (node v is saved). The rows are

$$\sum_v y_v \le \Phi, \qquad \sum_v x_v \le \Lambda, \qquad a_v + y_v \le 1 \;\; (v \in V),$$

$$a_v - a_u - x_v \le 0 \;\text{ for both orientations } (u,v) \text{ of every edge } uv \in E.$$

The first row is the leader's; the others are the follower's. The leader minimises $\sum_v w_v a_v$ and the follower minimises $-\sum_v w_v a_v$, with $w_v = 1$ in every instance (the arguments below only use $w \ge 0$). The reader of `bobi fire` checks every row, bound and objective coefficient of each file against this pattern and rejects anything else; a separate review session re-derived the pattern with its own parser. The game is zero-sum, so the optimistic tie-break plays no part. In this section Φ is the attack budget.

F.2 The follower's optimum as a closure

For an attack $Y \subseteq V$ and a protection set $X \subseteq V$, call $B \subseteq V$ *closed* if $Y \subseteq B$ and, for every edge and each ordering (u,v) of its two ends, $u \in B$ and $v \notin X$ imply $v \in B$. The intersection of closed sets is closed and V is closed, so there is a smallest closed set $B(Y,X)$: the nodes reached from Y by fire spreading through unprotected nodes.

Lemma F.1 (Closure). *For fixed $y = \mathbf{1}_Y$ and $x = \mathbf{1}_X$, a vector $a \in \{0,1\}^V$ satisfies the follower's rows if and only if $Z(a) = \{v : a_v = 0\}$ is closed. Hence the follower's best value for X is $w(V \setminus B(Y,X))$, and its optimum against the attack Y is*

$$\mathrm{sv}(Y) = \max\bigl\{w(V \setminus B(Y,X)) : X \subseteq V,\ |X| \le \Lambda\bigr\}.$$

Proof. The rows $a_v + y_v \le 1$ say $Y \subseteq Z(a)$. The row $a_v \le a_u + x_v$ for the orientation (u,v) says: if $a_u = 0$ and $x_v = 0$ then $a_v = 0$, that is, $u \in Z(a)$ and $v \notin X$ imply $v \in Z(a)$. So the feasible a are exactly the indicator vectors of complements of closed sets. Every closed set contains $B(Y,X)$, and $B(Y,X)$ is closed, so with $w \ge 0$ the best choice is $a = \mathbf{1}_{V \setminus B(Y,X)}$. □

Not formalised (checked against brute force over all (a,x) on 400 random graphs with up to 6 nodes, and used by the review's brute force).

Corollary F.2 (Monotonicity in the attack). *If $Y \subseteq Y'$ then $\mathrm{sv}(Y') \le \mathrm{sv}(Y)$. Hence, when $|V| \ge \Phi$, the minimum of* sv *over attacks with $|Y| \le \Phi$ is attained by an attack with exactly Φ nodes.*

Proof. $B(Y',X)$ is closed for Y (it contains $Y' \supseteq Y$), so $B(Y,X) \subseteq B(Y',X)$ and $w(V \setminus B(Y',X)) \le w(V \setminus B(Y,X))$ for every X. Any attack with fewer than Φ nodes can be enlarged. □

Not formalised. The enumeration below covers every attack size from 0 to Φ, so the certificate does not depend on this corollary. It explains why the least saved count falls with the attack size: for the instance `tree-50_1-5-5_006` the minima for attack sizes $0,\dots,5$ are $49, 48, 44, 38, 32, 26$.

F.3 The tree dynamic program

Suppose the graph is a forest. Root every component and hang the roots below a virtual root whose edges carry no constraint. Give every node v a state $\sigma_v = (x_v, a_v) \in \{0,1\}^2$; state σ is *allowed* at v unless $v \in Y$ and $a_v = 1$; states σ_p, σ_c of the ends of an edge pc are *compatible* if $a_c \le a_p + x_c$ and $a_p \le a_c + x_p$. For a node v with children $c_1, \dots, c_d$, a state σ and $0 \le q \le \Lambda$, let $T_v(\sigma, q)$ be the largest $\sum_u w_u a_u$ over the subtree of v among assignments of allowed states with v in state σ, compatible states on every edge of the subtree and exactly q protected nodes ($-\infty$ if there is none). With $T_v^{(0)}(\sigma, q) = w_v a(\sigma)$ if σ is allowed and $q = x(\sigma)$, and $-\infty$ otherwise,

$$T_v^{(i)}(\sigma, q) = \max_{0 \le q' \le q} \Bigl[T_v^{(i-1)}(\sigma, q - q') + \max_{\sigma' \text{ compatible with } \sigma} T_{c_i}(\sigma', q') \Bigr], \qquad T_v = T_v^{(d)}.$$

Proposition F.3 (Tree DP). *T_v satisfies the definition above for every node, and the follower's optimum against Y is the best value at the virtual root over $q \le \Lambda$.*

Proof. Induction over the nodes in post-order. The follower's rows other than the budget row involve one node ($a_v + y_v \le 1$) or the two ends of one edge (the two orientations of the arc row), so an assignment of the subtree of v is feasible iff all states are allowed and every edge is compatible. The edges of the subtree of v are the edges vc_i and the edges inside the subtrees of the c_i, which share no node. Hence a feasible assignment of the subtree of v with v in state σ and q protections is the same as: a state for v, and for each i a feasible assignment of the subtree of c_i whose root state is compatible with σ, with protection counts adding up to q; and its value is the sum of the parts. The recursion maximises over exactly these choices, one child at a time. At the virtual root every child state is compatible, and the budget row is $q \le \Lambda$. □

Not formalised. The DP is tested against brute force over all (a,x) on 300 random forests. A separate C implementation of a tree DP, written in the review session, was compared with ours value by value (through a hash of the sequence of values) at every attack of 0 to Φ nodes on each of the 40 `inter-fire` instances whose graph is a forest, once per instance: the 36 trees, with 9,340,760 attacks (Tables S8 and S9), and the 4 `rndgraph-25` instances that are forests, with 4,142 attacks; 9,344,902 attacks in all, with no disagreement.

F.4 Exhaustive enumeration and the certificate

Theorem F.4 (Certified optimum)**.** *Suppose that for every attack Y with $|Y| \le \Phi$ we have a follower point (x^Y, a^Y) that satisfies all follower rows at Y (checked against the raw model), with value* $\mathrm{dp}(Y) = w(a^Y)$. *Let Y^* minimise* dp*, and suppose the follower's optimum at Y^* is proven to equal* $\mathrm{dp}(Y^*)$. *Then the bilevel optimum is* $\mathrm{dp}(Y^*)$*, attained by Y^* and (x^{Y^*}, a^{Y^*}).*

Proof. For every attack, $\mathrm{sv}(Y) \ge \mathrm{dp}(Y) \ge \mathrm{dp}(Y^*)$, since the follower may play the checked point. The leader's feasible set is exactly $\{Y : |Y| \le \Phi\}$, so the optimum is at least $\mathrm{dp}(Y^*)$; and $\mathrm{sv}(Y^*) = \mathrm{dp}(Y^*)$ shows it is attained. □

Not formalised.

The certificate does not rest on the DP being optimal at any attack other than Y^*. There the follower's optimum is given by Proposition F.3, HiGHS re-solves the follower MILP built from the raw model, and the review's brute force over every protection set of at most Λ nodes (Lemma F.1) gives the same value. For `tree-50_1-5-5_006` and `_010` the graph has 49 nodes and $\Phi = \Lambda = 5$; the enumeration visits every k-subset for $k = 0, \ldots, 5$ (the count per size is checked against $\binom{49}{k}$), $\sum_{k=0}^{5} \binom{49}{k} = 2{,}138{,}410$ attacks per instance (1,906,884 of them with exactly 5 nodes), and both optima are 26. The review session obtained the same minima, the same number of optimal attacks and the same per-size minima with its own code. The attack counts of Tables S8 and S9 are of this kind, every attack of 0 to Φ nodes once per instance: 9,340,760 over the 36 trees (5,063,940 over the 34 of Table S9).

These optima are exact certificates (class (i), Section A.4): every witness is a 0/1 point of a model with integer coefficients, checked against every row with no tolerance in play, and the follower's optimum at Y^* rests on Proposition F.3 and on the review's exhaustive search; the HiGHS re-solve adds a solver check. The 12 `rndgraph-25` values that `bobi fire` re-derives with HiGHS as the follower at every attack (47,496 attacks on 28 instances, trees included) are solver-certified (class (ii)); the review's brute force re-derives them and 8 larger random graphs exactly.

G. FORMAL VERIFICATION MAP (LEAN)

The project `bobilib/lean`, written in Lean[4] (version 4.35.0) with Mathlib[5] (tag v4.35.0), proves the lemmas in Table S1. Lean file names are relative to `bobilib/lean`; Rust names are in `bobilib/src` (`Search`, `RobustTable` and `play_order` are in `robust_ss.rs`; `Kip` is in `kip.rs`). The Lean statements are about abstract models (items, weight classes, graphs); the Rust code is not verified, and the right-hand column of the table says which code relies on each statement.

Table S1. Results of this supplement with machine-checked proofs: the Lean theorems that state them and the Rust code that relies on them.

Result	Lean theorems (file)	Rust code that relies on it
Knapsack interdiction (Section B)		
Lemma B.1 (monotonicity)	`Bobi/Knapsack.lean`: `follower_antitone`, `bestAvoiding_antitone`, `bestAvoiding_eq_sup_sdiff`, `packings_downward_closed`, `K_eq_bestAvoiding`	`Kip::free_items`, `Kip::greedy`, `Kip::subset_sum_classes`; guaranteed copies in `Search::node`
Lemma B.2 (count formulation)	`Bobi/Knapsack.lean`: `K_map_equiv`, `swap_block`, `cheapest_first`	`Kip::subset_sum_classes`, `Kip::x_from_deletions`; branching on counts in `Search::node`
Lemmas B.5, B.6 (windows, stall test)	`Bobi/SubsetSum.lean`: `window_extend`, `window_extend_all`, `fits_survivors`, `window_survivors`, `SubsetSum.perm`, `Window.perm`	`Search::window_safe`

(continued)

Result	Lean theorems (file)	Rust code that relies on it
Theorem B.9 (online bound)	`Bobi/Online.lean`: `online_le_K`, `online_le_K_any`, `online_lower_bound`; example: `table1_online`, `table1_block`, `table1_opt`	`mkip::omega_root`
Theorem B.12 (table soundness)	`Bobi/RobustTable.lean`: `H_sound`, `H_sound_any`, `take_sum_le`, `counts_of_deletion`; `Bobi/BandTable.lean`: `G_sound`, `G_sound_any`, `G_one`, `G_sound_one`; `Bobi/EndTable.lean`: `GA_sound`, `GA_sound_any`, `G_iff_GA`, `H_iff_GA`, `G_sound_of_GA`, `H_sound_of_GA`	`RobustTable::build`, `RobustTable::build_band`
Proposition B.13 (play orders)	`Bobi/EndTable.lean`: `counts_perm`, `remaining_eq_flatten`, `totalCost_eq_sum`, `remaining_perm`, `totalCost_perm`, `GA_sound_perm`, `GA_sound_perm_any`	`play_order`, `RobustTable::build_suffix`, `Search::node_table_threshold`
Corollary B.14 (root-table rules)	`Bobi/RobustTable.lean`: `covers_sound`; `Bobi/BandTable.lean`: `band_covers_sound`, `covers_sound_of_band`; `Bobi/Threshold.lean`: `covers_threshold_sound`, `covers_threshold_sound_perm`, `band_covers_threshold_sound`, `band_covers_threshold_sound_perm`	`RobustTable::covers`, `RobustTable::covers_run`, `Search::tables_cover`
Lemma B.15 (antitone in the budget)	`Bobi/EndTable.lean`: `GA_antitone`, `G_antitone`, `H_antitone`, `GA_mono`	threshold storage in `RobustTable`; cache entries of `Search::remember`
Theorem B.17 (threshold form)	`Bobi/Threshold.lean`: `thr` (definition), `thr_iff`, `GA_iff_thr`, `thr_bounds`, `thr_eq_neg_one_iff`, `thr_greatest`, `G_iff_thr`, `H_iff_thr`, `le_minOver`, `le_maxOver`	`RobustTable::step`, `RobustTable::base`, `RobustTable::threshold`, `RobustTable::contains`, `RobustTable::step_range`
Theorem B.19 (per-node rule)	`Bobi/EndTable.lean`: `nodeEnd` (definition), `node_rule`, `node_table_sound`, `node_table_sound_perm`, `node_table_sound_any`; `Bobi/Threshold.lean`: `node_table_threshold_sound`, `node_table_threshold_sound_counts`	`Search::node_table_threshold`, the prune and cache entry in `Search::node`, `Search::remember`
Remark after Theorem B.19 (strength)	`Bobi/EndTable.lean`: `node_table_of_H`	none (strength, not soundness)
Edge clique interdiction (Section D)		
Theorem D.2 (Mantel, deletion form)	`Bobi/Clique.lean`: `mantel`, `mantel'`, `triangle_survives`, `mantel_40`, `mantel_50`, `mantel_60`, `clique40`, `clique50`, `clique60`	none; the arithmetic is also checked by `scripts/clique/lb.py`
Proposition D.3 (partition bound)	`Bobi/Clique.lean`: `cliqueFree_of_partition`	none; the certificates are written by `scripts/clique/ub.py`

All names in the middle column are in namespace `Bobi`. There is no `sorry`; `lean/Check.lean` prints the axioms of 72 main theorems, each of which uses at most `propext`, `Classical.choice` and `Quot.sound`.

Not formalised.

- *Subset-sum search.* The reduction of Proposition B.4 as a whole (free and unusable items, the gcd scaling; `Kip::subset_sum_classes`); the fractional deletion bound, Lemma B.7 (`Cumulative::min_cost`), the exact largest deletable mass (`Search::build_max_del`) and the reduction of the stall test to one test per class in Proposition B.8; restricted copies, Proposition B.21 (`restrict_copies`); Lemma B.22, Theorems B.23 and B.24 and Corollary B.25 as statements

about the code (`Search::node`, `Search::run`, `Kip::solve_subset_sum`); that the bitset holds exactly the decided subset sums up to *hi*; that the Rust loops compute thr, including the range shortcuts and early exit of `RobustTable::step_range` and `Search::node_table_threshold`, the fallback to `RobustTable::step` for classes with more than 63 copies, the `row_max` and `blockmax` skips, and the absence of overflow in the `i32` arithmetic (all tested against a direct transcription of thr and against brute force); and the containment of the band rules in the per-node rule.

- *Multidimensional knapsack interdiction.* Lemmas C.1 and C.3, Proposition C.2 and Theorem C.4 (`PackingInterdiction::lp_follower_bound` in `mkip.rs`); the merging, scaling and rounding of Theorem C.5, its properties (L) and (F) and its rounding analysis (`mkip::omega_root`, `mkip::tune_mu`, `mkip::omega_scales`); the local search; all floating-point evaluation.

- *Clique interdiction.* Lemma D.1; Brouwer's theorem (Theorem D.4) and Corollary D.5, which are neither proved nor assumed in Lean; Theorem D.6; Proposition D.7 and the 170 solver verdicts; that the instance files have the stated n, m and k and that each certificate partition has at most k inside edges (Lean takes these numbers as hypotheses).

- *Network interdiction.* Lemma F.1, Corollary F.2, Proposition F.3 (`fire::TreeDp::solve`) and Theorem F.4 (`fire::enumerate`).

- *General instances.* Theorem E.1 (`general::follower_unbounded_ray`, checked exactly by `scripts/verify_ray.py`) and Meyer's theorem; Lemma E.3 and the model comparison (`scripts/compare_hpr.py`); Proposition E.4 (`general::feasible_point`, `general::optimal_response`); Proposition E.5 and Lemma E.6 (`coupled::Search::evaluate`, `coupled::Search::best_response`, `coupled::certify`).

H. DATA PACKAGE AND FORMATS

Every result in the paper is backed by a file in the data package, the folder `bobilib/results/`. The package's index, `results/README.md`, follows the repository style of the INFORMS Journal on Computing (cite, description, data, file formats, verification, replication). The index, `instances.csv`, `bobilib_update.solu` and `SHA256SUMS` are written by `paper/scripts/build_data_package.py`; `make data` in `bobilib/paper/` regenerates them, together with the macros of every number quoted in the paper and the tables of Section I. BOBILib's statuses are those of its metadata snapshot of 18 October 2026.

H.1 Layout

Path	Contents
`instances.csv`	one row per BOBILib instance (2,654 rows); Section H.2
`bobilib_update.solu`	status changes and improved values in the MIPLIB/SCIP `.solu` format; Section H.3
`SHA256SUMS`	SHA-256 checksum of every other file of the package
`summary.json`	per-instance outcome and per-family counts
`kip/`	583 knapsack interdiction solution files at the published optima of Weninger and Fukasawa[8]
`ss/`	the 16 subset-sum optima proven here; `proofs.csv` (optimum, nodes, CPU seconds, nodes of the second prover, budget and capacity)
`or/`	690 verified points for the 810 OR instances; the bounds of each instance; `compare.csv` (published bounds, BOBILib's value, ours, gaps)
`clique/`	60 partition certificates, branch-and-bound certificates of the 170 part-size cases (`certificates/`, with the checker's `check.log`), solver logs of the part-size proofs, the separate review
`network/`	36 attack-protect trees certified by enumeration; `summary.csv`; the review
`rays/`	41 recession directions (the 40 open instances and one that BOBILib lists as infeasible); `ray_verify.jsonl` (exact checks)
`general/`	64 points for MIPLIB-derived instances without coupling rows (nine files over 1 MB in `large_points.tar.gz`); `hpr_optimal.csv`; `points.csv`
`general-coupled/`	29 points for MIPLIB-derived instances with coupling rows; `summary.csv`
`*/tolerance/`	the 26 superseded files of `general/` and `general-coupled/` (25 of them benchmark-accepted), replaced by points with the follower within 10^{-9} of its optimum, and the checks of the new files
`notes/`	`bobi verify --tight` on every file (`tight_verify.md`, `.tsv`); Lefebvre and Schmidt's run files checked against this package; BOBILib's file of `general20-30-20-20-20-2`; `instance_structure.csv`, the structure of the general and assignment instances that the paper describes

Each folder has its own `README.md` with the method, the columns of its tables, the commands that produced it and, where one was done, its separate review.

H.2 The instance table `instances.csv`

The file is UTF-8, comma separated, with one header row and one row per BOBILib instance. Its columns:

Column	Meaning
`instance`	BOBILib instance name
`collection`	BOBILib sub-collection (last directory of the instance path), e.g. `kp`, `or`, `miplib2017`
`family`	`knapsack`, `multidim-knapsack`, `clique`, `network`, `generalized`, `assignment` (interdiction), `general-pure`, `general-mixed`
`bobilib_status`, `bobilib_value`	BOBILib's status and best known objective value (empty: none)
`new_status`	`optimal` (proven, in published work or here), `infeasible_bilevel` (no bilevel feasible point), `feasible` (verified point here; BOBILib had none; the notes say whether it is the first point anywhere, or name the published point), `unchanged`, `bobilib_optimal`, `bobilib_infeasible`
`primal_value`	best upper-level value backed by a verified solution file (of this package or of BOBILib) or by a published optimum
`dual_bound`	best proven lower bound (the leader minimises), if any
`source`	citation key of the published result (Table S2), `this work`, or `ThuraufEtAl2026` (BOBILib's own value)
`method`	how the status or value was obtained
`certificate`	path, relative to `results/`, of the solution or certificate file; `ARCHIVE.tar.gz:FILE` names a member of an archive
`verified`	`yes`: the certificate passed its check (`bobi verify`, or the exact ray check); `no`: it did not (no row); `na`: no certificate in this package. The column says whether a check passed, not how rigorous it is
`certificate_class`	the certificate class of the row's result (Definition A.2): `exact`, `numerical` or `benchmark` for every row with `new_status` `optimal`, `infeasible_bilevel` or `feasible`, `na` for the others; a published optimum counts as `numerical`
`notes`	free text, separated by semicolons

Rows per `new_status`: 1,027 `bobilib_optimal`, 33 `bobilib_infeasible`, 610 `optimal`, 40 `infeasible_bilevel`, 228 `feasible` and 716 `unchanged`. The 610 optimal rows are the 528 published optima and the 82 proven here; together with the 40 rows without a bilevel feasible point they are the 650 settled instances. Of the 878 rows with a result, 115 are `exact` (the 75 new optima of the subset-sum, clique and network families and the 40 recession directions), 763 `numerical` (the 528 published optima, the 7 optima at the MIPLIB bound and the 228 verified points, all of which pass `bobi verify --tight`) and 0 `benchmark`. The keys of the `source` column map to the references as follows.

Table S2. Citation keys of the `source` column of `instances.csv`.

Key in `source`	Reference	Optima
`ThuraufEtAl2026`	BOBILib's own value[1]	
`WeningerFukasawa2025`	Weninger and Fukasawa[8]	376
`LozanoBergmanCire2026`	Lozano, Bergman and Cire[10]	97
`FischettiEtAl2019`	Fischetti, Ljubić, Monaci and Sinnl[9]	52
`FischettiEtAl2017`	Fischetti, Ljubić, Monaci and Sinnl[30]	1
`VasquezLozanoVanHoeve2025`	Vásquez, Lozano and van Hoeve[31]	1
`BOBILibEarlierRelease`	an earlier release of BOBILib's metadata[32]	1
`this work`	a certificate in this package	82

H.3 The status file `bobilib_update.solu`

The file follows the MIPLIB/SCIP convention, one line per instance whose status differs from BOBILib's metadata or whose best value improves:

`=opt=`	*name value*	proven optimal value (published, or proven here)
`=inf=`	*name*	no bilevel feasible point (the follower's problem is infeasible or unbounded for every leader choice)
`=best=`	*name value*	best verified bilevel feasible point (a solution file here), not proven optimal
`=bestdual=`	*name value*	best proven lower bound, where this work proves a better one than any published

It has 610 `=opt=` lines (528 published, 82 from this work), 40 `=inf=`, 693 `=best=` and 194 `=bestdual=` lines. The `=bestdual=` lines are the 191 OR instances with one follower knapsack and the 3 open clique instances; an instance can have both a `=best=` and a `=bestdual=` line. Every value is the upper-level objective $c^\top x + d^\top y$ of the minimising leader under the optimistic convention, printed at full precision (integers without a decimal point, other values as the shortest decimal that reads back as the same double). Comment lines start with `#`; the header gives the counts and states that every `=inf=` line rests on a recession direction checked in exact rational arithmetic.

H.4 Solution files

Solution files (`*.res.json`) are in BOBILib's own format, so BOBILib's checker and `bobi verify` read them unchanged. Each is a JSON object with the fields `name`, `bilevel_type` (always `optimistic`), `status` (`optimal` or `feasible`), `difficulty`, `objective_value` (the upper-level objective, which BOBILib minimises), and `upper_level_decisions` and `lower_level_decisions`, which map every variable name of the instance to its value. A file states `optimal` only when the instance's row in `instances.csv` has `new_status optimal`.

H.5 Certificates

Recession directions. `rays/*.ray.json` maps follower variable names to the entries of a direction d (its support only), as decimal numbers; `*.exact.ray.json` holds exact rationals written as `"p/q"`. The certificate is valid if d satisfies the conditions of Theorem E.1: for every follower row, with the leader terms and right-hand side dropped, $a^\top d \le 0$ for a row $a^\top y \le \cdot$, $a^\top d \ge 0$ for a row $a^\top y \ge \cdot$ and $a^\top d = 0$ for an equation; $d_j \ge 0$ for a finite lower bound and $d_j \le 0$ for a finite upper bound; and $f^\top d < 0$. `scripts/verify_ray.py` checks this in rational arithmetic and writes `rays/ray_verify.jsonl`.

Partitions. The clique solution files are the certificates: the leader deletes exactly the edges inside the parts of an r-partition (at most k of them), so the remainder is r-partite (Proposition D.3), and the follower part is a maximum clique of the remainder. `clique/summary.csv` gives per instance n, the density, m, k, BOBILib's value, the lower bound with its argument (Mantel, or Brouwer's theorem with the part-size proofs) and the upper bound with its partition. The logs of the solver runs behind the 170 part-size proofs are in `clique/logs/lb_solver_logs.tar.gz`, and the branch-and-bound certificates of the same cases, one file per density-0.9 instance, are in `clique/certificates/`; the integer checker `scripts/check_clique_certificates.py` verifies them from the instance files alone (Section D; its output is `check.log`, and `certificates/README.md` gives the format and the tests).

Subset-sum proofs. A proof of OPT $\ge \ell$ is a computation, not a file. `ss/proofs.csv` records, per instance, the optimum, the nodes and CPU seconds of the search and the nodes of the second prover `scripts/ss_check.py`, which re-proves the bound from the instance files (Section A.3).

Enumerations and HPR bounds. `network/summary.csv` gives per tree the attacks enumerated, the optimum, the number of optimal attacks and the cross-checks (Theorem F.4). `general/hpr_optimal.csv`

lists the points that meet the MIPLIB optimum of the HPR; `scripts/compare_hpr.py` re-checks that the HPR is the MIPLIB model (Lemma E.3).

OR bounds. `or/bounds_one_knapsack.jsonl` gives, per instance with one follower knapsack, the lower bound with its two parts (merged-row online bound, Theorem C.5, and LP bracket, Theorem C.4) and the upper bound; `or/points_several_knapsacks.jsonl` gives, per instance with several follower knapsacks, the bracket minimiser's LP bound and the certified follower value.

H.6 Checksums and how to check the package

`SHA256SUMS` has one line per file of the package other than itself, sorted by path, in the format of `sha256sum`. With BOBILib's instance files in `data/collection` (from `bobilib.org`) and the `bobi` crate built, the following commands, run from `bobilib/`, repeat the checks of Section A.3:

```
(cd results && sha256sum -c SHA256SUMS)
D=data/collection
./target/release/bobi verify --instances $D results/kip results/ss results/network results/or
./target/release/bobi verify --instances $D --generic results/clique results/general \
    results/general-coupled
python3 -I scripts/verify_ray.py INSTANCE.mps.gz INSTANCE.aux results/rays/INSTANCE.ray.json
python3 -I scripts/ss_check.py INSTANCE.mps.gz INSTANCE.aux - OPTIMUM
python3 -I scripts/compare_hpr.py INSTANCE.mps.gz INSTANCE.aux MIPLIB.mps.gz
python3 -I scripts/check_clique_certificates.py \
    $D/interdiction/clique-interdiction/clique results/clique/certificates
./target/release/bobi verify --instances $D --tight --time-limit 1200 -j 2 DIR
```

The nine large general points are checked after unpacking `general/large_points.tar.gz`.

I. PER-INSTANCE TABLES

The tables below are generated from the data package by `paper/scripts/gen_appendix_tables.py`. Values are upper-level objectives of the minimising leader; full-precision values and the certificate of every row are in `instances.csv` (Section H.2). The order follows the paper: published optima (Table S3); knapsack interdiction (Table S4); the OR set (Tables S5 and S6); clique interdiction (Table S7); network interdiction (Tables S8 and S9); general instances without a feasible point (Table S10), with an HPR-bound optimum (Table S11) and with a verified point where BOBILib has none (Table S12); and the instances whose status is unchanged (Table S13).

I.1 Published optima

Table S3. Published optima for instances BOBILib lists as feasible or open, by source (sub-collection and count). The instance names follow.

Source	Collections	Instances
Weninger and Fukasawa (2025), combinatorial branch and bound	kp 336, inter-kp 25, cclw 15	376
Lozano, Bergman and Cire (2026), decision-diagram relaxations	or 88, miplib3 5, generalized 4	97
Fischetti, Ljubić, Monaci and Sinnl (2019), interdiction cuts	imkp 52	52
earlier BOBILib release (listed optimal there)	miplib2017 1	1
Fischetti, Ljubić, Monaci and Sinnl (2017), MIX++	miplib3 1	1
Vásquez, Lozano and van Hoeve (2025), decision diagrams	miplib3 1	1
Total		528

Weninger and Fukasawa (2025), combinatorial branch and bound (376)

interdiction40-5
interdiction40-6
interdiction45-4
interdiction45-5
interdiction45-6
interdiction50-3
interdiction50-4
interdiction50-5
interdiction50-7
interdiction55-3
interdiction55-4
interdiction55-5
interdiction55-6
interdiction55-7
interdiction55-8
interKP-100-100-1-1
interKP-100-100-1-2
interKP-100-100-1-3
interKP-100-100-1-4
interKP-100-100-1-5
interKP-100-100-1-6
interKP-100-100-1-7
interKP-100-100-1-8
interKP-100-100-2-1
interKP-100-100-2-2
interKP-100-100-2-3
interKP-100-100-2-4
interKP-100-100-2-5
interKP-100-100-2-6
interKP-100-100-2-7
interKP-100-100-3-1
interKP-100-100-3-2
interKP-100-100-3-3
interKP-100-100-3-4
interKP-100-100-3-5
interKP-100-100-3-6
interKP-100-100-3-7
interKP-100-100-3-8
interKP-100-100-3-9
interKP-100-100-4-2
interKP-100-100-4-3
interKP-100-100-4-4
interKP-100-100-4-5
interKP-100-100-4-6
interKP-100-100-4-7
interKP-100-100-4-8
interKP-100-100-5-1
interKP-100-100-5-2
interKP-100-100-5-3
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interKP-100-100-5-5
interKP-100-100-5-6
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interKP-100-100-6-6
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interKP-200-100-1-1
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interKP-500-100-8-5
interKP-500-100-8-6
interKP-500-100-8-7
interKP-500-100-8-8
interKP-500-100-8-9
interKP-500-100-8-10
K5040W01.KNP
K5040W05.KNP
K5040W08.KNP
K5040W15.KNP
K5040W19.KNP
K5050W01.KNP
K5050W02.KNP
K5050W03.KNP
K5050W04.KNP
K5050W05.KNP
K5050W06.KNP
K5050W07.KNP
K5050W08.KNP
K5050W09.KNP
K5050W10.KNP
K5050W11.KNP
K5050W12.KNP
K5050W13.KNP
K5050W14.KNP
K5050W15.KNP
K5050W16.KNP
K5050W17.KNP
K5050W18.KNP
K5050W19.KNP
K5050W20.KNP

Lozano, Bergman and Cire (2026), decision-diagram relaxations (97)

general30-30-10-20-20-8
general30-30-20-20-20-9
general30-30-30-20-20-9
general30-30-30-20-20-10
lseu-0.500000
OR5x100-0.75_1.dat-0
OR5x100-0.75_1.dat-50
OR5x100-0.75_1.dat-100
OR5x100-0.75_2.dat-0
OR5x100-0.75_2.dat-50
OR5x100-0.75_2.dat-100
OR5x100-0.75_3.dat-0
OR5x100-0.75_3.dat-50
OR5x100-0.75_3.dat-100
OR5x100-0.75_4.dat-0
OR5x100-0.75_4.dat-50
OR5x100-0.75_4.dat-100
OR5x100-0.75_5.dat-0
OR5x100-0.75_5.dat-50
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OR5x100-0.75_6.dat-0
OR5x100-0.75_6.dat-50
OR5x100-0.75_6.dat-100
OR5x100-0.75_7.dat-0
OR5x100-0.75_7.dat-50
OR5x100-0.75_7.dat-100
OR5x100-0.75_8.dat-0
OR5x100-0.75_8.dat-50
OR5x100-0.75_8.dat-100
OR5x100-0.75_9.dat-0
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OR5x100-0.75_9.dat-100
OR5x100-0.75_10.dat-0
OR5x100-0.75_10.dat-50
OR5x100-0.75_10.dat-100
OR5x250-0.75_1.dat-0
OR5x250-0.75_1.dat-50
OR5x250-0.75_1.dat-100
OR5x250-0.75_2.dat-0
OR5x250-0.75_2.dat-50
OR5x250-0.75_2.dat-100
OR5x250-0.75_3.dat-0
OR5x250-0.75_3.dat-50
OR5x250-0.75_3.dat-100
OR5x250-0.75_4.dat-0
OR5x250-0.75_4.dat-50
OR5x250-0.75_4.dat-100
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OR5x250-0.75_5.dat-50
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OR5x250-0.75_6.dat-100
OR5x250-0.75_7.dat-0
OR5x250-0.75_7.dat-50
OR5x250-0.75_7.dat-100
OR5x250-0.75_8.dat-0
OR5x250-0.75_8.dat-50
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OR10x100-0.75_6.dat-100
OR10x100-0.75_7.dat-0
OR10x100-0.75_7.dat-100
OR10x100-0.75_8.dat-0
OR10x100-0.75_8.dat-100
OR10x100-0.75_9.dat-0
OR10x100-0.75_9.dat-100
OR10x100-0.75_10.dat-100
OR10x250-0.75_1.dat-100
OR10x250-0.75_6.dat-100
OR10x250-0.75_7.dat-100

OR30x100-0.75_1.dat-100	OR30x100-0.75_5.dat-0	OR30x100-0.75_8.dat-100	p0548-0.900000
OR30x100-0.75_2.dat-100	OR30x100-0.75_5.dat-100	p0201-0.100000	
OR30x100-0.75_3.dat-100	OR30x100-0.75_6.dat-100	p0282-0.900000	
OR30x100-0.75_4.dat-100	OR30x100-0.75_7.dat-100	p0548-0.100000	

Fischetti, Ljubić, Monaci and Sinnl (2019), interdiction cuts (52)

pet7.dat-0	weish17.dat-50	weish21.dat-0	weish24.dat-100	weish28.dat-50
pet7.dat-50	weish17.dat-100	weish21.dat-50	weish25.dat-0	weish28.dat-100
pet7.dat-100	weish18.dat-0	weish21.dat-100	weish25.dat-50	weish29.dat-0
sento2.dat-50	weish18.dat-50	weish22.dat-0	weish25.dat-100	weish29.dat-50
sento2.dat-100	weish18.dat-100	weish22.dat-50	weish26.dat-0	weish29.dat-100
weish14.dat-0	weish19.dat-0	weish22.dat-100	weish26.dat-50	weish30.dat-0
weish14.dat-50	weish19.dat-50	weish23.dat-0	weish26.dat-100	weish30.dat-50
weish14.dat-100	weish19.dat-100	weish23.dat-50	weish27.dat-0	weish30.dat-100
weish16.dat-0	weish20.dat-0	weish23.dat-100	weish27.dat-50	
weish16.dat-50	weish20.dat-50	weish24.dat-0	weish27.dat-100	
weish16.dat-100	weish20.dat-100	weish24.dat-50	weish28.dat-0	

Earlier BOBILib release (listed optimal there) (1)

neos-950242_0_100

Fischetti, Ljubić, Monaci and Sinnl (2017), MIX++ (1)

air03-0.100000

Vásquez, Lozano and van Hoeve (2025), decision diagrams (1)

p0282-0.500000

I.2 Knapsack interdiction

Table S4. The 16 knapsack interdiction instances without a published optimum, all proven optimal here by the robust subset-sum search. n: items; class 6: subset sum ($p = w$), class 7: even weights and odd capacity; I: the instance number of the name, which sets the capacity; β: the leader's budget; c: the follower's capacity. Every optimum equals c (class 6) or $c-1$ (class 7), the trivial upper bound, and BOBILib's value. Each lower bound is a computational proof supported by formally verified lemmas (an exact certificate, class (i) of Section A.4). Nodes and CPU seconds of `bobi ss-solve` on one thread of a shared machine; second prover: nodes of `scripts/ss_check.py`, which shares no code with the search.

Instance	n	Class	I	β	c	Optimum	Nodes	CPU (s)	Prover nodes
interKP-200-100-6-2	200	6	2	1,860	1,855	1,855	303	0.10	303
interKP-300-100-6-2	300	6	2	2,865	2,868	2,868	366	0.16	366
interKP-400-100-6-2	400	6	2	3,828	3,832	3,832	257	0.15	257
interKP-500-100-6-2	500	6	2	4,645	4,651	4,651	190	0.26	190
interKP-200-100-6-3	200	6	3	2,787	2,782	2,782	13,498	3.93	13,502
interKP-300-100-6-3	300	6	3	4,299	4,302	4,302	19,405	8.06	19,537
interKP-400-100-6-3	400	6	3	5,744	5,748	5,748	13,430	9.01	13,583
interKP-500-100-6-3	500	6	3	6,970	6,976	6,976	10,079	10.04	10,116
interKP-200-100-7-2	200	7	2	1,878	1,873	1,872	22	0.06	22
interKP-300-100-7-2	300	7	2	2,892	2,895	2,894	27	0.04	27
interKP-400-100-7-2	400	7	2	3,863	3,867	3,866	18	0.05	18
interKP-500-100-7-2	500	7	2	4,689	4,695	4,694	1	0.05	1
interKP-200-100-7-3	200	7	3	2,814	2,809	2,808	651	0.10	651
interKP-300-100-7-3	300	7	3	4,338	4,341	4,340	470	0.20	470

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Instance	n	Class	I	β	c	Optimum	Nodes	CPU (s)	Prover nodes
interKP-400-100-7-3	400	7	3	5,797	5,801	5,800	378	0.17	378
interKP-500-100-7-3	500	7	3	7,037	7,043	7,042	256	0.15	256

I.3 The OR set

Table S5. All 810 multidimensional knapsack interdiction instances of the OR set; names have the form `OR`*rows*`x`*items*`-`*budget*`_`*i*`.dat-`*variant*. Budget: the tightness α in percent (the right-hand side of every knapsack row, leader's and follower's, is α times the sum of its coefficients); follower rows: knapsack rows of the follower (variant -100: one; -0 and -50: several). Published LB: Lozano, Bergman and Cire (2026); published UB: best published point (Lozano, Bergman and Cire, or Lefebvre and Schmidt (2025); not verified here); BOBILib: BOBILib's best known value; our LB: dualised follower LP bracket or merged-row online bound (one follower knapsack only); our point: verified solution file; gap: (best upper bound − best lower bound)/best upper bound in percent, where the best upper bound may be the follower LP bound at our leader choice. n/a: no nontrivial lower bound is known (published or here; the only bound is the trivial 0), so no gap is given; these 143 instances, all with several follower knapsacks, are left out of every gap statistic of the paper and of this supplement. Blank: none. Published UB counts Lefebvre and Schmidt's values only from runs that ended before the time limit (Table S6). †: the published lower bound exceeds the published point; the instance is closed at 8,813, the authors' branch-and-cut bound.

Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR5x100-0.25_1.dat-0	25	4	16,014	20,321	21,879		19,956	19.75
OR5x100-0.25_2.dat-0	25	4	15,984	20,174	22,491		19,943	19.85
OR5x100-0.25_3.dat-0	25	4	15,180	19,538	21,738		19,312	21.40
OR5x100-0.25_4.dat-0	25	4	16,092	20,786	22,263		20,353	20.94
OR5x100-0.25_5.dat-0	25	4	15,645	20,356	21,965		19,957	21.61
OR5x100-0.25_6.dat-0	25	4	16,312	20,774	22,601		20,235	19.39
OR5x100-0.25_7.dat-0	25	4	16,257	20,848	22,905		20,484	20.64
OR5x100-0.25_8.dat-0	25	4	15,874	20,050	22,343		19,849	20.03
OR5x100-0.25_9.dat-0	25	4	16,103	20,661	22,294		20,382	20.99
OR5x100-0.25_10.dat-0	25	4	16,095	20,912	22,698		20,967	23.03
OR5x100-0.25_1.dat-50	25	2	17,464	21,887	23,044		21,569	19.03
OR5x100-0.25_2.dat-50	25	2	17,711	21,621	23,112		21,219	16.53
OR5x100-0.25_3.dat-50	25	2	17,082	21,405	22,880		20,930	18.39
OR5x100-0.25_4.dat-50	25	2	16,472	21,595	22,781		21,439	23.17
OR5x100-0.25_5.dat-50	25	2	16,576	21,031	21,738		20,621	19.62
OR5x100-0.25_6.dat-50	25	2	18,143	22,689	24,519		22,527	19.46
OR5x100-0.25_7.dat-50	25	2	17,106	22,366	23,302		22,125	22.68
OR5x100-0.25_8.dat-50	25	2	17,155	21,697	22,739		21,330	19.57
OR5x100-0.25_9.dat-50	25	2	17,061	21,121	22,661		20,882	18.30
OR5x100-0.25_10.dat-50	25	2	18,212	22,642	23,881		22,543	19.21
OR5x100-0.25_1.dat-100	25	1	20,026	24,034	24,489	23,062	23,482	1.79
OR5x100-0.25_2.dat-100	25	1	18,986	22,045	22,393	21,401	21,750	1.60
OR5x100-0.25_3.dat-100	25	1	19,494	22,455	23,303	21,768	22,110	1.55
OR5x100-0.25_4.dat-100	25	1	20,713	24,754	25,202	24,026	24,260	0.96
OR5x100-0.25_5.dat-100	25	1	19,603	23,666	24,344	22,599	23,026	1.85
OR5x100-0.25_6.dat-100	25	1	19,923	23,962	24,483	23,103	23,431	1.40
OR5x100-0.25_7.dat-100	25	1	21,157	25,569	26,905	24,642	24,958	1.27
OR5x100-0.25_8.dat-100	25	1	20,006	23,937	25,286	23,135	23,514	1.61
OR5x100-0.25_9.dat-100	25	1	20,298	24,653	25,199	23,862	24,184	1.33
OR5x100-0.25_10.dat-100	25	1	19,389	23,831	24,840	22,970	23,242	1.17
OR5x100-0.50_1.dat-0	50	4	18,975	23,672	27,489		19,785	4.09
OR5x100-0.50_2.dat-0	50	4	19,757	22,821	27,526		21,265	7.09

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR5x100-0.50_3.dat-0	50	4	19,204	23,929	25,707		21,226	9.53
OR5x100-0.50_4.dat-0	50	4	20,704	24,066	30,878		21,779	4.94
OR5x100-0.50_5.dat-0	50	4	18,942	22,742	26,240		21,172	10.53
OR5x100-0.50_6.dat-0	50	4	17,608	23,765	30,318		22,019	20.03
OR5x100-0.50_7.dat-0	50	4	18,963	22,721	29,946		21,030	9.83
OR5x100-0.50_8.dat-0	50	4	20,364	23,945	29,893		22,283	8.61
OR5x100-0.50_9.dat-0	50	4	19,037	24,888	32,863		23,465	18.87
OR5x100-0.50_10.dat-0	50	4	19,569	25,360	27,107		22,596	13.40
OR5x100-0.50_1.dat-50	50	2	21,207	30,496	33,475		29,651	28.48
OR5x100-0.50_2.dat-50	50	2	20,796	30,461	32,306		28,254	26.40
OR5x100-0.50_3.dat-50	50	2	20,033	30,123	31,051		28,000	28.45
OR5x100-0.50_4.dat-50	50	2	21,539	31,945	33,225		29,334	26.57
OR5x100-0.50_5.dat-50	50	2	20,100	29,003	31,008		27,882	27.91
OR5x100-0.50_6.dat-50	50	2	19,414	27,930	30,410		27,331	28.97
OR5x100-0.50_7.dat-50	50	2	19,888	28,098	31,826		26,901	26.07
OR5x100-0.50_8.dat-50	50	2	21,341	30,971	34,206		29,952	28.75
OR5x100-0.50_9.dat-50	50	2	19,847	31,902	33,819		29,032	31.64
OR5x100-0.50_10.dat-50	50	2	19,879	31,436	31,881		29,152	31.81
OR5x100-0.50_1.dat-100	50	1	20,708	30,100	31,172	28,108	29,437	4.51
OR5x100-0.50_2.dat-100	50	1	20,894	29,084	30,477	27,660	28,530	3.05
OR5x100-0.50_3.dat-100	50	1	22,308	29,545	29,887	27,960	28,767	2.81
OR5x100-0.50_4.dat-100	50	1	21,678	31,632	32,359	29,107	30,499	4.56
OR5x100-0.50_5.dat-100	50	1	20,966	28,955	29,142	27,322	28,269	3.35
OR5x100-0.50_6.dat-100	50	1	19,588	28,598	28,746	26,699	27,621	3.34
OR5x100-0.50_7.dat-100	50	1	22,087	28,623	30,947	26,727	27,915	4.26
OR5x100-0.50_8.dat-100	50	1	21,521	31,740	33,174	29,358	30,424	3.50
OR5x100-0.50_9.dat-100	50	1	20,773	30,871	31,618	28,251	29,163	3.13
OR5x100-0.50_10.dat-100	50	1	23,071	30,367	30,448	28,292	29,232	3.22
OR5x100-0.75_1.dat-0	75	4	8,855	8,855	9,350		8,855	0.00
OR5x100-0.75_2.dat-0	75	4	11,323	11,323	12,841		11,323	0.00
OR5x100-0.75_3.dat-0	75	4	9,344	9,344	10,231		9,344	0.00
OR5x100-0.75_4.dat-0	75	4	9,098	9,098	10,126		9,098	0.00
OR5x100-0.75_5.dat-0	75	4	8,492	8,492	8,895		8,492	0.00
OR5x100-0.75_6.dat-0	75	4	8,775	8,775	10,651		8,775	0.00
OR5x100-0.75_7.dat-0	75	4	8,596	8,596	9,311		8,596	0.00
OR5x100-0.75_8.dat-0	75	4	9,712	9,712	10,945		9,712	0.00
OR5x100-0.75_9.dat-0	75	4	9,212	9,212	9,964		9,212	0.00
OR5x100-0.75_10.dat-0	75	4	8,785	8,785	9,089		8,785	0.00
OR5x100-0.75_1.dat-50	75	2	12,477	12,477	13,783		12,477	0.00
OR5x100-0.75_2.dat-50	75	2	13,354	13,354	14,941		13,354	0.00
OR5x100-0.75_3.dat-50	75	2	12,774	12,774	14,365		12,774	0.00
OR5x100-0.75_4.dat-50	75	2	13,086	13,086	15,331		13,086	0.00
OR5x100-0.75_5.dat-50	75	2	12,907	12,907	15,056		12,907	0.00
OR5x100-0.75_6.dat-50	75	2	12,060	12,060	13,522		12,060	0.00
OR5x100-0.75_7.dat-50	75	2	13,344	13,344	14,571		13,344	0.00
OR5x100-0.75_8.dat-50	75	2	13,364	13,364	15,170		13,364	0.00
OR5x100-0.75_9.dat-50	75	2	12,077	12,077	13,567		12,077	0.00
OR5x100-0.75_10.dat-50	75	2	12,824	12,824	14,457		12,824	0.00
OR5x100-0.75_1.dat-100	75	1	13,104	13,104	14,018	11,988	13,104	0.00
OR5x100-0.75_2.dat-100	75	1	14,603	14,603	16,122	13,373	14,603	0.00
OR5x100-0.75_3.dat-100	75	1	13,214	13,214	14,408	12,092	13,214	0.00
OR5x100-0.75_4.dat-100	75	1	13,555	13,555	15,053	12,414	13,555	0.00
OR5x100-0.75_5.dat-100	75	1	13,604	13,604	15,116	12,379	13,604	0.00
OR5x100-0.75_6.dat-100	75	1	12,929	12,929	14,960	11,795	12,929	0.00

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR5x100-0.75_7.dat-100	75	1	14,103	14,103	15,902	12,877	14,103	0.00
OR5x100-0.75_8.dat-100	75	1	14,004	14,004	15,169	12,821	14,004	0.00
OR5x100-0.75_9.dat-100	75	1	12,783	12,783	14,027	11,708	12,783	0.00
OR5x100-0.75_10.dat-100	75	1	13,322	13,322	14,702	12,136	13,322	0.00
OR5x250-0.25_1.dat-0	25	4	34,494	51,621	54,798		50,291	31.41
OR5x250-0.25_2.dat-0	25	4	36,022	53,453	56,258			31.34
OR5x250-0.25_3.dat-0	25	4	35,700	53,506	56,308			31.79
OR5x250-0.25_4.dat-0	25	4	35,239	52,573	55,907			32.97
OR5x250-0.25_5.dat-0	25	4	33,896	51,603	54,740			31.28
OR5x250-0.25_6.dat-0	25	4	34,604	51,849	55,007			33.26
OR5x250-0.25_7.dat-0	25	4	35,740	54,014	55,294		52,020	31.30
OR5x250-0.25_8.dat-0	25	4	36,632	55,028	57,526		53,918	32.06
OR5x250-0.25_9.dat-0	25	4	35,282	52,910	55,996			32.14
OR5x250-0.25_10.dat-0	25	4	35,280	52,031	55,196			30.58
OR5x250-0.25_1.dat-50	25	2	38,082	53,836	55,192		52,384	27.30
OR5x250-0.25_2.dat-50	25	2	39,285	55,748	57,851		54,668	28.14
OR5x250-0.25_3.dat-50	25	2	38,444	55,392	57,004		53,485	28.12
OR5x250-0.25_4.dat-50	25	2	38,716	56,262	57,428		54,753	29.29
OR5x250-0.25_5.dat-50	25	2	38,560	55,651	55,884		53,890	28.45
OR5x250-0.25_6.dat-50	25	2	40,088	56,250	58,016		54,875	26.95
OR5x250-0.25_7.dat-50	25	2	40,961	57,130	59,607		56,511	27.52
OR5x250-0.25_8.dat-50	25	2	39,968	58,725	60,600		57,002	29.88
OR5x250-0.25_9.dat-50	25	2	38,919	56,112	57,882		55,402	29.75
OR5x250-0.25_10.dat-50	25	2	37,815	53,686	55,473		52,645	28.17
OR5x250-0.25_1.dat-100	25	1	46,075	60,613	61,601	56,913	58,410	2.56
OR5x250-0.25_2.dat-100	25	1	45,538	59,501	60,499	56,239	57,402	2.03
OR5x250-0.25_3.dat-100	25	1	43,947	58,421	59,396	54,982	56,343	2.42
OR5x250-0.25_4.dat-100	25	1	44,916	59,212	59,212	55,270	56,933	2.92
OR5x250-0.25_5.dat-100	25	1	45,370	59,118	59,118	55,404	56,943	2.70
OR5x250-0.25_6.dat-100	25	1	46,901	62,072	63,066	57,987	59,284	2.19
OR5x250-0.25_7.dat-100	25	1	48,069	63,569	63,569	59,937	61,611	2.72
OR5x250-0.25_8.dat-100	25	1	45,860	60,675	60,675	57,505	58,769	2.15
OR5x250-0.25_9.dat-100	25	1	46,112	60,663	60,911	57,097	58,528	2.44
OR5x250-0.25_10.dat-100	25	1	47,795	61,775	61,847	57,576	59,133	2.63
OR5x250-0.50_1.dat-0	50	4	40,069	64,651	84,711		57,509	30.33
OR5x250-0.50_2.dat-0	50	4	38,908	62,186	85,349		54,525	28.64
OR5x250-0.50_3.dat-0	50	4	37,954	59,247	76,226		52,500	27.71
OR5x250-0.50_4.dat-0	50	4	40,005	65,156	84,713		59,307	32.55
OR5x250-0.50_5.dat-0	50	4	40,847	64,714	88,188		58,188	29.80
OR5x250-0.50_6.dat-0	50	4	40,393	63,384	84,965		55,904	27.75
OR5x250-0.50_7.dat-0	50	4	39,146	62,593	87,174		55,731	29.76
OR5x250-0.50_8.dat-0	50	4	39,927	62,174	88,969		56,029	28.74
OR5x250-0.50_9.dat-0	50	4	39,676	63,075	87,710		56,625	29.93
OR5x250-0.50_10.dat-0	50	4	38,309	60,764	90,119		54,001	29.06
OR5x250-0.50_1.dat-50	50	2	47,812	81,492	83,288		73,732	35.15
OR5x250-0.50_2.dat-50	50	2	51,097	79,739	81,981		70,644	27.67
OR5x250-0.50_3.dat-50	50	2	46,295	79,097	83,230		72,948	36.54
OR5x250-0.50_4.dat-50	50	2	48,764	80,365	82,044		72,591	32.82
OR5x250-0.50_5.dat-50	50	2	48,263	81,400	87,609		75,882	36.40
OR5x250-0.50_6.dat-50	50	2	48,037	80,543	83,800		74,496	35.52
OR5x250-0.50_7.dat-50	50	2	45,898	79,433	84,028		71,471	35.78
OR5x250-0.50_8.dat-50	50	2	47,169	81,476	85,274		73,949	36.21
OR5x250-0.50_9.dat-50	50	2	47,712	80,901	83,871		74,801	36.21
OR5x250-0.50_10.dat-50	50	2	46,116	77,653	83,708		71,580	35.57

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR5x250-0.50_1.dat-100	50	1	51,511	78,004	78,004	69,671	73,838	5.64
OR5x250-0.50_2.dat-100	50	1	51,277	74,726	74,792	66,618	70,903	6.04
OR5x250-0.50_3.dat-100	50	1	49,591	77,700	78,199	66,098	73,113	9.59
OR5x250-0.50_4.dat-100	50	1	52,652	78,077	78,738	67,268	73,498	8.48
OR5x250-0.50_5.dat-100	50	1	55,375	79,729	80,304	70,969	75,888	6.48
OR5x250-0.50_6.dat-100	50	1	52,942	78,894	78,894	68,667	75,355	8.88
OR5x250-0.50_7.dat-100	50	1	53,948	74,755	76,937	67,519	72,397	6.74
OR5x250-0.50_8.dat-100	50	1	51,525	78,290	78,290	64,771	73,554	11.94
OR5x250-0.50_9.dat-100	50	1	53,259	78,999	78,999	68,884	74,135	7.08
OR5x250-0.50_10.dat-100	50	1	52,454	74,272	75,944	65,988	72,064	8.43
OR5x250-0.75_1.dat-0	75	4	24,032	24,032	26,953		24,032	0.00
OR5x250-0.75_2.dat-0	75	4	21,799	21,799	25,425		21,799	0.00
OR5x250-0.75_3.dat-0	75	4	25,330	25,330	31,757		25,330	0.00
OR5x250-0.75_4.dat-0	75	4	23,637	23,637	28,816		23,637	0.00
OR5x250-0.75_5.dat-0	75	4	24,011	24,011	32,918		24,011	0.00
OR5x250-0.75_6.dat-0	75	4	19,766	19,766	22,525		19,766	0.00
OR5x250-0.75_7.dat-0	75	4	21,914	21,914	25,534		21,914	0.00
OR5x250-0.75_8.dat-0	75	4	23,874	23,874	30,417		23,874	0.00
OR5x250-0.75_9.dat-0	75	4	25,048	25,048	30,699		25,048	0.00
OR5x250-0.75_10.dat-0	75	4	23,965	23,965	30,455		23,965	0.00
OR5x250-0.75_1.dat-50	75	2	31,006	31,006	37,195		31,006	0.00
OR5x250-0.75_2.dat-50	75	2	31,726	31,726	36,581		31,726	0.00
OR5x250-0.75_3.dat-50	75	2	32,286	32,286	39,738		32,286	0.00
OR5x250-0.75_4.dat-50	75	2	31,989	31,989	38,905		31,989	0.00
OR5x250-0.75_5.dat-50	75	2	31,009	31,009	38,480		31,009	0.00
OR5x250-0.75_6.dat-50	75	2	29,841	29,841	35,169		29,841	0.00
OR5x250-0.75_7.dat-50	75	2	31,421	31,421	37,858		31,421	0.00
OR5x250-0.75_8.dat-50	75	2	30,763	30,763	36,473		30,763	0.00
OR5x250-0.75_9.dat-50	75	2	32,198	32,198	37,947		32,198	0.00
OR5x250-0.75_10.dat-50	75	2	32,569	32,569	36,519		32,569	0.00
OR5x250-0.75_1.dat-100	75	1	33,318	33,318	40,373	32,049	33,358	0.00
OR5x250-0.75_2.dat-100	75	1	33,912	33,912	38,176	32,705	33,912	0.00
OR5x250-0.75_3.dat-100	75	1	34,439	34,439	41,500	33,242	34,471	0.00
OR5x250-0.75_4.dat-100	75	1	34,306	34,306	39,445	32,967	34,339	0.00
OR5x250-0.75_5.dat-100	75	1	33,649	33,649	38,720	32,411	33,665	0.00
OR5x250-0.75_6.dat-100	75	1	31,670	31,670	36,294	30,383	31,670	0.00
OR5x250-0.75_7.dat-100	75	1	33,373	33,373	38,127	32,165	33,373	0.00
OR5x250-0.75_8.dat-100	75	1	32,417	32,417	37,313	31,260	32,417	0.00
OR5x250-0.75_9.dat-100	75	1	34,576	34,576	38,511	33,256	34,607	0.00
OR5x250-0.75_10.dat-100	75	1	33,891	33,891	39,129	32,687	33,928	0.00
OR5x500-0.25_1.dat-0	25	4	69,835	110,214	117,265			34.49
OR5x500-0.25_2.dat-0	25	4	67,484	107,643	111,973			36.71
OR5x500-0.25_3.dat-0	25	4	70,067	111,192	118,046			36.52
OR5x500-0.25_4.dat-0	25	4	69,097	110,303	120,882			35.98
OR5x500-0.25_5.dat-0	25	4	68,932	110,666	117,279			36.44
OR5x500-0.25_6.dat-0	25	4	67,351	108,756				36.14
OR5x500-0.25_7.dat-0	25	4	68,077	108,402				35.11
OR5x500-0.25_8.dat-0	25	4	67,539	108,202	112,423			34.93
OR5x500-0.25_9.dat-0	25	4	69,004	110,234	113,726			35.77
OR5x500-0.25_10.dat-0	25	4	68,019	108,009				35.73
OR5x500-0.25_1.dat-50	25	2	76,052	113,261	114,719		111,010	31.49
OR5x500-0.25_2.dat-50	25	2	75,218	111,681	114,348		112,174	32.65
OR5x500-0.25_3.dat-50	25	2	78,945	118,460	119,657		114,278	30.92
OR5x500-0.25_4.dat-50	25	2	79,508	116,473	116,473		111,381	28.62

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR5x500-0.25_5.dat-50	25	2	75,546	114,868	116,012		110,268	31.49
OR5x500-0.25_6.dat-50	25	2	77,490	114,252	116,134		111,698	30.63
OR5x500-0.25_7.dat-50	25	2	77,511	112,684	113,888		109,462	29.19
OR5x500-0.25_8.dat-50	25	2	76,353	113,134	113,597		111,252	31.37
OR5x500-0.25_9.dat-50	25	2	77,845	115,734	115,734		113,693	31.53
OR5x500-0.25_10.dat-50	25	2	76,071	113,094	114,455		109,429	30.48
OR5x500-0.25_1.dat-100	25	1	90,594	123,807	123,813	112,053	117,212	4.40
OR5x500-0.25_2.dat-100	25	1	90,110	120,379	120,379	110,468	115,579	4.42
OR5x500-0.25_3.dat-100	25	1	90,361	122,897	122,897	111,471	116,685	4.47
OR5x500-0.25_4.dat-100	25	1	93,316	126,374	126,374	113,621	119,915	5.25
OR5x500-0.25_5.dat-100	25	1	89,472	120,760	120,760	110,348	115,732	4.65
OR5x500-0.25_6.dat-100	25	1	90,508	117,619	117,619	108,350	113,962	4.92
OR5x500-0.25_7.dat-100	25	1	91,222	118,599	118,599	109,038	114,264	4.57
OR5x500-0.25_8.dat-100	25	1	90,045	117,971	117,971	106,979	112,554	4.95
OR5x500-0.25_9.dat-100	25	1	93,289	123,411	123,411	113,207	118,639	4.58
OR5x500-0.25_10.dat-100	25	1	91,103	120,024	120,024	109,797	115,683	5.09
OR5x500-0.50_1.dat-0	50	4	77,109	123,841	214,120		108,216	28.75
OR5x500-0.50_2.dat-0	50	4	74,534	124,608			109,574	31.98
OR5x500-0.50_3.dat-0	50	4	71,925	123,468	211,434		108,233	33.55
OR5x500-0.50_4.dat-0	50	4	76,000	129,661			113,945	33.30
OR5x500-0.50_5.dat-0	50	4	74,417	126,925	201,516		110,730	32.79
OR5x500-0.50_6.dat-0	50	4	76,099	125,246	197,735		111,703	31.87
OR5x500-0.50_7.dat-0	50	4	74,173	127,351	204,545		110,612	32.94
OR5x500-0.50_8.dat-0	50	4	74,840	127,420	218,814		111,627	32.96
OR5x500-0.50_9.dat-0	50	4	72,145	121,479			106,432	32.21
OR5x500-0.50_10.dat-0	50	4	76,215	129,471	196,699		115,998	34.30
OR5x500-0.50_1.dat-50	50	2	93,703	161,392	164,286		144,300	35.06
OR5x500-0.50_2.dat-50	50	2	94,588	163,727	163,727		147,077	35.69
OR5x500-0.50_3.dat-50	50	2	91,607	159,366	163,792		144,492	36.60
OR5x500-0.50_4.dat-50	50	2	94,797	167,239	168,310		150,448	36.99
OR5x500-0.50_5.dat-50	50	2	92,434	162,645	165,910		144,796	36.16
OR5x500-0.50_6.dat-50	50	2	91,819	160,755	167,066		147,365	37.69
OR5x500-0.50_7.dat-50	50	2	91,349	159,842	168,265		145,558	37.24
OR5x500-0.50_8.dat-50	50	2	93,238	161,161	161,684		146,795	36.48
OR5x500-0.50_9.dat-50	50	2	91,421	159,578	166,726		144,903	36.91
OR5x500-0.50_10.dat-50	50	2	92,710	164,190	169,033		147,553	37.17
OR5x500-0.50_1.dat-100	50	1	95,743	152,050	152,050	122,445	143,472	14.66
OR5x500-0.50_2.dat-100	50	1	96,224	152,674	152,674	123,075	145,663	15.51
OR5x500-0.50_3.dat-100	50	1	95,193	150,648	150,648	122,060	142,500	14.34
OR5x500-0.50_4.dat-100	50	1	98,696	158,065	158,065	123,483	148,736	16.98
OR5x500-0.50_5.dat-100	50	1	95,783	154,977	154,977	118,933	146,297	18.70
OR5x500-0.50_6.dat-100	50	1	96,634	154,727	154,727	121,111	146,071	17.09
OR5x500-0.50_7.dat-100	50	1	95,596	156,375	156,375	119,143	145,735	18.25
OR5x500-0.50_8.dat-100	50	1	96,324	151,145	151,145	121,654	144,375	15.74
OR5x500-0.50_9.dat-100	50	1	94,932	153,060	153,060	115,963	144,891	19.97
OR5x500-0.50_10.dat-100	50	1	95,786	152,964	155,612	116,185	147,142	21.04
OR5x500-0.75_1.dat-0	75	4	33,280	70,213	275,421		42,843	22.32
OR5x500-0.75_2.dat-0	75	4	48,611	66,550	83,975		48,617	0.01
OR5x500-0.75_3.dat-0	75	4	35,724	71,632	97,517		47,256	24.40
OR5x500-0.75_4.dat-0	75	4	35,106	71,978	100,542		48,597	27.76
OR5x500-0.75_5.dat-0	75	4	35,165	69,985	96,801		48,644	27.71
OR5x500-0.75_6.dat-0	75	4	36,713	65,198	92,928		46,868	21.67
OR5x500-0.75_7.dat-0	75	4	45,651	66,476	94,972		45,668	0.04
OR5x500-0.75_8.dat-0	75	4	34,991	66,585	82,364		46,099	24.10

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR5x500-0.75_9.dat-0	75	4	37,192	46,817	100,248		46,817	20.56
OR5x500-0.75_10.dat-0	75	4	45,895	65,644	83,529		45,921	0.06
OR5x500-0.75_1.dat-50	75	2	44,989	61,770	97,526		61,697	27.08
OR5x500-0.75_2.dat-50	75	2	59,252	62,479	100,543		62,391	5.03
OR5x500-0.75_3.dat-50	75	2	59,953	61,213	91,905		61,034	1.77
OR5x500-0.75_4.dat-50	75	2	46,017	63,661	90,979		63,550	27.59
OR5x500-0.75_5.dat-50	75	2	46,155	62,096	96,309		62,048	25.61
OR5x500-0.75_6.dat-50	75	2	46,315	63,547	91,224		63,437	26.99
OR5x500-0.75_7.dat-50	75	2	61,642	61,745	99,090		61,693	0.08
OR5x500-0.75_8.dat-50	75	2	63,201	63,342	108,200		63,248	0.07
OR5x500-0.75_9.dat-50	75	2	45,302	62,819	101,695		62,717	27.77
OR5x500-0.75_10.dat-50	75	2	44,977	61,063	102,403		60,990	26.26
OR5x500-0.75_1.dat-100	75	1	53,711	64,823	91,492	63,500	64,837	2.04
OR5x500-0.75_2.dat-100	75	1	64,903	68,349	96,443	66,954	68,329	2.01
OR5x500-0.75_3.dat-100	75	1	62,448	65,458	90,500	64,135	65,432	1.98
OR5x500-0.75_4.dat-100	75	1	67,838	68,073	95,257	66,774	68,060	0.33
OR5x500-0.75_5.dat-100	75	1	62,361	65,516	93,947	64,199	65,544	2.01
OR5x500-0.75_6.dat-100	75	1	63,429	66,482	93,609	65,136	66,459	1.99
OR5x500-0.75_7.dat-100	75	1	64,910	65,076	97,773	63,795	65,065	0.24
OR5x500-0.75_8.dat-100	75	1	68,064	68,254	96,401	66,884	68,282	0.28
OR5x500-0.75_9.dat-100	75	1	63,244	66,562	91,808	65,292	66,541	1.88
OR5x500-0.75_10.dat-100	75	1	62,937	66,051	92,698	64,720	66,070	2.02
OR10x100-0.25_1.dat-0	25	9	14,537	19,029	21,555		18,761	22.51
OR10x100-0.25_2.dat-0	25	9	14,481	18,822	20,668			23.06
OR10x100-0.25_3.dat-0	25	9	14,432	18,681	20,698		18,323	21.24
OR10x100-0.25_4.dat-0	25	9	14,720	19,808	21,661		19,703	25.29
OR10x100-0.25_5.dat-0	25	9	14,857	19,281	21,437			22.94
OR10x100-0.25_6.dat-0	25	9	14,875	19,330	21,616		19,372	23.05
OR10x100-0.25_7.dat-0	25	9	13,644	18,104	20,751			24.64
OR10x100-0.25_8.dat-0	25	9	14,546	18,738	21,025		18,854	22.37
OR10x100-0.25_9.dat-0	25	9	13,989	18,348	19,917		18,154	22.94
OR10x100-0.25_10.dat-0	25	9	14,119	18,673	20,543		18,565	23.95
OR10x100-0.25_1.dat-50	25	5	15,584	20,784	23,483		20,673	24.62
OR10x100-0.25_2.dat-50	25	5	15,054	20,134	21,586		20,000	24.73
OR10x100-0.25_3.dat-50	25	5	14,995	20,097	21,085		19,736	24.02
OR10x100-0.25_4.dat-50	25	5	15,804	20,754	22,771			23.85
OR10x100-0.25_5.dat-50	25	5	15,826	20,765	22,055		20,480	22.72
OR10x100-0.25_6.dat-50	25	5	16,183	21,415	22,920		21,121	23.38
OR10x100-0.25_7.dat-50	25	5	14,361	19,438	20,736		19,009	24.45
OR10x100-0.25_8.dat-50	25	5	14,996	20,322	21,458		19,872	24.54
OR10x100-0.25_9.dat-50	25	5	14,717	19,527	21,027		19,106	22.97
OR10x100-0.25_10.dat-50	25	5	14,825	20,264	22,328		19,924	25.59
OR10x100-0.25_1.dat-100	25	1	21,158	25,534	26,730	24,632	24,994	1.45
OR10x100-0.25_2.dat-100	25	1	21,782	25,551	26,754	24,132	24,767	2.56
OR10x100-0.25_3.dat-100	25	1	20,762	25,175	27,090	23,863	24,515	2.66
OR10x100-0.25_4.dat-100	25	1	20,157	23,669	24,052	23,011	23,239	0.98
OR10x100-0.25_5.dat-100	25	1	20,096	23,916	24,424	23,016	23,417	1.71
OR10x100-0.25_6.dat-100	25	1	21,247	26,003	27,184	25,036	25,612	2.25
OR10x100-0.25_7.dat-100	25	1	20,146	24,079	25,884	23,051	23,787	3.09
OR10x100-0.25_8.dat-100	25	1	21,219	25,907	26,857	24,635	25,403	3.02
OR10x100-0.25_9.dat-100	25	1	19,077	22,758	23,084	22,091	22,525	1.93
OR10x100-0.25_10.dat-100	25	1	20,546	25,246	26,228	24,062	24,599	2.18
OR10x100-0.50_1.dat-0	50	9	15,216	22,209	26,516		20,476	25.69
OR10x100-0.50_2.dat-0	50	9	15,426	21,266	25,742		20,545	24.92

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR10x100-0.50_3.dat-0	50	9	19,112	22,787	34,196		20,916	8.62
OR10x100-0.50_4.dat-0	50	9	17,568	26,422	32,127		24,455	28.16
OR10x100-0.50_5.dat-0	50	9	15,358	22,399	29,498		20,730	25.91
OR10x100-0.50_6.dat-0	50	9	15,440	23,188	31,185		21,633	28.63
OR10x100-0.50_7.dat-0	50	9	18,533	24,346	30,828		22,981	19.36
OR10x100-0.50_8.dat-0	50	9	18,216	23,791	33,462		21,906	16.84
OR10x100-0.50_9.dat-0	50	9	16,056	23,364	28,760		21,609	25.70
OR10x100-0.50_10.dat-0	50	9	15,718	21,319	25,989		19,557	19.63
OR10x100-0.50_1.dat-50	50	5	18,370	28,595	31,221		25,614	28.28
OR10x100-0.50_2.dat-50	50	5	20,276	29,586	33,445		27,913	27.36
OR10x100-0.50_3.dat-50	50	5	19,767	31,145	33,428		28,340	30.25
OR10x100-0.50_4.dat-50	50	5	21,026	32,837	35,267		29,808	29.46
OR10x100-0.50_5.dat-50	50	5	18,568	29,650	32,485		27,197	31.73
OR10x100-0.50_6.dat-50	50	5	19,411	30,498	33,486		27,518	29.46
OR10x100-0.50_7.dat-50	50	5	19,257	29,791	33,101		27,821	30.78
OR10x100-0.50_8.dat-50	50	5	18,556	30,003	32,252		26,704	30.51
OR10x100-0.50_9.dat-50	50	5	18,731	30,808	31,770		27,713	32.41
OR10x100-0.50_10.dat-50	50	5	18,813	29,318	30,758		26,340	28.58
OR10x100-0.50_1.dat-100	50	1	19,018	29,329	29,722	26,565	27,500	3.40
OR10x100-0.50_2.dat-100	50	1	20,461	29,499	30,910	27,630	28,430	2.81
OR10x100-0.50_3.dat-100	50	1	21,243	30,415	32,149	28,023	29,356	4.54
OR10x100-0.50_4.dat-100	50	1	23,916	30,985	31,201	28,913	29,876	3.22
OR10x100-0.50_5.dat-100	50	1	20,743	28,407	29,681	26,550	27,632	3.92
OR10x100-0.50_6.dat-100	50	1	20,724	29,761	30,654	27,684	28,708	3.57
OR10x100-0.50_7.dat-100	50	1	21,211	30,595	30,595	28,508	29,565	3.58
OR10x100-0.50_8.dat-100	50	1	20,788	30,178	30,901	27,463	28,648	4.14
OR10x100-0.50_9.dat-100	50	1	19,814	30,015	30,340	27,070	28,147	3.83
OR10x100-0.50_10.dat-100	50	1	18,980	27,905	27,905	25,774	26,650	3.29
OR10x100-0.75_1.dat-0	75	9	9,389	10,212	10,334		9,426	0.39
OR10x100-0.75_2.dat-0	75	9	9,057	9,057	9,436		9,057	0.00
OR10x100-0.75_3.dat-0	75	9	8,747	9,310	9,462		8,749	0.02
OR10x100-0.75_4.dat-0[†]	75	9	8,814	8,813	9,447		8,813	0.00
OR10x100-0.75_5.dat-0	75	9	8,616	9,027	9,850		8,685	0.79
OR10x100-0.75_6.dat-0	75	9	9,638	9,638	10,426		9,638	0.00
OR10x100-0.75_7.dat-0	75	9	8,374	8,374	9,661		8,374	0.00
OR10x100-0.75_8.dat-0	75	9	8,745	8,745	9,631		8,745	0.00
OR10x100-0.75_9.dat-0	75	9	8,364	8,364	8,853		8,364	0.00
OR10x100-0.75_10.dat-0	75	9	7,632	8,183	8,950		7,671	0.51
OR10x100-0.75_1.dat-50	75	5	12,090	13,507	14,330		12,339	2.02
OR10x100-0.75_2.dat-50	75	5	12,747	13,752	14,781		12,970	1.72
OR10x100-0.75_3.dat-50	75	5	12,421	13,957	13,957		12,597	1.40
OR10x100-0.75_4.dat-50	75	5	13,358	14,742	15,110		13,571	1.57
OR10x100-0.75_5.dat-50	75	5	12,750	14,444	14,866		12,954	1.57
OR10x100-0.75_6.dat-50	75	5	14,202	15,594	16,039		14,401	1.38
OR10x100-0.75_7.dat-50	75	5	12,214	13,308	14,590		12,375	1.30
OR10x100-0.75_8.dat-50	75	5	12,851	14,951	14,993		12,996	1.12
OR10x100-0.75_9.dat-50	75	5	12,896	14,548	15,142		13,085	1.44
OR10x100-0.75_10.dat-50	75	5	11,983	13,232	13,232		12,240	2.10
OR10x100-0.75_1.dat-100	75	1	13,599	13,599	14,962	12,582	13,599	0.00
OR10x100-0.75_2.dat-100	75	1	14,155	14,155	15,768	12,977	14,317	0.00
OR10x100-0.75_3.dat-100	75	1	14,216	14,216	15,618	13,062	14,216	0.00
OR10x100-0.75_4.dat-100	75	1	15,290	15,290	16,627	13,954	15,332	0.00
OR10x100-0.75_5.dat-100	75	1	14,439	14,439	15,413	13,359	14,439	0.00
OR10x100-0.75_6.dat-100	75	1	15,347	15,347	16,867	14,256	15,347	0.00

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR10x100-0.75_7.dat-100	75	1	13,934	13,934	14,996	12,749	13,984	0.00
OR10x100-0.75_8.dat-100	75	1	14,137	14,137	15,508	13,006	14,137	0.00
OR10x100-0.75_9.dat-100	75	1	14,185	14,185	15,078	13,088	14,252	0.00
OR10x100-0.75_10.dat-100	75	1	13,814	13,814	15,169	12,728	13,814	0.00
OR10x250-0.25_1.dat-0	25	9	29,491	52,379				40.38
OR10x250-0.25_2.dat-0	25	9	28,790	51,927				41.88
OR10x250-0.25_3.dat-0	25	9	31,377	51,223				34.80
OR10x250-0.25_4.dat-0	25	9	33,546	53,025				35.61
OR10x250-0.25_5.dat-0	25	9						n/a
OR10x250-0.25_6.dat-0	25	9	32,947	52,205				34.00
OR10x250-0.25_7.dat-0	25	9	32,666	53,579				37.05
OR10x250-0.25_8.dat-0	25	9	31,028	51,707				38.65
OR10x250-0.25_9.dat-0	25	9	31,302	51,439				36.94
OR10x250-0.25_10.dat-0	25	9	33,079	52,856				35.43
OR10x250-0.25_1.dat-50	25	5	33,844	53,502	54,589			33.89
OR10x250-0.25_2.dat-50	25	5	34,108	53,382	56,458			35.76
OR10x250-0.25_3.dat-50	25	5	33,876	52,024	54,475			33.43
OR10x250-0.25_4.dat-50	25	5	35,545	54,623	57,372			33.41
OR10x250-0.25_5.dat-50	25	5	34,172	53,530	56,657			33.51
OR10x250-0.25_6.dat-50	25	5	35,036	53,835	57,033			32.52
OR10x250-0.25_7.dat-50	25	5	35,749	55,317	58,507			32.44
OR10x250-0.25_8.dat-50	25	5	34,152	53,363	55,998			33.62
OR10x250-0.25_9.dat-50	25	5	33,469	53,427	56,737			35.30
OR10x250-0.25_10.dat-50	25	5	34,721	54,167	58,614			33.22
OR10x250-0.25_1.dat-100	25	1	45,733	60,741	60,741	57,006	58,509	2.57
OR10x250-0.25_2.dat-100	25	1	45,395	61,073	62,057	56,593	58,389	3.08
OR10x250-0.25_3.dat-100	25	1	47,383	62,082	63,543	57,819	60,158	3.89
OR10x250-0.25_4.dat-100	25	1	45,811	61,629	61,629	57,008	58,720	2.92
OR10x250-0.25_5.dat-100	25	1	47,078	63,531	64,835	58,258	60,139	3.13
OR10x250-0.25_6.dat-100	25	1	46,276	61,727	62,410	56,557	58,847	3.89
OR10x250-0.25_7.dat-100	25	1	47,040	61,136	61,691	57,270	58,419	1.97
OR10x250-0.25_8.dat-100	25	1	45,924	60,648	61,592	56,036	57,691	2.87
OR10x250-0.25_9.dat-100	25	1	47,463	61,503	61,503	57,741	59,446	2.87
OR10x250-0.25_10.dat-100	25	1	46,128	62,297	63,736	57,528	59,535	3.37
OR10x250-0.50_1.dat-0	50	9					57,818	n/a
OR10x250-0.50_2.dat-0	50	9	38,126	62,818			54,748	30.36
OR10x250-0.50_3.dat-0	50	9	37,856	61,853			54,326	30.32
OR10x250-0.50_4.dat-0	50	9	37,582	59,368			52,717	28.71
OR10x250-0.50_5.dat-0	50	9	38,115	62,987			58,115	34.41
OR10x250-0.50_6.dat-0	50	9	39,300	62,837			56,345	30.25
OR10x250-0.50_7.dat-0	50	9	37,201	61,372			53,302	30.21
OR10x250-0.50_8.dat-0	50	9	36,697	60,043			52,577	30.20
OR10x250-0.50_9.dat-0	50	9	36,004	57,088			49,519	27.29
OR10x250-0.50_10.dat-0	50	9	35,602	57,134			49,964	28.74
OR10x250-0.50_1.dat-50	50	5	46,783	83,614	90,802		73,119	36.02
OR10x250-0.50_2.dat-50	50	5	45,093	79,698	87,145		71,142	36.62
OR10x250-0.50_3.dat-50	50	5	45,192	80,303	87,484		71,263	36.58
OR10x250-0.50_4.dat-50	50	5	46,428	83,324	86,634		72,536	35.99
OR10x250-0.50_5.dat-50	50	5	45,521	81,995	85,921		69,837	34.82
OR10x250-0.50_6.dat-50	50	5	46,130	78,957	88,255		71,000	35.03
OR10x250-0.50_7.dat-50	50	5	44,281	78,565	84,387		68,581	35.43
OR10x250-0.50_8.dat-50	50	5	44,289	79,996	86,485		68,859	35.68
OR10x250-0.50_9.dat-50	50	5	45,700	81,071	90,949		70,068	34.78
OR10x250-0.50_10.dat-50	50	5	43,973	77,887	81,253		67,139	34.50

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR10x250-0.50_1.dat-100	50	1	54,353	80,190	80,190	70,254	75,854	7.38
OR10x250-0.50_2.dat-100	50	1	50,538	74,852	74,852	66,529	70,788	6.02
OR10x250-0.50_3.dat-100	50	1	47,200	75,120	75,120	66,152	70,219	5.79
OR10x250-0.50_4.dat-100	50	1	53,571	76,382	76,382	67,648	72,297	6.43
OR10x250-0.50_5.dat-100	50	1	48,841	76,790	76,790	68,015	72,938	6.75
OR10x250-0.50_6.dat-100	50	1	47,428	76,052	76,052	67,528	71,923	6.11
OR10x250-0.50_7.dat-100	50	1	48,261	73,743	73,743	65,056	69,311	6.14
OR10x250-0.50_8.dat-100	50	1	49,499	73,301	73,403	64,555	68,763	6.12
OR10x250-0.50_9.dat-100	50	1	52,821	79,259	79,836	69,811	74,526	6.33
OR10x250-0.50_10.dat-100	50	1	46,410	71,374	71,374	63,747	67,988	6.24
OR10x250-0.75_1.dat-0	75	9	22,839	25,765			22,888	0.21
OR10x250-0.75_2.dat-0	75	9	21,009	23,182			21,041	0.15
OR10x250-0.75_3.dat-0	75	9	22,398	25,879			22,460	0.28
OR10x250-0.75_4.dat-0	75	9	23,056	25,999			23,083	0.12
OR10x250-0.75_5.dat-0	75	9	23,022	26,190			23,031	0.04
OR10x250-0.75_6.dat-0	75	9	23,154	25,506			23,173	0.08
OR10x250-0.75_7.dat-0	75	9	23,248	25,986			23,276	0.12
OR10x250-0.75_8.dat-0	75	9	22,123	25,242			22,156	0.15
OR10x250-0.75_9.dat-0	75	9	22,775	25,768	143,153		22,794	0.08
OR10x250-0.75_10.dat-0	75	9	22,613	25,872			22,631	0.08
OR10x250-0.75_1.dat-50	75	5	32,053	36,970	38,491		32,193	0.43
OR10x250-0.75_2.dat-50	75	5	31,550	37,354	38,526		31,710	0.50
OR10x250-0.75_3.dat-50	75	5	32,078	36,456	38,500		32,237	0.49
OR10x250-0.75_4.dat-50	75	5	32,823	38,496	40,143		32,914	0.28
OR10x250-0.75_5.dat-50	75	5	32,161	34,949	39,199		32,305	0.45
OR10x250-0.75_6.dat-50	75	5	32,547	36,905	38,195		32,655	0.33
OR10x250-0.75_7.dat-50	75	5	31,906	37,389	37,389		32,012	0.33
OR10x250-0.75_8.dat-50	75	5	32,573	38,048	39,745		32,696	0.38
OR10x250-0.75_9.dat-50	75	5	32,563	37,021	40,293		32,743	0.55
OR10x250-0.75_10.dat-50	75	5	32,629	37,946	39,600		32,753	0.38
OR10x250-0.75_1.dat-100	75	1	34,846	34,846	38,358	33,580	34,846	0.00
OR10x250-0.75_2.dat-100	75	1	34,304	34,351	40,332	33,059	34,419	0.14
OR10x250-0.75_3.dat-100	75	1	35,647	35,699	39,734	34,368	35,711	0.15
OR10x250-0.75_4.dat-100	75	1	35,302	35,355	41,019	34,020	35,501	0.15
OR10x250-0.75_5.dat-100	75	1	35,887	35,920	40,366	34,676	36,042	0.09
OR10x250-0.75_6.dat-100	75	1	35,101	35,101	39,878	33,748	35,188	0.00
OR10x250-0.75_7.dat-100	75	1	35,021	35,021	40,501	33,732	35,071	0.00
OR10x250-0.75_8.dat-100	75	1	36,089	36,139	41,139	34,889	36,209	0.14
OR10x250-0.75_9.dat-100	75	1	34,584	34,625	38,456	33,281	34,714	0.12
OR10x250-0.75_10.dat-100	75	1	34,852	34,889	40,957	33,611	34,935	0.11
OR10x500-0.25_1.dat-0	25	9						n/a
OR10x500-0.25_2.dat-0	25	9						n/a
OR10x500-0.25_3.dat-0	25	9						n/a
OR10x500-0.25_4.dat-0	25	9						n/a
OR10x500-0.25_5.dat-0	25	9						n/a
OR10x500-0.25_6.dat-0	25	9						n/a
OR10x500-0.25_7.dat-0	25	9						n/a
OR10x500-0.25_8.dat-0	25	9						n/a
OR10x500-0.25_9.dat-0	25	9						n/a
OR10x500-0.25_10.dat-0	25	9						n/a
OR10x500-0.25_1.dat-50	25	5	67,440	113,396				36.39
OR10x500-0.25_2.dat-50	25	5	67,265	112,795				38.06
OR10x500-0.25_3.dat-50	25	5	67,129	113,842				38.00
OR10x500-0.25_4.dat-50	25	5	67,267	113,364				36.00

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR10x500-0.25_5.dat-50	25	5	67,175	112,086				36.89
OR10x500-0.25_6.dat-50	25	5	66,729	112,484				37.63
OR10x500-0.25_7.dat-50	25	5	66,534	113,266				38.07
OR10x500-0.25_8.dat-50	25	5	67,532	113,484				37.37
OR10x500-0.25_9.dat-50	25	5	66,584	112,885				37.55
OR10x500-0.25_10.dat-50	25	5	66,920	113,424				38.21
OR10x500-0.25_1.dat-100	25	1	94,743	127,585	127,585	115,162	121,246	5.02
OR10x500-0.25_2.dat-100	25	1	95,933	127,453	127,453	115,492	122,627	5.82
OR10x500-0.25_3.dat-100	25	1	94,994	125,471	125,471	113,509	119,730	5.20
OR10x500-0.25_4.dat-100	25	1	89,911	123,364	123,364	111,864	117,808	5.05
OR10x500-0.25_5.dat-100	25	1	91,545	120,705	120,705	108,352	114,902	5.70
OR10x500-0.25_6.dat-100	25	1	87,880	117,664	117,664	107,695	112,855	4.57
OR10x500-0.25_7.dat-100	25	1	89,663	117,580	117,580	108,463	115,125	5.79
OR10x500-0.25_8.dat-100	25	1	88,272	118,222	118,222	108,534	113,884	4.70
OR10x500-0.25_9.dat-100	25	1	94,528	123,911	123,911	112,103	118,877	5.70
OR10x500-0.25_10.dat-100	25	1	91,633	125,687	125,687	113,444	119,347	4.95
OR10x500-0.50_1.dat-0	50	9					110,228	n/a
OR10x500-0.50_2.dat-0	50	9					108,994	n/a
OR10x500-0.50_3.dat-0	50	9					109,258	n/a
OR10x500-0.50_4.dat-0	50	9					109,277	n/a
OR10x500-0.50_5.dat-0	50	9					109,764	n/a
OR10x500-0.50_6.dat-0	50	9					109,294	n/a
OR10x500-0.50_7.dat-0	50	9					108,537	n/a
OR10x500-0.50_8.dat-0	50	9					111,621	n/a
OR10x500-0.50_9.dat-0	50	9					107,753	n/a
OR10x500-0.50_10.dat-0	50	9					111,864	n/a
OR10x500-0.50_1.dat-50	50	5	90,553	168,280			143,800	37.03
OR10x500-0.50_2.dat-50	50	5	89,644	168,134			144,652	38.03
OR10x500-0.50_3.dat-50	50	5	88,559	167,065			142,099	37.68
OR10x500-0.50_4.dat-50	50	5	89,058	165,223			138,428	35.66
OR10x500-0.50_5.dat-50	50	5	88,025	165,182			141,199	37.66
OR10x500-0.50_6.dat-50	50	5	88,666	163,977			140,895	37.07
OR10x500-0.50_7.dat-50	50	5	89,541	165,363			142,685	37.25
OR10x500-0.50_8.dat-50	50	5	90,548	171,282			139,449	35.07
OR10x500-0.50_9.dat-50	50	5	88,168	163,683			136,827	35.56
OR10x500-0.50_10.dat-50	50	5	91,352	171,052			144,345	36.71
OR10x500-0.50_1.dat-100	50	1	94,989	151,252	151,252	126,056	143,544	12.18
OR10x500-0.50_2.dat-100	50	1	95,347	152,805	152,805	126,165	144,228	12.52
OR10x500-0.50_3.dat-100	50	1	93,960	150,071	150,081	124,877	141,697	11.87
OR10x500-0.50_4.dat-100	50	1	92,771	144,641	144,641	124,705	139,035	10.31
OR10x500-0.50_5.dat-100	50	1	93,142	149,765	149,765	124,880	140,684	11.23
OR10x500-0.50_6.dat-100	50	1	92,805	147,967	148,618	123,818	139,239	11.08
OR10x500-0.50_7.dat-100	50	1	93,245	149,764	149,764	124,042	141,572	12.38
OR10x500-0.50_8.dat-100	50	1	94,131	147,014	147,020	124,989	140,147	10.82
OR10x500-0.50_9.dat-100	50	1	91,805	149,977	149,977	123,180	140,613	12.40
OR10x500-0.50_10.dat-100	50	1	95,673	155,420	155,420	128,690	146,691	12.27
OR10x500-0.75_1.dat-0	75	9	33,983	52,026			45,235	24.87
OR10x500-0.75_2.dat-0	75	9					45,759	n/a
OR10x500-0.75_3.dat-0	75	9					46,872	n/a
OR10x500-0.75_4.dat-0	75	9					43,498	n/a
OR10x500-0.75_5.dat-0	75	9					46,681	n/a
OR10x500-0.75_6.dat-0	75	9					47,558	n/a
OR10x500-0.75_7.dat-0	75	9					47,506	n/a
OR10x500-0.75_8.dat-0	75	9					42,652	n/a

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR10x500-0.75_9.dat-0	75	9					44,192	n/a
OR10x500-0.75_10.dat-0	75	9					45,448	n/a
OR10x500-0.75_1.dat-50	75	5	63,789	74,925			63,888	0.15
OR10x500-0.75_2.dat-50	75	5	63,960	74,542			64,075	0.18
OR10x500-0.75_3.dat-50	75	5	46,714	77,304	290,174		65,792	29.00
OR10x500-0.75_4.dat-50	75	5	45,938	74,701			63,697	27.88
OR10x500-0.75_5.dat-50	75	5	45,616	75,575			65,907	30.79
OR10x500-0.75_6.dat-50	75	5	64,635	74,937			64,738	0.16
OR10x500-0.75_7.dat-50	75	5	63,946	74,679	101,020		64,068	0.19
OR10x500-0.75_8.dat-50	75	5	63,071	74,193	299,817		63,159	0.14
OR10x500-0.75_9.dat-50	75	5	45,646	74,809			63,140	27.71
OR10x500-0.75_10.dat-50	75	5	46,380	77,141	96,962		65,197	28.86
OR10x500-0.75_1.dat-100	75	1	69,335	69,713	96,504	68,281	69,743	0.54
OR10x500-0.75_2.dat-100	75	1	68,643	69,027	98,003	67,630	69,069	0.56
OR10x500-0.75_3.dat-100	75	1	54,281	70,802	94,624	69,499	70,802	1.84
OR10x500-0.75_4.dat-100	75	1	68,445	68,843	95,396	67,377	68,881	0.58
OR10x500-0.75_5.dat-100	75	1	69,744	70,096	94,674	68,551	69,998	0.36
OR10x500-0.75_6.dat-100	75	1	70,268	70,641	95,464	69,250	70,720	0.53
OR10x500-0.75_7.dat-100	75	1	69,661	70,123	97,355	68,679	70,134	0.66
OR10x500-0.75_8.dat-100	75	1	68,849	68,975	93,518	67,671	69,081	0.18
OR10x500-0.75_9.dat-100	75	1	67,835	68,188	95,135	66,827	68,228	0.52
OR10x500-0.75_10.dat-100	75	1	69,841	70,117	98,288	68,733	70,190	0.39
OR30x100-0.25_1.dat-0	25	29	12,896	18,176	20,967			28.58
OR30x100-0.25_2.dat-0	25	29	11,887	19,013	20,883			34.52
OR30x100-0.25_3.dat-0	25	29	12,408	17,619	20,050			29.58
OR30x100-0.25_4.dat-0	25	29		19,822				n/a
OR30x100-0.25_5.dat-0	25	29		20,626				n/a
OR30x100-0.25_6.dat-0	25	29	13,153	18,387				28.47
OR30x100-0.25_7.dat-0	25	29	12,125	18,688				32.94
OR30x100-0.25_8.dat-0	25	29		19,967				n/a
OR30x100-0.25_9.dat-0	25	29		21,618				n/a
OR30x100-0.25_10.dat-0	25	29	974	18,151	19,939			94.58
OR30x100-0.25_1.dat-50	25	15	14,358	19,539	21,678			26.52
OR30x100-0.25_2.dat-50	25	15	13,955	19,468	20,351			28.32
OR30x100-0.25_3.dat-50	25	15	13,826	18,956	20,892			27.06
OR30x100-0.25_4.dat-50	25	15	13,997	19,452	20,591			27.69
OR30x100-0.25_5.dat-50	25	15	14,772	20,202	22,158			26.88
OR30x100-0.25_6.dat-50	25	15	14,488	19,788	21,075			26.78
OR30x100-0.25_7.dat-50	25	15	14,106	19,498	21,055		19,443	27.45
OR30x100-0.25_8.dat-50	25	15	14,328	19,699	21,345			27.27
OR30x100-0.25_9.dat-50	25	15	15,173	20,935	21,651			27.52
OR30x100-0.25_10.dat-50	25	15	13,710	18,470	20,780			25.77
OR30x100-0.25_1.dat-100	25	1	19,857	24,023	24,689	22,442	23,431	4.22
OR30x100-0.25_2.dat-100	25	1	21,423	25,500	26,280	23,845	24,863	4.09
OR30x100-0.25_3.dat-100	25	1	18,927	23,033	23,988	21,604	22,607	4.44
OR30x100-0.25_4.dat-100	25	1	19,203	23,311	24,900	22,150	23,073	4.00
OR30x100-0.25_5.dat-100	25	1	20,638	25,265	27,414	24,017	24,955	3.76
OR30x100-0.25_6.dat-100	25	1	20,272	24,540	25,644	23,099	24,116	4.22
OR30x100-0.25_7.dat-100	25	1	22,418	27,396	28,746	25,916	26,993	3.99
OR30x100-0.25_8.dat-100	25	1	19,527	23,856	25,096	22,336	23,319	4.22
OR30x100-0.25_9.dat-100	25	1	23,302	27,345	28,174	25,753	26,760	3.76
OR30x100-0.25_10.dat-100	25	1	19,848	24,084	24,908	22,825	23,762	3.94
OR30x100-0.50_1.dat-0	50	29	14,360	22,276	28,821		20,246	29.07
OR30x100-0.50_2.dat-0	50	29	15,060	23,891	37,927		21,279	29.23

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR30x100-0.50_3.dat-0	50	29	15,798	25,271			23,024	31.38
OR30x100-0.50_4.dat-0	50	29	15,637	24,998	29,193		22,792	31.39
OR30x100-0.50_5.dat-0	50	29	14,858	22,171			20,805	28.58
OR30x100-0.50_6.dat-0	50	29	15,306	23,182	35,501		21,594	29.12
OR30x100-0.50_7.dat-0	50	29	15,351	24,028	31,149		21,739	29.38
OR30x100-0.50_8.dat-0	50	29	16,082	24,989	39,030		23,398	31.27
OR30x100-0.50_9.dat-0	50	29	15,390	24,269	31,571		22,260	30.86
OR30x100-0.50_10.dat-0	50	29	14,678	22,218	29,171		20,776	29.35
OR30x100-0.50_1.dat-50	50	15	18,108	30,596	32,058		27,226	33.49
OR30x100-0.50_2.dat-50	50	15	18,572	31,150	34,004		27,576	32.65
OR30x100-0.50_3.dat-50	50	15	18,267	30,402	32,721		27,358	33.23
OR30x100-0.50_4.dat-50	50	15	18,029	30,630	31,866		26,504	31.98
OR30x100-0.50_5.dat-50	50	15	18,421	31,891	33,177		27,982	34.17
OR30x100-0.50_6.dat-50	50	15	17,915	29,987	30,968		27,098	33.89
OR30x100-0.50_7.dat-50	50	15	18,097	30,614	32,053		27,330	33.78
OR30x100-0.50_8.dat-50	50	15	18,584	31,674	32,923		28,026	33.69
OR30x100-0.50_9.dat-50	50	15	18,432	31,308	32,481		27,966	34.09
OR30x100-0.50_10.dat-50	50	15	18,020	29,750	30,905		27,420	34.28
OR30x100-0.50_1.dat-100	50	1	19,936	29,647	29,647	25,597	27,647	7.41
OR30x100-0.50_2.dat-100	50	1	20,863	29,888	31,795	27,075	28,969	6.54
OR30x100-0.50_3.dat-100	50	1	20,153	29,784	30,364	26,400	28,282	6.65
OR30x100-0.50_4.dat-100	50	1	19,939	28,972	28,972	25,836	27,529	6.15
OR30x100-0.50_5.dat-100	50	1	21,653	30,673	30,673	27,834	29,571	5.87
OR30x100-0.50_6.dat-100	50	1	21,042	30,180	31,040	27,033	28,618	5.54
OR30x100-0.50_7.dat-100	50	1	20,283	30,451	30,452	26,681	28,877	7.60
OR30x100-0.50_8.dat-100	50	1	20,947	29,461	30,154	27,156	28,624	5.13
OR30x100-0.50_9.dat-100	50	1	21,885	29,984	30,731	27,090	28,534	5.06
OR30x100-0.50_10.dat-100	50	1	20,037	30,229	30,229	27,100	28,530	5.01
OR30x100-0.75_1.dat-0	75	29	9,652	10,302	10,772		9,657	0.05
OR30x100-0.75_2.dat-0	75	29	9,301	9,756	10,657		9,397	1.02
OR30x100-0.75_3.dat-0	75	29	9,041	9,545	10,543		9,157	1.27
OR30x100-0.75_4.dat-0	75	29	10,023	10,623	10,847		10,119	0.95
OR30x100-0.75_5.dat-0	75	29	7,334	7,334	8,411		7,334	0.00
OR30x100-0.75_6.dat-0	75	29	8,783	9,699	9,906		8,785	0.02
OR30x100-0.75_7.dat-0	75	29	9,261	9,757	9,757		9,307	0.49
OR30x100-0.75_8.dat-0	75	29	9,225	9,562	10,097		9,274	0.53
OR30x100-0.75_9.dat-0	75	29	9,660	10,089	10,937		9,733	0.75
OR30x100-0.75_10.dat-0	75	29	8,612	9,138	9,138		8,688	0.87
OR30x100-0.75_1.dat-50	75	15	13,858	15,421	15,421		14,319	3.22
OR30x100-0.75_2.dat-50	75	15	14,431	16,556	16,626		14,835	2.72
OR30x100-0.75_3.dat-50	75	15	14,312	16,042	16,731		14,622	2.12
OR30x100-0.75_4.dat-50	75	15	14,154	16,050	16,716		14,599	3.05
OR30x100-0.75_5.dat-50	75	15	13,870	15,596	16,815		14,321	3.15
OR30x100-0.75_6.dat-50	75	15	14,160	15,919	15,919		14,557	2.73
OR30x100-0.75_7.dat-50	75	15	10,328	15,766	16,527		14,263	27.59
OR30x100-0.75_8.dat-50	75	15	13,826	15,902	16,438		14,368	3.77
OR30x100-0.75_9.dat-50	75	15	14,221	15,871	15,871		14,668	3.05
OR30x100-0.75_10.dat-50	75	15	14,605	16,851	17,182		15,050	2.96
OR30x100-0.75_1.dat-100	75	1	15,145	15,145	16,706	13,801	15,235	0.00
OR30x100-0.75_2.dat-100	75	1	15,901	15,901	17,154	14,507	15,931	0.00
OR30x100-0.75_3.dat-100	75	1	15,501	15,501	16,795	14,070	15,561	0.00
OR30x100-0.75_4.dat-100	75	1	15,693	15,693	16,687	14,497	15,704	0.00
OR30x100-0.75_5.dat-100	75	1	15,066	15,066	16,563	13,816	15,066	0.00
OR30x100-0.75_6.dat-100	75	1	15,608	15,608	17,235	14,275	15,608	0.00

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR30x100-0.75_7.dat-100	75	1	14,810	14,810	16,254	13,448	14,815	0.00
OR30x100-0.75_8.dat-100	75	1	15,300	15,300	16,672	13,908	15,339	0.00
OR30x100-0.75_9.dat-100	75	1	12,626	15,543	16,977	14,162	15,588	8.89
OR30x100-0.75_10.dat-100	75	1	12,507	16,354	17,842	14,963	16,383	8.51
OR30x250-0.25_1.dat-0	25	29						n/a
OR30x250-0.25_2.dat-0	25	29						n/a
OR30x250-0.25_3.dat-0	25	29						n/a
OR30x250-0.25_4.dat-0	25	29						n/a
OR30x250-0.25_5.dat-0	25	29						n/a
OR30x250-0.25_6.dat-0	25	29						n/a
OR30x250-0.25_7.dat-0	25	29						n/a
OR30x250-0.25_8.dat-0	25	29						n/a
OR30x250-0.25_9.dat-0	25	29						n/a
OR30x250-0.25_10.dat-0	25	29						n/a
OR30x250-0.25_1.dat-50	25	15						n/a
OR30x250-0.25_2.dat-50	25	15						n/a
OR30x250-0.25_3.dat-50	25	15						n/a
OR30x250-0.25_4.dat-50	25	15						n/a
OR30x250-0.25_5.dat-50	25	15						n/a
OR30x250-0.25_6.dat-50	25	15						n/a
OR30x250-0.25_7.dat-50	25	15						n/a
OR30x250-0.25_8.dat-50	25	15						n/a
OR30x250-0.25_9.dat-50	25	15						n/a
OR30x250-0.25_10.dat-50	25	15						n/a
OR30x250-0.25_1.dat-100	25	1	47,685	62,454	62,454	58,091	59,568	2.48
OR30x250-0.25_2.dat-100	25	1	47,281	63,145	63,145	58,939	60,467	2.53
OR30x250-0.25_3.dat-100	25	1	44,762	61,389	61,552	55,280	58,423	5.38
OR30x250-0.25_4.dat-100	25	1	44,104	59,622	59,622	54,631	57,031	4.21
OR30x250-0.25_5.dat-100	25	1	49,566	64,027	64,027	59,421	62,358	4.71
OR30x250-0.25_6.dat-100	25	1	46,755	62,882	64,184	56,954	59,705	4.61
OR30x250-0.25_7.dat-100	25	1	48,882	63,143	63,143	57,442	60,532	5.10
OR30x250-0.25_8.dat-100	25	1	45,023	60,008	60,008	55,218	57,832	4.52
OR30x250-0.25_9.dat-100	25	1	49,081	65,036	65,036	58,335	61,704	5.46
OR30x250-0.25_10.dat-100	25	1	44,752	60,515	60,515	55,118	57,781	4.61
OR30x250-0.50_1.dat-0	50	29					57,878	n/a
OR30x250-0.50_2.dat-0	50	29					55,862	n/a
OR30x250-0.50_3.dat-0	50	29					53,747	n/a
OR30x250-0.50_4.dat-0	50	29					53,794	n/a
OR30x250-0.50_5.dat-0	50	29					51,556	n/a
OR30x250-0.50_6.dat-0	50	29					54,634	n/a
OR30x250-0.50_7.dat-0	50	29					52,579	n/a
OR30x250-0.50_8.dat-0	50	29					55,431	n/a
OR30x250-0.50_9.dat-0	50	29					54,788	n/a
OR30x250-0.50_10.dat-0	50	29					53,757	n/a
OR30x250-0.50_1.dat-50	50	15					69,476	n/a
OR30x250-0.50_2.dat-50	50	15					71,164	n/a
OR30x250-0.50_3.dat-50	50	15					69,324	n/a
OR30x250-0.50_4.dat-50	50	15					70,897	n/a
OR30x250-0.50_5.dat-50	50	15					68,846	n/a
OR30x250-0.50_6.dat-50	50	15					71,186	n/a
OR30x250-0.50_7.dat-50	50	15					67,987	n/a
OR30x250-0.50_8.dat-50	50	15					67,765	n/a
OR30x250-0.50_9.dat-50	50	15					69,846	n/a
OR30x250-0.50_10.dat-50	50	15					68,031	n/a

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR30x250-0.50_1.dat-100	50	1	47,439	76,643	76,643	66,200	71,552	7.48
OR30x250-0.50_2.dat-100	50	1	51,684	78,437	78,796	66,953	72,889	8.14
OR30x250-0.50_3.dat-100	50	1	47,608	77,047	77,047	66,136	72,226	8.43
OR30x250-0.50_4.dat-100	50	1	46,976	74,221	74,221	65,658	70,800	7.26
OR30x250-0.50_5.dat-100	50	1	46,896	75,191	75,191	65,974	71,547	7.79
OR30x250-0.50_6.dat-100	50	1	52,961	75,033	75,033	66,853	71,874	6.99
OR30x250-0.50_7.dat-100	50	1	51,425	74,651	74,651	65,133	70,073	7.05
OR30x250-0.50_8.dat-100	50	1	45,524	72,609	72,609	63,845	68,983	7.45
OR30x250-0.50_9.dat-100	50	1	51,372	73,939	73,939	65,269	70,341	7.21
OR30x250-0.50_10.dat-100	50	1	51,170	72,859	72,859	64,963	69,569	6.62
OR30x250-0.75_1.dat-0	75	29					20,531	n/a
OR30x250-0.75_2.dat-0	75	29					23,922	n/a
OR30x250-0.75_3.dat-0	75	29					22,432	n/a
OR30x250-0.75_4.dat-0	75	29					24,409	n/a
OR30x250-0.75_5.dat-0	75	29	17,158	24,747			21,944	21.81
OR30x250-0.75_6.dat-0	75	29					22,434	n/a
OR30x250-0.75_7.dat-0	75	29					22,141	n/a
OR30x250-0.75_8.dat-0	75	29					22,793	n/a
OR30x250-0.75_9.dat-0	75	29					22,874	n/a
OR30x250-0.75_10.dat-0	75	29	17,213	27,191			23,629	27.15
OR30x250-0.75_1.dat-50	75	15	23,195	40,774			34,678	33.11
OR30x250-0.75_2.dat-50	75	15	23,642	41,815			34,967	32.39
OR30x250-0.75_3.dat-50	75	15	23,943	40,383			34,184	29.96
OR30x250-0.75_4.dat-50	75	15	23,596	41,271			35,785	34.06
OR30x250-0.75_5.dat-50	75	15	24,119	41,200			35,026	31.14
OR30x250-0.75_6.dat-50	75	15	23,627	40,846			35,722	33.86
OR30x250-0.75_7.dat-50	75	15	23,374	40,010			34,015	31.28
OR30x250-0.75_8.dat-50	75	15	24,081	41,760			35,598	32.35
OR30x250-0.75_9.dat-50	75	15	23,681	41,135			35,634	33.54
OR30x250-0.75_10.dat-50	75	15	23,516	41,566			34,659	32.15
OR30x250-0.75_1.dat-100	75	1	35,895	36,063	40,966	34,582	36,237	0.47
OR30x250-0.75_2.dat-100	75	1	36,220	36,430	40,815	34,976	36,504	0.58
OR30x250-0.75_3.dat-100	75	1	36,391	36,609	41,476	35,115	36,684	0.60
OR30x250-0.75_4.dat-100	75	1	36,846	37,032	41,307	35,512	37,261	0.50
OR30x250-0.75_5.dat-100	75	1	36,130	36,271	41,647	34,891	36,418	0.39
OR30x250-0.75_6.dat-100	75	1	36,439	36,648	41,492	35,114	36,771	0.57
OR30x250-0.75_7.dat-100	75	1	35,243	35,461	40,829	33,992	35,508	0.61
OR30x250-0.75_8.dat-100	75	1	37,215	37,353	42,512	35,966	37,562	0.37
OR30x250-0.75_9.dat-100	75	1	36,702	36,862	41,515	35,298	37,109	0.43
OR30x250-0.75_10.dat-100	75	1	36,601	36,808	41,351	35,367	36,915	0.56
OR30x500-0.25_1.dat-0	25	29						n/a
OR30x500-0.25_2.dat-0	25	29						n/a
OR30x500-0.25_3.dat-0	25	29						n/a
OR30x500-0.25_4.dat-0	25	29						n/a
OR30x500-0.25_5.dat-0	25	29						n/a
OR30x500-0.25_6.dat-0	25	29						n/a
OR30x500-0.25_7.dat-0	25	29						n/a
OR30x500-0.25_8.dat-0	25	29						n/a
OR30x500-0.25_9.dat-0	25	29						n/a
OR30x500-0.25_10.dat-0	25	29						n/a
OR30x500-0.25_1.dat-50	25	15						n/a
OR30x500-0.25_2.dat-50	25	15						n/a
OR30x500-0.25_3.dat-50	25	15						n/a
OR30x500-0.25_4.dat-50	25	15						n/a

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR30x500-0.25_5.dat-50	25	15						n/a
OR30x500-0.25_6.dat-50	25	15						n/a
OR30x500-0.25_7.dat-50	25	15						n/a
OR30x500-0.25_8.dat-50	25	15						n/a
OR30x500-0.25_9.dat-50	25	15						n/a
OR30x500-0.25_10.dat-50	25	15						n/a
OR30x500-0.25_1.dat-100	25	1	89,570	122,025	122,025	110,666	117,538	5.85
OR30x500-0.25_2.dat-100	25	1	91,797	121,733	121,733	109,040	116,234	6.19
OR30x500-0.25_3.dat-100	25	1	95,947	127,974	127,974	115,167	123,627	6.84
OR30x500-0.25_4.dat-100	25	1	92,444	123,519	123,519	110,414	117,614	6.12
OR30x500-0.25_5.dat-100	25	1	91,811	121,916	121,916	111,642	116,721	4.35
OR30x500-0.25_6.dat-100	25	1	92,537	127,402	127,402	112,511	120,018	6.25
OR30x500-0.25_7.dat-100	25	1	86,807	118,761	118,761	106,181	113,072	6.09
OR30x500-0.25_8.dat-100	25	1	88,265	123,281	123,281	109,231	114,185	4.34
OR30x500-0.25_9.dat-100	25	1	88,718	121,264	121,264	108,776	115,608	5.91
OR30x500-0.25_10.dat-100	25	1	93,353	124,000	124,000	111,669	119,793	6.78
OR30x500-0.50_1.dat-0	50	29					108,021	n/a
OR30x500-0.50_2.dat-0	50	29					107,585	n/a
OR30x500-0.50_3.dat-0	50	29					103,067	n/a
OR30x500-0.50_4.dat-0	50	29					108,935	n/a
OR30x500-0.50_5.dat-0	50	29					108,370	n/a
OR30x500-0.50_6.dat-0	50	29					109,852	n/a
OR30x500-0.50_7.dat-0	50	29					101,196	n/a
OR30x500-0.50_8.dat-0	50	29					104,091	n/a
OR30x500-0.50_9.dat-0	50	29					109,002	n/a
OR30x500-0.50_10.dat-0	50	29					109,755	n/a
OR30x500-0.50_1.dat-50	50	15					142,560	n/a
OR30x500-0.50_2.dat-50	50	15					140,696	n/a
OR30x500-0.50_3.dat-50	50	15					140,130	n/a
OR30x500-0.50_4.dat-50	50	15					140,690	n/a
OR30x500-0.50_5.dat-50	50	15					139,185	n/a
OR30x500-0.50_6.dat-50	50	15					140,397	n/a
OR30x500-0.50_7.dat-50	50	15					137,570	n/a
OR30x500-0.50_8.dat-50	50	15					138,542	n/a
OR30x500-0.50_9.dat-50	50	15					143,174	n/a
OR30x500-0.50_10.dat-50	50	15					138,253	n/a
OR30x500-0.50_1.dat-100	50	1	92,736	148,902	148,902	121,643	140,962	13.71
OR30x500-0.50_2.dat-100	50	1	92,207	147,816	147,816	123,890	138,905	10.81
OR30x500-0.50_3.dat-100	50	1	92,248	148,928	148,928	123,199	139,553	11.72
OR30x500-0.50_4.dat-100	50	1	94,098	147,747	147,747	125,054	141,383	11.55
OR30x500-0.50_5.dat-100	50	1	92,187	145,957	145,957	123,385	138,983	11.22
OR30x500-0.50_6.dat-100	50	1	93,745	147,913	147,913	125,045	142,158	12.04
OR30x500-0.50_7.dat-100	50	1	92,239	147,833	147,833	125,268	140,414	10.79
OR30x500-0.50_8.dat-100	50	1	92,092	146,197	146,197	120,078	137,682	12.79
OR30x500-0.50_9.dat-100	50	1	92,908	147,501	147,501	123,625	139,773	11.55
OR30x500-0.50_10.dat-100	50	1	91,157	146,327	146,435	121,197	137,127	11.62
OR30x500-0.75_1.dat-0	75	29					43,478	n/a
OR30x500-0.75_2.dat-0	75	29					44,990	n/a
OR30x500-0.75_3.dat-0	75	29					42,837	n/a
OR30x500-0.75_4.dat-0	75	29					44,294	n/a
OR30x500-0.75_5.dat-0	75	29					46,644	n/a
OR30x500-0.75_6.dat-0	75	29					45,373	n/a
OR30x500-0.75_7.dat-0	75	29					45,390	n/a
OR30x500-0.75_8.dat-0	75	29					48,711	n/a

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Instance	Budget	Foll. rows	Publ. LB	Publ. UB	BOBILib	Our LB	Our point	Gap (%)
OR30x500-0.75_9.dat-0	75	29					43,827	n/a
OR30x500-0.75_10.dat-0	75	29					45,775	n/a
OR30x500-0.75_1.dat-50	75	15					68,720	n/a
OR30x500-0.75_2.dat-50	75	15					69,640	n/a
OR30x500-0.75_3.dat-50	75	15					67,961	n/a
OR30x500-0.75_4.dat-50	75	15					68,523	n/a
OR30x500-0.75_5.dat-50	75	15					70,609	n/a
OR30x500-0.75_6.dat-50	75	15					68,309	n/a
OR30x500-0.75_7.dat-50	75	15					70,946	n/a
OR30x500-0.75_8.dat-50	75	15					69,553	n/a
OR30x500-0.75_9.dat-50	75	15					69,181	n/a
OR30x500-0.75_10.dat-50	75	15					69,553	n/a
OR30x500-0.75_1.dat-100	75	1	70,260	71,066	102,997	69,454	71,091	1.13
OR30x500-0.75_2.dat-100	75	1	54,615	72,321	102,633	70,624	72,265	2.27
OR30x500-0.75_3.dat-100	75	1	69,924	70,774	97,761	69,122	70,777	1.20
OR30x500-0.75_4.dat-100	75	1	70,645	71,580	95,980	69,315	71,456	1.13
OR30x500-0.75_5.dat-100	75	1	72,234	73,108	103,856	70,847	73,245	1.20
OR30x500-0.75_6.dat-100	75	1	70,312	71,195	95,892	69,402	71,025	1.00
OR30x500-0.75_7.dat-100	75	1	72,604	74,658	92,638	71,534	73,314	0.97
OR30x500-0.75_8.dat-100	75	1	70,708	71,683	103,231	69,893	71,592	1.23
OR30x500-0.75_9.dat-100	75	1	71,479	72,484	96,781	70,710	72,384	1.25
OR30x500-0.75_10.dat-100	75	1	71,384	72,146	91,017	70,457	72,101	0.99

Table S6. OR instances whose published bounds need care. Publ. LB: Lozano, Bergman and Cire's lower bound; L&S, time limit: the smallest value in Lefebvre and Schmidt's run files from a run stopped at the time limit, a kind of row their own analysis does not count and that can carry a relaxation value; it would be the best published point if it counted (blank: none). Best publ. point: the best published point used in the paper (Lozano, Bergman and Cire's, Lefebvre and Schmidt's from a run that ended in time, or BOBILib's). Our point: the verified solution file of this work (blank: none). For `OR10x100-0.75_4.dat-0` the published lower bound exceeds the published point; it is closed at 8,813 and left out of the paired median gaps.

Instance	Publ. LB	L&S, time limit	Best publ. point	Our point	Note
OR5x100-0.25_7.dat-50	17,106	16,951	22,366	22,125	below the published LB
OR5x100-0.25_8.dat-0	15,874	10,859	20,050	19,849	below the published LB
OR10x100-0.25_10.dat-50	14,825	16,399	20,264	19,924	not counted
OR10x100-0.25_4.dat-50	15,804	8,090	20,754		below the published LB
OR10x100-0.25_5.dat-100	20,096	23,151	23,916	23,417	not counted
OR10x100-0.25_6.dat-0	14,875	7,726	19,330	19,372	below the published LB
OR10x100-0.25_8.dat-0	14,546	17,533	18,738	18,854	not counted
OR10x100-0.75_4.dat-0	8,814	45,902	8,813	8,813	LB above point; closed (Section C.5)
OR10x250-0.25_4.dat-50	35,545	40,735	54,623		not counted
OR10x250-0.25_7.dat-50	35,749	36,279	55,317		not counted
OR10x250-0.50_3.dat-0	37,856	6,575	61,853	54,326	below the published LB
OR10x250-0.50_3.dat-50	45,192	46,801	80,303	71,263	not counted
OR10x250-0.50_7.dat-0	37,201	52,992	61,372	53,302	not counted
OR10x500-0.25_2.dat-50	67,265	7,083	112,795		below the published LB
OR10x500-0.25_5.dat-50	67,175	20,909	112,086		below the published LB
OR30x100-0.25_2.dat-50	13,955	4,917	19,468		below the published LB
OR30x100-0.25_5.dat-50	14,772	13,127	20,202		below the published LB
OR30x100-0.25_6.dat-50	14,488	14,393	19,788		below the published LB
OR30x100-0.25_8.dat-50	14,328	16,674	19,699		not counted

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Instance	Publ. LB	L&S, time limit	Best publ. point	Our point	Note
OR30x100-0.25_9.dat-50	15,173	19,139	20,935		not counted
OR30x100-0.50_7.dat-0	15,351	16,014	24,028	21,739	not counted
OR30x250-0.50_1.dat-100	47,439	66,391	76,643	71,552	not counted

I.4 Clique interdiction

Table S7. Edge clique interdiction, BOBILib set `clique`: n vertices, density d, m edges, budget k (the leader deletes at most k edges). Lower bound: Mantel's theorem ($m-k > \lfloor n^2/4 \rfloor$, so a triangle survives) or Brouwer's 1981 Theorem 2 ($m-k \geq t_3(n) - \lfloor n/3 \rfloor + 2$, so a K_4-free remainder is 3-partite) together with the proof that every 3-partition has more than k edges inside its parts ($w_3(G) > k$): 170 part-size cases, each decided by solvers and by exhaustive integer searches, and each with a branch-and-bound certificate that the integer checker `scripts/check_clique_certificates.py` verifies. Upper bound: the solution file deletes the edges inside an r-partition (certificate: r and the number of inside edges, at most k), leaving an r-partite graph. Both bounds are exact certificates (class (i) of Section A.4). Open: the best 3-partitions found for normalClique60-70-2, normalClique60-70-3, normalClique60-70-5 have 311, 310, 310 inside edges, more than $k = 309$, so only $3 \leq$ value ≤ 4 is known.

Instance	n	d	m	k	BOBILib	LB	Argument	UB	Certificate	Status
normalClique40-70-1	40	0.7	546	136	6	3	Mantel	3	3-partition, 125	optimal
normalClique40-70-2	40	0.7	546	136	6	3	Mantel	3	3-partition, 125	optimal
normalClique40-70-3	40	0.7	546	136	6	3	Mantel	3	3-partition, 126	optimal
normalClique40-70-4	40	0.7	546	136	5	3	Mantel	3	3-partition, 125	optimal
normalClique40-70-5	40	0.7	546	136	7	3	Mantel	3	3-partition, 128	optimal
normalClique40-70-6	40	0.7	546	136	6	3	Mantel	3	3-partition, 126	optimal
normalClique40-70-7	40	0.7	546	136	5	3	Mantel	3	3-partition, 123	optimal
normalClique40-70-8	40	0.7	546	136	6	3	Mantel	3	3-partition, 126	optimal
normalClique40-70-9	40	0.7	546	136	6	3	Mantel	3	3-partition, 125	optimal
normalClique40-70-10	40	0.7	546	136	6	3	Mantel	3	3-partition, 125	optimal
normalClique40-90-1	40	0.9	702	175	10	4	Brouwer $+ w_3(G) > k$	4	4-partition, 131	optimal
normalClique40-90-2	40	0.9	702	175	10	4	Brouwer $+ w_3(G) > k$	4	4-partition, 129	optimal
normalClique40-90-3	40	0.9	702	175	9	4	Brouwer $+ w_3(G) > k$	4	4-partition, 128	optimal
normalClique40-90-4	40	0.9	702	175	9	4	Brouwer $+ w_3(G) > k$	4	4-partition, 127	optimal
normalClique40-90-5	40	0.9	702	175	10	4	Brouwer $+ w_3(G) > k$	4	4-partition, 128	optimal
normalClique40-90-6	40	0.9	702	175	10	4	Brouwer $+ w_3(G) > k$	4	4-partition, 127	optimal
normalClique40-90-7	40	0.9	702	175	9	4	Brouwer $+ w_3(G) > k$	4	4-partition, 127	optimal
normalClique40-90-8	40	0.9	702	175	11	4	Brouwer $+ w_3(G) > k$	4	4-partition, 129	optimal
normalClique40-90-9	40	0.9	702	175	10	4	Brouwer $+ w_3(G) > k$	4	4-partition, 130	optimal
normalClique40-90-10	40	0.9	702	175	10	4	Brouwer $+ w_3(G) > k$	4	4-partition, 130	optimal
normalClique50-70-1	50	0.7	857	214	7	3	Mantel	3	3-partition, 205	optimal
normalClique50-70-2	50	0.7	857	214	7	3	Mantel	3	3-partition, 205	optimal
normalClique50-70-3	50	0.7	857	214	7	3	Mantel	3	3-partition, 208	optimal
normalClique50-70-4	50	0.7	857	214	7	3	Mantel	3	3-partition, 203	optimal
normalClique50-70-5	50	0.7	857	214	7	3	Mantel	3	3-partition, 205	optimal
normalClique50-70-6	50	0.7	857	214	8	3	Mantel	3	3-partition, 209	optimal
normalClique50-70-7	50	0.7	857	214	8	3	Mantel	3	3-partition, 205	optimal
normalClique50-70-8	50	0.7	857	214	7	3	Mantel	3	3-partition, 204	optimal
normalClique50-70-9	50	0.7	857	214	7	3	Mantel	3	3-partition, 210	optimal
normalClique50-70-10	50	0.7	857	214	8	3	Mantel	3	3-partition, 208	optimal
normalClique50-90-1	50	0.9	1,102	275	11	4	Brouwer $+ w_3(G) > k$	4	4-partition, 210	optimal
normalClique50-90-2	50	0.9	1,102	275	11	4	Brouwer $+ w_3(G) > k$	4	4-partition, 212	optimal
normalClique50-90-3	50	0.9	1,102	275	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 210	optimal
normalClique50-90-4	50	0.9	1,102	275	11	4	Brouwer $+ w_3(G) > k$	4	4-partition, 210	optimal
normalClique50-90-5	50	0.9	1,102	275	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 212	optimal

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Instance	n	d	m	k	BOBILib	LB	Argument	UB	Certificate	Status
normalClique50-90-6	50	0.9	1,102	275	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 212	optimal
normalClique50-90-7	50	0.9	1,102	275	11	4	Brouwer $+ w_3(G) > k$	4	4-partition, 210	optimal
normalClique50-90-8	50	0.9	1,102	275	13	4	Brouwer $+ w_3(G) > k$	4	4-partition, 211	optimal
normalClique50-90-9	50	0.9	1,102	275	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 209	optimal
normalClique50-90-10	50	0.9	1,102	275	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 209	optimal
normalClique60-70-1	60	0.7	1,239	309	8	3	Mantel	3	3-partition, 306	optimal
normalClique60-70-2	60	0.7	1,239	309	9	3	Mantel	4	4-partition, 202	**open**
normalClique60-70-3	60	0.7	1,239	309	9	3	Mantel	4	4-partition, 200	**open**
normalClique60-70-4	60	0.7	1,239	309	9	3	Mantel	3	3-partition, 309	optimal
normalClique60-70-5	60	0.7	1,239	309	8	3	Mantel	4	4-partition, 202	**open**
normalClique60-70-6	60	0.7	1,239	309	9	3	Mantel	3	3-partition, 306	optimal
normalClique60-70-7	60	0.7	1,239	309	8	3	Mantel	3	3-partition, 307	optimal
normalClique60-70-8	60	0.7	1,239	309	8	3	Mantel	3	3-partition, 308	optimal
normalClique60-70-9	60	0.7	1,239	309	8	3	Mantel	3	3-partition, 307	optimal
normalClique60-70-10	60	0.7	1,239	309	8	3	Mantel	3	3-partition, 308	optimal
normalClique60-90-1	60	0.9	1,593	398	13	4	Brouwer $+ w_3(G) > k$	4	4-partition, 314	optimal
normalClique60-90-2	60	0.9	1,593	398	13	4	Brouwer $+ w_3(G) > k$	4	4-partition, 312	optimal
normalClique60-90-3	60	0.9	1,593	398	13	4	Brouwer $+ w_3(G) > k$	4	4-partition, 311	optimal
normalClique60-90-4	60	0.9	1,593	398	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 313	optimal
normalClique60-90-5	60	0.9	1,593	398	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 311	optimal
normalClique60-90-6	60	0.9	1,593	398	13	4	Brouwer $+ w_3(G) > k$	4	4-partition, 314	optimal
normalClique60-90-7	60	0.9	1,593	398	13	4	Brouwer $+ w_3(G) > k$	4	4-partition, 313	optimal
normalClique60-90-8	60	0.9	1,593	398	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 310	optimal
normalClique60-90-9	60	0.9	1,593	398	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 312	optimal
normalClique60-90-10	60	0.9	1,593	398	12	4	Brouwer $+ w_3(G) > k$	4	4-partition, 317	optimal

I.5 Network interdiction

Table S8. The two attack-protect games (`inter-fire`) BOBILib lists as feasible, proven optimal by enumerating every attack of at most Φ nodes, with the follower (protect at most Λ nodes) solved exactly by a tree dynamic program and the minimiser re-solved by HiGHS. Attacks: every attack of 0 to Φ nodes, $\sum_{k=0}^{\Phi}\binom{49}{k} = 2{,}138{,}410$ per instance (1,906,884 of them with exactly $\Phi = 5$ nodes), each enumerated once. Optimal attacks: attacks at which the follower saves only the optimum. Time: wall seconds of the enumeration on one thread of a loaded machine. The optima are exact certificates (class (i) of Section A.4).

Instance	Nodes	Edges	Φ	Λ	Attacks	Optimum	Optimal attacks	BOBILib	Time (s)
tree-50_1-5-5_006	49	42	5	5	2,138,410	26	4	26	80
tree-50_1-5-5_010	49	44	5	5	2,138,410	26	28	26	70

Table S9. The 34 tree instances BOBILib lists as optimal, re-derived by the same enumeration: instances per group, nodes (after removing the vaccinated ones), budgets, attacks (every attack of 0 to Φ nodes, $\sum_{k=0}^{\Phi}\binom{n}{k}$ for n nodes) per instance and in all over the group's instances, the optimal values found, and how many equal BOBILib's value with a certified solution file. The 34 trees have 5,063,940 attacks; with the two trees of Table S8 the 36 trees have 9,340,760, and with the 4 `rndgraph-25` instances that are forests (4,142 attacks) the 40 forests of `inter-fire` have 9,344,902, the attacks at which the review compared its dynamic program with ours (Section F).

Group	Inst.	Nodes	Φ	Λ	Attacks per instance	Attacks in all	Optima	Agree
tree-25_1-3-3	4	24	3	3	2,325	9,300	12, 13, 13, 13	4/4
tree-25_2-2-2	4	23	2	2	277	1,108	17, 17, 17, 18	4/4

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Group	Inst.	Nodes	Φ	Λ	Attacks per instance	Attacks in all	Optima	Agree
tree-25_3-3-3	4	22	3	3	1,794	7,176	15, 16, 16, 17	4/4
tree-50_1-3-3	4	49	3	3	19,650	78,600	28, 28, 30, 32	4/4
tree-50_1-5-5	2	49	5	5	2,138,410	4,276,820	24, 25	2/2
tree-50_3-3-1	4	47	3	1	17,344	69,376	26, 27, 30, 31	4/4
tree-80_2-2-2	4	78	2	2	3,082	12,328	56, 57, 61, 63	4/4
tree-80_3-3-1	4	77	3	1	76,154	304,616	45, 45, 48, 48	4/4
tree-80_3-3-3	4	77	3	3	76,154	304,616	55, 56, 56, 58	4/4
Total	34					5,063,940		34/34

I.6 General instances

Table S10. The 40 open MIPLIB-derived instances with no bilevel feasible point: the follower's own rows admit a recession direction r (follower variables only) with $f^\top r < 0$, for every leader choice (Meyer 1974). Support: nonzero entries of r; $f^\top r$ rounded to four decimals; exact: checked in exact rational arithmetic by `scripts/verify_ray.py`, an exact certificate (class (i) of Section A.4). For `gen-ip054_0_100`, `gen-ip054_50_50`, `supportcase42_0_100` and `supportcase42_50_50` the certificate is the exact direction of `rays/`*name*`.exact.ray.json` (repaired from a floating-point direction, or a single follower variable found by `scripts/exact_ray.py`).

Instance	Collection	Support	$f^\top r$	Exact
biella1_50_50	miplib2010	621	−444,178,600	yes
binkar10_1_50_50	miplib2010	208	−90,000,000	yes
brazil3_0_100	miplib2017	3	−1	yes
brazil3_50_50	miplib2017	11	−1	yes
buildingenergy_0_100	miplib2017	17	−10,766.212	yes
buildingenergy_50_50	miplib2017	8,777	−11,236.372	yes
core2536-691_50_50	miplib2010	9	−200.9	yes
cost266-UUE_50_50	miplib2017	2,024	−13,041.23	yes
gen-ip054_0_100	miplib2017	9	−1,475.2477	yes
gen-ip054_50_50	miplib2017	13	−2,718.065	yes
highschool1-aigio_0_100	miplib2017	3	−1	yes
irish-electricity_0_100	miplib2017	5,206	−5,206	yes
irish-electricity_50_50	miplib2017	5,206	−5,206	yes
lotsize_50_50	miplib2017	297	−871	yes
mad_0_100	miplib2017	20	−20	yes
mad_50_50	miplib2017	20	−20	yes
neos-476283_0_100	miplib2010	369	−369	yes
neos-476283_50_50	miplib2010	369	−369	yes
neos-1396125_0_100	miplib2010	3	−240	yes
neos-1396125_50_50	miplib2010	3	−240	yes
neos-3627168-kasai_50_50	miplib2017	3	−3	yes
neos-5093327-huahum_0_100	miplib2017	3,968	−3,968	yes
neos-5093327-huahum_50_50	miplib2017	3,968	−3,968	yes
nu25-pr12_0_100	miplib2017	36	−9,000	yes
rail507_0_100	miplib2010	10	−201	yes
rail507_50_50	miplib2010	10	−201	yes
rmatr100-p5_0_100	miplib2010	4,392	−117,792	yes
rmatr100-p5_50_50	miplib2010	4,392	−117,792	yes
rmatr100-p10_0_100	miplib2010	3,679	−61,179	yes
rmatr100-p10_50_50	miplib2010	3,679	−61,179	yes
rmatr200-p5_0_100	miplib2017	18,908	−988,708	yes
rmatr200-p5_50_50	miplib2017	18,908	−988,708	yes

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Instance	Collection	Support	$f^\top r$	Exact
snp-02-004-104_0_100	miplib2017	104	−10,400,000	yes
snp-02-004-104_50_50	miplib2017	104	−10,400,000	yes
supportcase40_0_100	miplib2017	3,360	−338,410.0838	yes
supportcase40_50_50	miplib2017	3,360	−338,410.0838	yes
supportcase42_0_100	miplib2017	1	−1	yes
supportcase42_50_50	miplib2017	1	−1	yes
tr12-30_50_50	miplib2017	180	−405	yes
trento1_50_50	miplib2017	655	−33,097,000	yes

Table S11. Seven MIPLIB-derived instances whose high point relaxation is exactly the MIPLIB model they were built from (checked in exact arithmetic by `scripts/compare_hpr.py`), so the published MIPLIB optimum (MIPLIB 2017's solution file) is a lower bound; a verified bilevel feasible point at that value is optimal by accepted benchmark conventions. The MIPLIB optima are floating-point results of MILP solvers (class (ii) of Section A.4), not exact-arithmetic proofs: the objectives of noswot, ex10 and neos-1445765 take integer values only (integer coefficients on integer variables), and those of pigeon-10 multiples of 1,000 only, so their published bounds round up to −41, 100, −17,783 and −9,000 and the results are safe if those bounds are valid; for timtab1 the agreement of point and bound is numerical. BOBILib: the instance's status before this work. Lefebvre and Schmidt's run files also report ex10_0_100 optimal at 100; their paper does not state it, so it is not counted as published. (neos-950242_0_100 meets its bound too, but was already listed as optimal in an earlier BOBILib release and is counted as published.)

Instance	BOBILib	MIPLIB instance	MIPLIB optimum	Bilevel optimum	Point from
noswot_0_100	feasible	noswot	−41.00000885	−41	BOBILib's solution file
noswot_50_50	feasible	noswot	−41.00000885	−41	BOBILib's solution file
pigeon-10_0_100	open	pigeon-10	−9,000	−9,000	`bobi general`
timtab1_0_100	open	timtab1	764,771.99999978	764,771.99999978	`bobi general`
pigeon-10_50_50	open	pigeon-10	−9,000	−9,000	`bobi coupled`
ex10_0_100	open	ex10	100	100	`bobi general`
neos-1445765_0_100	open	neos-1445765	−17,783	−17,783	`bobi general`

Table S12. The 88 verified bilevel feasible points for open MIPLIB-derived instances, for which BOBILib lists no point. ‡: a point had been published elsewhere (its value and source are in the notes of `instances.csv`); for the other 79 this is the first point known. Coupling: whether leader rows contain follower variables. Method: `hpr-f-t`, the high point relaxation, then the follower solved with the leader fixed, then an optimistic tie-break (`bobi general`); `hpr`, `penalty`, `follow`: the part of the `bobi coupled` portfolio that found the point (the HPR's own leader values, a penalised HPR, best-response dynamics). Values rounded to four decimals (full precision in `instances.csv`); the leader minimises, so does the follower. Every file passes `bobi verify --tight` (the follower's value within $\max(10^{-6}, 10^{-9}|\varphi|)$ of its proven optimum), so every point is solver-certified (class (ii) of Section A.4). §: one of the 26 points re-derived under that strict tolerance by `bobi coupled-tighten`, at the leader values of a superseded file that had used BOBILib's tolerance of 10^{-4} on the follower's value (kept in `tolerance/`, Section E). Five further points are optimal (Table S11).

Instance	Collection	Coupling	Method	Leader value	Follower value
air05-0.900000‡	miplib3	no	hpr-f-t	64,955	−63,283
beasleyC3_0_100	miplib2010	no	hpr-f-t	6,844	−6,844
beasleyC3_50_50	miplib2010	yes	hpr	6,844	−6,844
binkar10_1_0_100§	miplib2010	no	hpr-f-t	5,006,811.08	−5,000,741.82
cap6000-0.500000‡	miplib3	no	hpr-f-t	−1,169,217	0
cap6000-0.900000‡	miplib3	no	hpr-f-t	−100,780	45,829
cmflsp50-24-8-8_0_100§	miplib2017	no	hpr-f-t	59,480,214.6542	−12,534,898.0379

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Instance	Collection	Coupling	Method	Leader value	Follower value
core2536-691_0_100§	miplib2010	no	hpr-f-t	10,000,000,000	−9,999,999,544
cost266-UUE_0_100§	miplib2017	no	hpr-f-t	35,801,082	−10,431,022.02
exp-1-500-5-5_0_100§	miplib2017	no	hpr-f-t	912,511	−879,352
h80x6320d_0_100	miplib2017	no	hpr-f-t	12,461.7042	−10,960.0714
ic97_potential_0_100§	miplib2017	no	hpr-f-t	4,003	−353
ic97_potential_50_50§	miplib2017	yes	penalty	4,197	−592
lotsize_0_100	miplib2017	no	hpr-f-t	3,691,725	−3,059,545
mc11_0_100	miplib2017	no	hpr-f-t	128,960	−128,960
mc11_50_50	miplib2017	yes	hpr	128,960	−128,960
mik-250-1-100-1_0_100	miplib2010	no	hpr-f-t	890,671	−1,000,000
mik-250-1-100-1_50_50	miplib2010	yes	hpr	890,671	−1,000,000
mik-250-20-75-4_0_100	miplib2017	no	hpr-f-t	663,799	−750,000
mik-250-20-75-4_50_50	miplib2017	yes	hpr	663,799	−750,000
mitre-0.500000‡	miplib3	no	hpr-f-t	147,075	−96,600
mitre-0.900000‡	miplib3	no	hpr-f-t	169,500	−169,500
mushroom-best_0_100	miplib2017	no	hpr-f-t	4,230.0553	−4,230
mushroom-best_50_50	miplib2017	yes	hpr	4,230.0553	−4,230
n4-3_0_100§	miplib2010	no	hpr-f-t	34,802,896	−34,801,026
n4-3_50_50§	miplib2010	yes	penalty	50,621,410	−42,710,515
n5-3_0_100§	miplib2017	no	hpr-f-t	30,001,635	−30,000,665
n5-3_50_50§	miplib2017	yes	hpr	30,001,775	−30,000,805
n9-3_0_100§	miplib2017	no	hpr-f-t	50,404,633	−50,401,629
n9-3_50_50§	miplib2017	yes	penalty	74,423,855	−62,411,805
neos17_0_100	miplib2017	no	hpr-f-t	0.6872	−0.564
neos17_50_50§	miplib2017	yes	hpr	0.6872	−0.564
neos-631710_0_100‡	miplib2017	no	hpr-f-t	556	−556
neos-827175_0_100	miplib2017	no	hpr-f-t	122.0015	−122
neos-827175_50_50	miplib2017	yes	hpr	122.0015	−122
neos-860300_0_100	miplib2017	no	hpr-f-t	5,920	−3,702
neos-873061_0_100§	miplib2017	no	hpr-f-t	258,644.4199	−258,644.1933
neos-916792_0_100	miplib2010	no	hpr-f-t	32.6951	−5.6656
neos-933966_0_100	miplib2017	no	hpr-f-t	326	−189
neos-933966_50_50	miplib2017	yes	hpr	47,506	−224
neos-934278_0_100	miplib2010	no	hpr-f-t	6,339	−267
neos-934278_50_50	miplib2010	yes	hpr	32,490	−299
neos-950242_50_50	miplib2017	yes	follow	24	−24
neos-1122047_0_100	miplib2017	no	hpr-f-t	472	−472
neos-1337307_50_50	miplib2010	yes	hpr	−201,272	27,257
neos-1445765_50_50	miplib2017	yes	penalty	0	0
neos-1456979_0_100	miplib2017	no	hpr-f-t	250	−157
neos-2978193-inde_0_100§	miplib2017	no	hpr-f-t	61.454	−62.3292
neos-2978193-inde_50_50§	miplib2017	yes	hpr	69.2751	−69.5863
neos-3216931-puriri_0_100	miplib2017	no	hpr-f-t	81,430	900
neos-3627168-kasai_0_100	miplib2017	no	hpr-f-t	1,591,085.62	−1,591,085.62
neos-3754480-nidda_0_100	miplib2017	no	hpr-f-t	13,973.2905	−2,052,644.3591
neos-4300652-rahue_0_100	miplib2017	no	hpr-f-t	6.4075	−2.3316
neos-4722843-widden_0_100	miplib2017	no	hpr-f-t	25,109.6634	−7,400
neos-4763324-toguru_0_100	miplib2017	no	hpr-f-t	17,296.7271	−13,725.617
neos-4954672-berkel_0_100	miplib2017	no	hpr-f-t	8,891,491	−7,334,836
neos-4954672-berkel_50_50	miplib2017	yes	hpr	8,861,094	−7,334,836
neos-5188808-nattai_0_100	miplib2017	no	hpr-f-t	6.1216	−6.1216
neos-5188808-nattai_50_50	miplib2017	yes	hpr	8.1556	−8.1556
netdiversion_0_100§	miplib2010	no	hpr-f-t	19,850,250	−19,850,250
ns1830653_0_100	miplib2010	no	hpr-f-t	23,622	−1,910

continued on the next page

(continued)

Instance	Collection	Coupling	Method	Leader value	Follower value
opm2-z10-s4_0_100	miplib2017	no	hpr-f-t	−11,630	4,812
p200x1188c_0_100	miplib2017	no	hpr-f-t	3,584,430	−3,584,430
p200x1188c_50_50	miplib2017	yes	hpr	3,584,430	−3,584,430
p0548-0.500000‡	miplib3	no	hpr-f-t	22,197	−17,695
pg5_34_0_100§	miplib2010	no	hpr-f-t	−11,516.2869	1,824.44
pg5_34_50_50	miplib2010	yes	penalty	−11,591.1699	0
pg_0_100§	miplib2017	no	hpr-f-t	−5,364.2592	2,018.342
pg_50_50§	miplib2017	yes	follow	−6,061.0681	575.8461
ran16x16_0_100	miplib2010	no	hpr-f-t	43,000	−42,126
ran16x16_50_50	miplib2010	yes	hpr	43,000	−42,126
rmine6_0_100‡	miplib2010	no	hpr-f-t	−440.6503	201.3995
rmine6_50_50‡	miplib2010	yes	follow	−153.7498	119.897
s100_0_100	miplib2017	no	hpr-f-t	−0.0758	0
s250r10_0_100	miplib2017	no	hpr-f-t	−0.0608	0.0035
satellites1-25_0_100	miplib2010	no	hpr-f-t	32	−32
satellites2-40_0_100	miplib2017	no	hpr-f-t	0	0
seymour1_0_100§	miplib2017	no	hpr-f-t	959.536	−686
seymour1_50_50§	miplib2017	yes	hpr	960.6332	−686
sing44_0_100	miplib2017	no	hpr-f-t	10,059,257.3791	−112,109.371
sing326_0_100	miplib2017	no	hpr-f-t	20,652,331.0689	−181,252.322
sp150x300d_0_100	miplib2017	no	hpr-f-t	300	−300
sp150x300d_50_50	miplib2017	yes	hpr	300	−300
supportcase12_0_100	miplib2017	no	hpr-f-t	−1,811.9515	0
timtab1_50_50	miplib2010	yes	hpr	764,774	−20,655.996
tr12-30_0_100§	miplib2017	no	hpr-f-t	201,019	−92,641
uct-subprob_0_100§	miplib2017	no	hpr-f-t	828	−682
uct-subprob_50_50§	miplib2017	yes	hpr	848	−682

I.7 Unchanged instances

Table S13. Instances whose BOBILib status is unchanged, by family: BOBILib status (feasible or open), and how many now have a verified point better than BOBILib's (marked * in the lists below).

Family	Collections	Feasible	Open	Better point	Total
multidimensional knapsack interdiction	or 582	503	79	462	582
general, pure integer	miplib2017 42, miplib3 20, miplib2010 13, misc 1	60	16	0	76
general, mixed integer	miplib2017 41, miplib2010 6	11	36	0	47
generalized interdiction	generalized 8	8	0	0	8
clique interdiction	clique 3	3	0	3	3
Total		585	131	465	716

Multidimensional knapsack interdiction (582)

OR5x100-0.25_1.dat-0*	OR5x100-0.25_3.dat-100*	OR5x100-0.25_6.dat-50*	OR5x100-0.25_9.dat-0*
OR5x100-0.25_1.dat-50*	OR5x100-0.25_4.dat-0*	OR5x100-0.25_6.dat-100*	OR5x100-0.25_9.dat-50*
OR5x100-0.25_1.dat-100*	OR5x100-0.25_4.dat-50*	OR5x100-0.25_7.dat-0*	OR5x100-0.25_9.dat-100*
OR5x100-0.25_2.dat-0*	OR5x100-0.25_4.dat-100*	OR5x100-0.25_7.dat-50*	OR5x100-0.25_10.dat-0*
OR5x100-0.25_2.dat-50*	OR5x100-0.25_5.dat-0*	OR5x100-0.25_7.dat-100*	OR5x100-0.25_10.dat-50*
OR5x100-0.25_2.dat-100*	OR5x100-0.25_5.dat-50*	OR5x100-0.25_8.dat-0*	OR5x100-0.25_10.dat-100*
OR5x100-0.25_3.dat-0*	OR5x100-0.25_5.dat-100*	OR5x100-0.25_8.dat-50*	OR5x100-0.50_1.dat-0*
OR5x100-0.25_3.dat-50*	OR5x100-0.25_6.dat-0*	OR5x100-0.25_8.dat-100*	OR5x100-0.50_1.dat-50*

OR5x100-0.50_1.dat-100*
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OR30x500-0.75_8.dat-100*
OR30x500-0.75_9.dat-100*
OR30x500-0.75_10.dat-100*

General, pure integer (76)

air03-0.500000
air03-0.900000
air04-0.100000
air04-0.500000
air04-0.900000
air05-0.500000
bab2_0_100
bab5_0_100
bab6_0_100
co-100_0_100
co-100_50_50
comp07-2idx_0_100
comp21-2idx_0_100
cvs16r128-89_0_100
cvs16r128-89_50_50
eilA101-2_0_100
ex10_50_50
gen-ip002_0_100
gen-ip002_50_50
harp2-0.100000
harp2-0.500000
harp2-0.900000
irp_0_100
l152lav-0.500000
l152lav-0.900000
lectsched-5-obj_50_50
milp_10_20_50_2310
mine-90-10_0_100
mine-166-5_0_100
mine-166-5_50_50
mitre-0.100000
mod010-0.500000
mod010-0.900000
neos-631710_50_50
neos-662469_0_100
neos-957323_0_100
neos-1109824_0_100
neos-1337307_0_100
neos-1582420_0_100
neos-1582420_50_50
neos-2657525-crna_0_100
neos-3024952-loue_0_100
neos-3083819-nubu_0_100
neos-5052403-cygnet_0_100
neos-5052403-cygnet_50_50
nursesched-medium-hint03_0_100
nw04-0.500000
nw04-0.900000
opm2-z7-s2_0_100
p2756-0.100000
p2756-0.500000
p2756-0.900000
reblock67_0_100
reblock115_0_100
s100_50_50
seymour-0.100000
sorrell3_0_100
sorrell3_50_50
sp97ar_0_100
sp97ar_50_50
sp98ar_0_100
sp98ar_50_50
sp98ir_0_100
square41_0_100
square41_50_50
square47_0_100
square47_50_50
supportcase6_0_100
supportcase6_50_50
supportcase19_0_100
tanglegram1_0_100
tanglegram1_50_50
tanglegram2_0_100
tanglegram2_50_50
toll-like_0_100
toll-like_50_50

General, mixed integer (47)

atlanta-ip_0_100
biella1_0_100
cmflsp50-24-8-8_50_50
drayage-25-23_0_100
drayage-100-23_0_100
exp-1-500-5-5_50_50
fiball_0_100
gmu-35-40_0_100
gmu-35-40_50_50
gmu-35-50_0_100
gmu-35-50_50_50
h80x6320d_50_50
momentum1_0_100
momentum1_50_50
neos-848589_0_100
neos-848589_50_50
neos-873061_50_50
neos-916792_50_50
neos-1122047_50_50
neos-1456979_50_50
neos-2746589-doon_0_100

neos-3216931-puriri_50_50
neos-3754224-navua_0_100
neos-3754224-navua_50_50
neos-3754480-nidda_50_50
neos-4300652-rahue_50_50
neos-4387871-tavua_0_100
neos-4387871-tavua_50_50
neos-4722843-widden_50_50
neos-4763324-toguru_50_50
neos-5049753-cuanza_0_100
neos-5049753-cuanza_50_50
neos-5104907-jarama_0_100
neos-5104907-jarama_50_50
neos-5114902-kasavu_0_100
neos-5114902-kasavu_50_50
nexp-150-20-8-5_0_100
ns1830653_50_50
ran14x18-disj-8_0_100
satellites1-25_50_50
satellites2-40_50_50
satellites2-60-fs_0_100
satellites2-60-fs_50_50
sing44_50_50
sing326_50_50
supportcase12_50_50
trento1_0_100

Generalized interdiction (8)

general20-30-20-20-20-2
general20-30-20-20-20-9
general20-30-20-20-20-10
general30-30-20-20-20-1
general30-30-20-20-20-2
general30-30-30-20-20-1
general30-30-30-20-20-2
general30-30-30-20-20-6

Clique interdiction (3)

normalClique60-70-2*
normalClique60-70-3*
normalClique60-70-5*

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